MA4266 Introduction to Algebraic Topology

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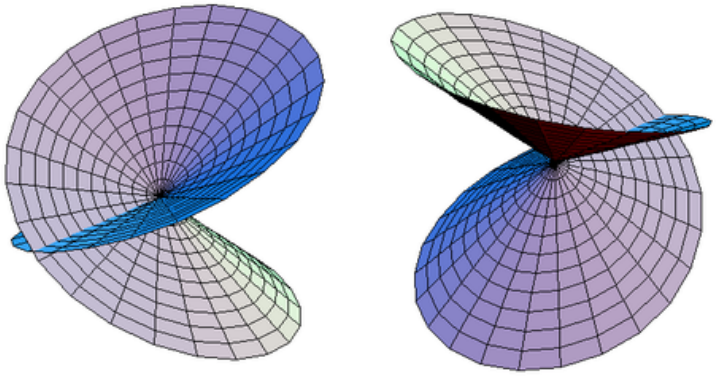
These notes are based off **Prof. Chin Chee Whye's** MA4266 Introduction to Algebraic Topology materials.

This set of notes was last updated on May 10, 2026. If you would like to contribute a nice discussion to the notes or point out a typo, please send me an email at `thangpangern@u.nus.edu`.



A self-intersecting disk

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Chapter 1

Fundamental Groups

1.1 Homotopy and Path Homotopy

Definition 1.1 (homotopy and nulhomotopy). Let $I = [0,1]$ be the closed unit interval and X and Y be topological spaces. If f and f' are continuous maps from X to Y, then f is homotopic to f' if there exists a continuous map $F : X \times I \to Y$ such that

$$F(x,0) = f(x) \text{ and } F(x,1) = f'(x) \quad \text{for all } x \in X.$$

The map F is a homotopy between f and f' and we write $f \simeq f'$. If $f \simeq f'$ where $f$is a constant map, then f is null-homotopic.

In Definition 1.1, think of a homotopy between f and f' as a continuous deformation from f to f'.

Example 1.1. Let A be a convex subspace of $\mathbb{R}^n$ and f and g be two continuous maps from X to A. Then, f and g are homotopic and the map

$$F(x,t) = (1-t)f(x) + tg(x) \quad \text{is a homotopy between them.}$$

Lemma 1.1. Suppose $f : I \to X$ is a path. Then, it is homotopic to some constant map. In fact, the constant map can be chosen to be the map that maps I to x, where x is some point in $f([0,1])$.

Proof. Consider the map

$$F : I \times I \to X \quad \text{given by} \quad F(s,t) = f((1-t)s).$$

Then, F is continuous, and $F(s,0) = f(s)$ and $F(s,1) = f(0)$. given by $F(s,t) = f((1-t)s)$. Hence, F is a desired homotopy. □

Lemma 1.2. If X is path-connected, then for any paths $f, g \in X$, $f \simeq g$.

Note that if X is path-connected, then by definition, there exists a path in X from x_0 to x_1, i.e. there exists a continuous map $f : [0,1] \to X$ such that $f(0) = x_0$ and $f(1) = x_1$. We say that x_0 is the initial point and x_1 is the final point of f

Definition 1.2 (path homotopy). Two paths $f, f' : I \to X$ are path homotopic if they have the same initial point x_0 and final point x_1 and if there exists a continuous map $F : I \times I \to X$ such that

$$F(s,0) = f(s) \text{ and } F(s,1) = f'(s) \quad \text{and} \quad F(0,t) = x_0 \text{ and } F(1,t) = x_1$$

for all $s,t \in I$. We say that F is a path homotopy between f and f' and we write $f \simeq_p f'$ (Figure 1.1).

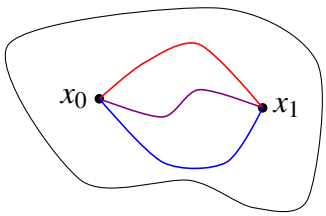


Figure 1.1: Homotopy of paths f and f'

Definition 1.3 (homotopy equivalence class). If f is a path, we denote its path homotopy equivalence class by $[f]$.

So, $[f]$ is the set of all paths that are path homotopic to f. In other words, this set (or equivalence class) consists of all paths that can be continuously deformed into f while keeping the endpoints fixed. Also, to jump the gun, when we encounter the fundamental group in Definition 1.9, we would be particularly interested in paths that are based at a specific point x_0 and share the same initial point and final point. These would be grouped into[1] path homotopy equivalence classes. For any topological space X, the fundamental group $\pi_1(X, x_0)$ would then be defined as the set of equivalence classes under the operation of path concatenation.

Lemma 1.3. The relations $\simeq$ and $\simeq_p$ are equivalence relations.

Proof. We will prove the results concurrently. First, we prove reflexivity. Let f be a path. Then, it is homotopic to itself, i.e. $f \simeq f$. Note that $F(x,t) = f(x)$ is the desired homotopy. To see why, at $= 0$, the homotopy $F(x,0) = f(x)$ is exactly the original path; at time $t = 1$, the homotopy $F(x,1) = f(x)$ is still the original path.

As for symmetry, suppose $f \simeq f'$, i.e. f is homotopic to f'. Let F be a homotopy between f to f'. Define $G(x,t) = F(x, 1-t)$. Then, G is a homotopy between f' and f. To see why, $G(x,0) = F(x,1)$ and $G(x,1) = F(x,0)$ which describe the situations at $t = 0$ and

[1] No pun intended.

$t = 1$ respectively. Hence, if F is a path homotopy, so is G.

Lastly, we prove transitivity. Suppose $f \simeq f'$ and $f' \simeq f''$. We wish to prove that $f \simeq f''$. Let F be a homotopy between f and f', and F' be a homotopy between f' and f''. Define $G : X \times I \to Y$ via

$$G(x,t) = \begin{cases} F(x,2t) & \text{for } 0 \leq x \leq 1/2; \\ F'(x,2t-1) & \text{for } 1/2 \leq x \leq 1. \end{cases}$$

Then, G is well-defined, continuous by the pasting lemma[2], and is indeed a homotopy between f and f''. Lastly, if F and F' are path homotopies, then so is G. □

Lemma 1.4 (Munkres p. 330 Question 1). If

$$h, h' : X \to Y \text{ are homotopic} \quad \text{and} \quad k, k' : Y \to Z \text{ are homotopic},$$

then $k \circ h$ and $k' \circ h'$ are homotopic.

Proof. Let F be a homotopy between h and h', and G be a homotopy between k and k'. We claim that $H(x,t) = G(F(x,t),t)$ is a homotopy between $k \circ h$ and $k' \circ h'$. To see why, we need to verify that for all $x \in X$, we have

$$H(x,0) = (k \circ h)(x) \quad \text{and} \quad H(x,1) = \left(k' \circ h'\right)(x).$$

This is trivial to see. □

Definition 1.4 (contractible space). A space X is contractible if the identity map $i_X : X \to X$ is null-homotopic.

Lemma 1.5 (Munkres p. 330 Question 3). Every contractible space is path-connected.

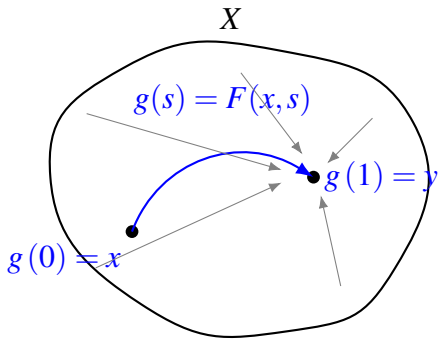
Proof. Let X be a contractible space. Let

$$c_y : X \to X \quad \text{where} \quad c_y : x \mapsto y$$

be a constant map such that i_X is homotopic to, and F be the homotopy between i_X and c_y. It suffices to show that for every $x \in X$, there exists a path between x and y. Since F is a homotopy between i_X and c_y, then $F(x,0) = x$ and $F(x,1) = y$. Define $g : [0,1] \to X$ via $g(s) = F(x,s)$. Then, $g(0) = x$ and $g(1) = y$. So, g is the composition of F and $s \mapsto (x,s)$,

[2]Recall this from MA3209 Metric and Topological Spaces.

which is continuous.



Hence, given $x, y \in X$, we can find a continuous path $g(s)$ connecting them, making X path-connected. □

Definition 1.5 (homotopy classes). Let X and Y be spaces. Let

$$[X,Y] \quad \text{denote the set of homotopy classes of maps of } X \text{ into } Y.$$

Geometrically, $[X,Y]$ is the set of different ways that the space X can be continuously mapped into the space Y, where two maps are said to be homotopic if they can be continuously deformed into each other.

Lemma 1.6 (Munkres p. 366 Question 3). Let $\mathcal{C}$ be a collection of spaces. Then,

$$\text{the relation of homotopy equivalence} \quad \text{is} \quad \text{an equivalence relation on } \mathcal{C}.$$

Proof. We need to check the three properties of an equivalence relation. We first prove reflexivity, i.e. need to show that a space $X \in \mathcal{C}$ is homotopy equivalent to itself, i.e. there exists a continuous map $f : X \to X$ such that f is homotopic to the identity map on X, denoted by id_X. To see why, consider the identity map $\mathrm{id}_X : X \to X$, for which $\mathrm{id}_X \circ \mathrm{id}_X = \mathrm{id}_X$, and id_X is trivially homotopic to itself via the constant homotopy.

We then prove symmetry, i.e. show that

$$\text{if } X \in \mathcal{C} \text{ is homotopy equivalent to } Y \in \mathcal{C} \quad \text{then} \quad Y \text{ is homotopy equivalent to } X.$$

Since X is homotopy equivalent to Y, then by definition, there exist continuous maps

$$f : X \to Y \text{ and } g : Y \to X \quad \text{such that} \quad g \circ f \simeq \mathrm{id}_X \text{ and } f \circ g \simeq \mathrm{id}_Y\,.$$

Recall that $\simeq$ means homotopic. We showed that $\simeq$ is an equivalence relation in the proof of Lemma 1.3. Hence, $\simeq$ is symmetric, i.e. the roles of f and g can be reversed. We conclude that homotopy equivalence is symmetric. Lastly, we prove transitivity, i.e.

$$\text{if } X \in \mathcal{C} \text{ is homotopy equivalent to } Y \in \mathcal{C} \quad \text{and} \quad Y \in \mathcal{C} \text{ is homotopy equivalent to } Z \in \mathcal{C},$$

then X is homotopy equivalent to Z. Since X is homotopy equivalent to Y, then by definition, there exist continuous maps

$$f : X \to Y \text{ and } g : Y \to X \quad \text{such that} \quad g \circ f \simeq \mathrm{id}_X \text{ and } f \circ g \simeq \mathrm{id}_Y .$$

Similarly, since Y is homotopy equivalent to Z, then there exist continuous maps

$$h : Y \to Z \text{ and } k : Z \to Y \quad \text{such that} \quad k \circ h \simeq \mathrm{id}_Y \text{ and } h \circ k \simeq \mathrm{id}_Z .$$

We need to construct maps $f' : X \to Z$ and $g' : Z \to X$, for which we use composition, i.e.

$$f' = h \circ f : X \to Z \quad \text{and} \quad g' = g \circ k : Z \to X.$$

One checks that $g' \circ f' \simeq \mathrm{id}_X$ and $f' \circ g' \sim \mathrm{id}_Z$. The result follows. □

Proposition 1.1 (Munkres p. 330 Question 3). Let X and Y be spaces. Then, the following hold:

(i) If Y is contractible, then for any X, $[X,Y]$ has a single element.

(ii) If X is contractible and Y is path-connected, then $[X,Y]$ has a single element.

Proof. We will only prove **(i)** as **(ii)** follows by joining several homotopies. To prove **(i)**, say $f : X \to Y$ and suppose i_Y is null-homotopic to the map $c_y : Y \to Y$ given by $q \mapsto y$, where $y \in Y$ is a fixed element. It suffices to prove that f is homotopic to

$$\text{the constant map } c : X \to Y \quad \text{where} \quad c : p \mapsto y.$$

Let F be the homotopy between i_Y and c_y. Then, $F(q,0) = q$ and $F(q,1) = y$. Then, consider the map $G : X \times I \to Y$ given by $G(p,t) = F(f(p),t)$. Note that G is continuous, $G(p,0) = f(p)$ and $G(p,1) = y$. So, $G(p,t)$ is a homotopy between f and the constant map c. □

Example 1.2 (Munkres p. 330 Question 2). Prove the following results:

(a) for any set X, the set $[X,I]$ has a single element;

(b) if Y is path-connected, then the set $[I,Y]$ has a single element

Solution.

(a) $[X,I]$ denotes the set of homotopy classes of maps of X into the unit interval $I = [0,1]$. It suffices to prove that I is contractible, i.e. we can continuously deform it to a singleton. We need to define a continuous family of maps (also known as a homotopy) that continuously shrinks every point $t \in I$ to 0 over time. In particular, this homotopy H can be written as

$$H(s,t) = (1-s)t \quad \text{which implies} \quad H(0,t) = t \text{ and } H(1,t) = 0.$$

We see that $H(0,t)$ is the identity map, and $H(1,t)$ is the constant map sending everything to 0.

(b) Similarly, $[I,Y]$ denotes the set of homotopy classes of maps of the unit interval I into a path-connected set Y. Our goal is to prove that any map $f : I \to Y$ is homotopic to a constant map. This can be done by considering the homotopy

$$F(s,t) = f((1-s)t) \quad \text{which implies} \quad F(0,t) = f(t) \text{ and } F(1,t) = f(0) \text{ is constant.}$$

So, f_1 is homotopic to the constant map equal to $f_1(0)$ and f_2 is homotopic to the constant map equal to $f_2(0)$. Since Y is path-connected, then any two points in Y can be joined by a single path.

So, there exists a path $\gamma : I \to Y$ such that $\gamma(0) = f_1(0)$ and $\gamma_1 = f_2(0)$. Let $F(s,t) = \gamma(s)$, which is a homotopy between the constant maps $t \mapsto f_1(0)$ and $t \mapsto f_2(0)$. Since $\simeq$ is an equivalence relation, i.e.

$$f_1 \sim (t \mapsto f_1(0)) \sim (t \mapsto f_2(0)) \sim f_2,$$

we obtain a homotopy between f_1 and f_2. □

Theorem 1.1 (Munkres p. 330 Question 3). I is contractible.

Proof. To prove that I is contractible, we need to construct a homotopy between the identity map on I and a constant map. Define the homotopy $H : I \times I \to I$ by the formula

$$H(x,t) = (1-t)x \quad \text{so} \quad H(x,0) = x \text{ and } H(x,1) = 0.$$

As such, $H(x,0)$ is the identity map on I and $H(x,1)$ is the constant map that sends every $x \in I$ to 0. Hence, H continuously deforms the identity map on I to a constant map. □

Theorem 1.2 (Munkres p. 330 Question 3). $\mathbb{R}$ is contractible.

Proof. Similar to our proof that I is contractible (Theorem 1.1), we can consider the homotopy $H : I \times I \to I$ defined by $H(x,t) = (1-t)x$. □

1.2 Fundamental Group

Definition 1.6 (path concatenation). Let f be a path in X from x_0 to x_1 and g be a path in X from x_1 to x_2. We define the product $f * g$ of f and g to be the path h given by the following equations:

$$h(s) = \begin{cases} f(2s) & \text{for } 0 \le s \le 1/2; \\ g(2s-1) & \text{for } 1/2 \le s \le 1. \end{cases}$$

$h = f * g$ is well-defined and continuous by the pasting lemma. Moreover, it is a path in X from x_0 to x_2. We can think of h as the path whose first half is the path f and whose second half is the path g (Figure 1.2).

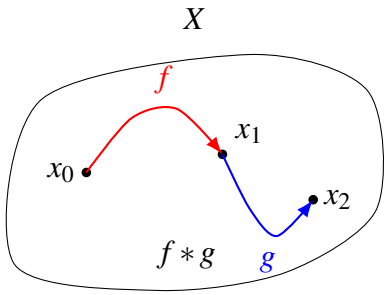


Figure 1.2: The path concatenation $h = f * g$

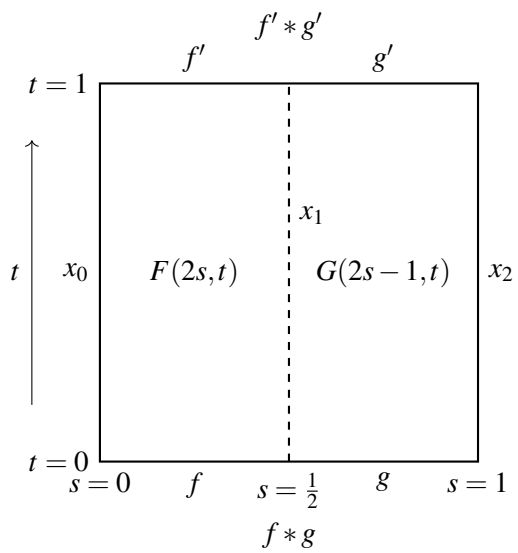
Proposition 1.2. The product operation on paths induces a well-defined operation on path homotopy classes, which is still denoted by $*$. This is defined by the operation $[f] * [g] = [f * g]$.

Proof. We prove that the product operation on paths is well-defined. Suppose $f \simeq_p f'$ and $g \simeq_p g'$, where f, f' are paths from x_0 to x_1, and g, g' are paths from x_1 to x_2. Let $F : I \times I \to X$ be a path homotopy from f to f', and let $G : I \times I \to X$ be a path homotopy from g to g'. Thus $F(s,0) = f(s)$, $F(s,1) = f'(s)$, $G(s,0) = g(s)$, and $G(s,1) = g'(s)$. Moreover, $F(0,t) = x_0$, $F(1,t) = x_1$, $G(0,t) = x_1$, and $G(1,t) = x_2$ for all $t \in I$.

Define $H : I \times I \to X$ by

$$H(s,t) = \begin{cases} F(2s,t) & 0 \le s \le \frac{1}{2}; \\ G(2s-1,t) & \frac{1}{2} \le s \le 1. \end{cases}$$

This is well-defined, since on the overlap $s = \frac{1}{2}$ we have $F(1,t) = x_1$ and $G(0,t) = x_1$. Therefore the two definitions agree along $s = \frac{1}{2}$. Since F and G are continuous, and the two pieces agree on their intersection, H is continuous by the pasting lemma.



Now, at $t = 0$, we get

$$H(s,0) = \begin{cases} F(2s,0) & 0 \le s \le \frac{1}{2}; \\ G(2s-1,0) & \frac{1}{2} \le s \le 1, \end{cases} = \begin{cases} f(2s) & 0 \le s \le \frac{1}{2}; \\ g(2s-1) & \frac{1}{2} \le s \le 1, \end{cases} = (f * g)(s).$$

Similarly, at $t = 1$, we have

$$H(s,1) = \begin{cases} F(2s,1) & 0 \le s \le \frac{1}{2}; \\ G(2s-1,1) & \frac{1}{2} \le s \le 1, \end{cases} = \begin{cases} f'(2s) & 0 \le s \le \frac{1}{2}; \\ g'(2s-1) & \frac{1}{2} \le s \le 1, \end{cases} = (f' * g')(s).$$

Finally, for all $t \in I$, the endpoints are fixed:

$$H(0,t) = F(0,t) = x_0 \quad \text{and} \quad H(1,t) = G(1,t) = x_2.$$

Hence H is a path homotopy from $f * g$ to $f' * g'$. Therefore $f * g \simeq_p f' * g'$, so the product operation on path homotopy classes is well-defined. □

Definition 1.7 (constant path). Let X be a topological space. Given $x \in X$, let e_x denote the constant path at x (Figure 1.3), i.e.

$$\text{the path } e_x : I \to X \quad \text{which carries all of } I \text{ to the point } x.$$

Then, $[e_x]$ is the path homotopy equivalence class of e_x.

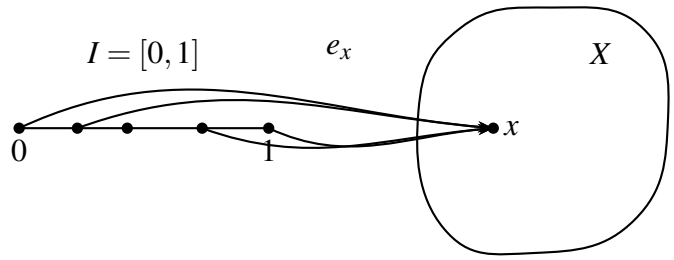


Figure 1.3: The constant path $e_x : I \to X$ at the point $x \in X$

Definition 1.8 (reverse path). Let f be a path in X from x_0 to x_1, then we define

$$\overline{f} \text{ to be the reverse of } f \quad \text{where} \quad \overline{f}(s) = f(1-s).$$

Proposition 1.3. The reverse operation on paths induces a well-defined operation (inverse) on path homotopy classes defined by the equation $[f]^{-1} = [\overline{f}]$.

Proof. Let $f \simeq_p g$. Then, it suffices to prove that $\overline{f} \simeq_p \overline{g}$. Let F be a path homotopy between f and g. Then, we define G via $G(x,t) = F(1-x,t)$, which is a path homotopy between f' and g'. □

Definition 1.9 (fundamental group). Let X be a topological space and $x_0 \in X$ be a point. We say that (X, x_0) is a pointed topological space.

(i) A path in X that begins and ends at x_0 is called a loop based at x_0

(ii) The set of path homotopy class of loops based at x_0 with the operation $*$ is called the fundamental group of X relative to the base point x_0, which is

denoted by $\pi_1(X, x_0)$

The elements of $\pi_1(X, x_0)$ are the path homotopy equivalence classes and the group operation is defined by $[f] * [g] = [f * g]$, the identity element is $[e_{x_0}]$ and the inverse of $[f]$ is $[\overline{f}]$.

Sometimes, $\pi_1(X, x_0)$ is called the first homotopy group of X, which term implies that there is a second homotopy group, and so on[3]. There are indeed groups $\pi_n(X, x_0)$ for all positive integers n, but we shall not study them here. They are part of the general subject called homotopy theory.

Example 1.3 (fundamental group of $\mathbb{R}^n$ and convex subsets of $\mathbb{R}^n$). Let $\mathbb{R}^n$ denote the Euclidean n-space. Then, $\pi_1(\mathbb{R}^n, x_0)$ is the trivial group. More generally, more any convex subset of $\mathbb{R}^n$, $\pi_1(X, x_0)$ is the trivial group. In particular, the unit ball

$$B^n = \left\{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_1^2 + \ldots + x_n^2 \leq 1\right\} \quad \text{has trivial fundamental group.}$$

To see why, let f be a loop based at x_0, then the straight-line homotopy is a path homotopy between f and the constant path at x_0. See the illustrations in Figure 1.4 for the fundamental group of the unit balls B^2 and B^3.

$B^2 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$ $\qquad$ $B^3 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1\}$

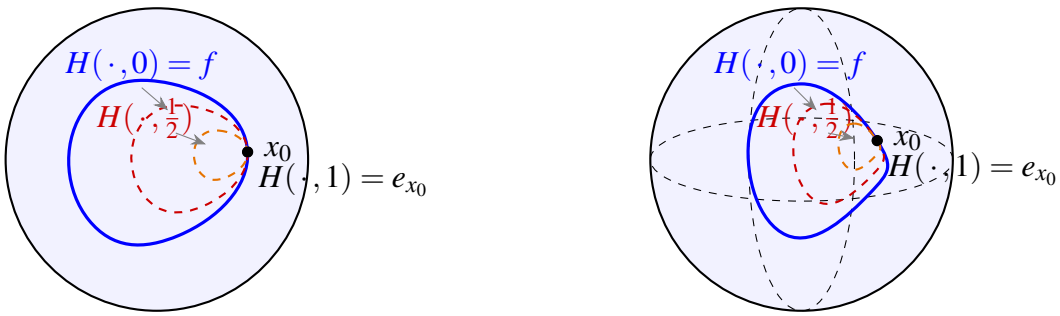


Figure 1.4: Examples in B^2 and B^3 where the straight-line homotopy contracts a loop f based at x_0 to the constant loop e_{x_0}

Since $\pi_1(X, x_0)$ depends on the path component of X containing x_0, it is usual to deal with only path-connected spaces when studying the fundamental group. Also, by constructing the path homotopy explicitly, we can derive Theorem 1.3.

Theorem 1.3 (fundamental group is a group).

- **Associativity:** If f, g, h are paths for which $f(1) = g(0)$ and $g(1) = h(0)$, then

$$[f] * ([g] * [h]) = ([f] * [g]) * [h].$$

[3]The composite function $f \circ g$ involves applying g first, then f. Unlike function composition, for path homotopy equivalence classes, the concatenation $[f] * [g]$ implies that we apply $[f]$ first, then $[g]$.

- **Right and left identities:** If f is a path in X from x_0 to x_1, then

$$[f]*[e_{x_1}]=[f] \quad \text{and} \quad [e_{x_0}]*[f]=[f].$$

- **Inverse:** If f is a path in X from x_0 to x_1, then

$$[f]*[\overline{f}]=[e_{x_0}] \quad \text{and} \quad [\overline{f}]*[f]=[e_{x_1}].$$

Definition 1.10 (conjugation map). Let α be a path in X from x_0 to x_1. We define the conjugation map

$$\widehat{\alpha}:\pi_1(X,x_0)\to\pi_1(X,x_1) \quad \text{via} \quad \widehat{\alpha}([f])=[\overline{\alpha}]*[f]*[\alpha].$$

Note that the map $\widehat{\alpha}(\gamma)=[\alpha]^{-1}*\gamma*[\alpha]$ depends only on the path homotopy class of α and the map is well-defined (Figure 1.5). Refer to Figure 1.5. Suppose we have two base points x_0 and x_1 and a loop γ at x_0. We can turn this into a loop based at x_1 easily. To do this, pick a path α in X from x_0 to x_1. Then, we can produce a new loop at x_1 by going along α^{-1} to x_0, take the path γ, then return to x_0 by α, i.e. consider $\alpha^{-1}*\gamma*\alpha$.

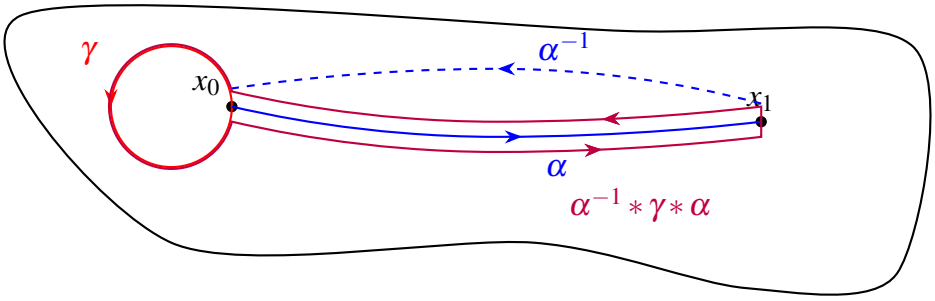


Figure 1.5: Transporting the base point of a loop from x_0 to x_1.

Theorem 1.4. $\widehat{\alpha}$ is a group isomorphism.

Proof. Note that $\widehat{\alpha}$ is a group homomorphism. It is easy to construct the inverse of $\widehat{\alpha}$, which implies $\widehat{\alpha}$ is invertible, and hence, a bijective map. □

Example 1.4 (Munkres p. 334 Question 2). Let α be a path in X from x_0 to x_1; let β be a path in X from x_1 to x_2. Show that if

$$\gamma=\alpha*\beta \quad \text{then} \quad \widehat{\gamma}=\widehat{\beta}\circ\widehat{\alpha}.$$

Solution. Since α is a path in X from x_0 to x_1 and β is a path in X from x_1 to x_2, then $\gamma=\alpha*\beta$ is a path in X from x_0 to x_2. The conjugation map $\widehat{\gamma}$ is defined as follows:

$$\widehat{\gamma}:\pi_1(X,x_0)\to\pi_1(X,x_2) \quad \text{via} \quad \widehat{\gamma}([f])=[\overline{\gamma}]*[f]*[\gamma]$$

Here, f is a loop based at x_0 and $[\overline{\gamma}]$ is the homotopy class of the reverse of the path γ. Let g and h be loops based at x_1 and x_0 respectively. Writing γ as $\alpha * \beta$, we have

$$\begin{aligned}
\widehat{\gamma}([f]) &= [\overline{\gamma}] * [f] * [\gamma] \quad \text{by definition} \\
&= [\overline{\alpha * \beta}] * [f] * [\alpha * \beta] \quad \text{by writing } \gamma = \alpha * \beta \\
&= \left([\overline{\beta}] * [\overline{\alpha}]\right) * [f] * [\alpha * \beta] \quad \text{since } \overline{\alpha * \beta} = [\overline{\beta}] * [\overline{\alpha}] \\
&= [\overline{\beta}] * ([\overline{\alpha}] * [f] * [\alpha]) * [\beta] \quad \text{by associativity} \\
&= [\overline{\beta}] * [\overline{\alpha}] * [f] * [\alpha] * [\beta] \\
&= \widehat{\beta}\,(\widehat{\alpha}\,(f))
\end{aligned}$$

The result follows. □

Corollary 1.1. If X is path-connected and x_0, x_1 are two points of X, then $\pi_1(X, x_0) \cong \pi_1(X, x_1)$. In particular, if X is path-connected, for any $x \in X$, all the groups in $\pi_1(X, x)$ are isomorphic.

Proof. Since X is path-connected, there exists a path $\alpha : I \to X$ such that $\alpha(0) = x_0$ and $\alpha(1) = x_1$. Let $\overline{\alpha} : I \to X$ denote the reverse path, defined by $\overline{\alpha}(t) = \alpha(1-t)$. Thus $\overline{\alpha}$ is a path from x_1 to x_0.

Define a map

$$\Phi_\alpha : \pi_1(X, x_0) \to \pi_1(X, x_1) \quad \text{by} \quad \Phi_\alpha([f]) = [\overline{\alpha} * f * \alpha],$$

where f is a loop based at x_0. Indeed, $\overline{\alpha} * f * \alpha$ starts at x_1, travels to x_0, goes around the loop f, and then travels back to x_1. Hence it is a loop based at x_1.

This map is well-defined. If $[f] = [g]$ in $\pi_1(X, x_0)$, then $f \simeq_p g$. By concatenating the same paths $\overline{\alpha}$ and α at the beginning and end, we obtain

$$\overline{\alpha} * f * \alpha \simeq_p \overline{\alpha} * g * \alpha.$$

Hence $[\overline{\alpha} * f * \alpha] = [\overline{\alpha} * g * \alpha]$, so Φ_α is well-defined.

Next, Φ_α is a group homomorphism. For $[f], [g] \in \pi_1(X, x_0)$, we have

$$\Phi_\alpha([f] \cdot [g]) = \Phi_\alpha([f * g]) = [\overline{\alpha} * f * g * \alpha].$$

On the other hand,

$$\Phi_\alpha([f]) \cdot \Phi_\alpha([g]) = [\overline{\alpha} * f * \alpha] \cdot [\overline{\alpha} * g * \alpha] = [(\overline{\alpha} * f * \alpha) * (\overline{\alpha} * g * \alpha)].$$

By associativity up to path homotopy (Theorem 1.3),

$$(\overline{\alpha} * f * \alpha) * (\overline{\alpha} * g * \alpha) \simeq_p \overline{\alpha} * f * (\alpha * \overline{\alpha}) * g * \alpha.$$

Since $\alpha * \overline{\alpha}$ is path-homotopic to the constant path e_{x_0}, we get

$$\overline{\alpha} * f * (\alpha * \overline{\alpha}) * g * \alpha \simeq_p \overline{\alpha} * f * e_{x_0} * g * \alpha \simeq_p \overline{\alpha} * f * g * \alpha.$$

Therefore

$$\Phi_\alpha([f] \cdot [g]) = \Phi_\alpha([f]) \cdot \Phi_\alpha([g]).$$

Thus Φ_α is a homomorphism.

Now define

$$\Psi_\alpha : \pi_1(X, x_1) \to \pi_1(X, x_0) \quad \text{by} \quad \Psi_\alpha([h]) = [\alpha * h * \overline{\alpha}],$$

where h is a loop based at x_1. We show that Ψ_α is the inverse of Φ_α.

For $[f] \in \pi_1(X, x_0)$,

$$\Psi_\alpha(\Phi_\alpha([f])) = \Psi_\alpha([\overline{\alpha} * f * \alpha]) = [\alpha * \overline{\alpha} * f * \alpha * \overline{\alpha}].$$

Since $\alpha * \overline{\alpha} \simeq_p e_{x_0}$ and $\alpha * \overline{\alpha} \simeq_p e_{x_0}$ again at the end after the loop f, we obtain

$$[\alpha * \overline{\alpha} * f * \alpha * \overline{\alpha}] = [e_{x_0} * f * e_{x_0}] = [f].$$

Hence $\Psi_\alpha \circ \Phi_\alpha = \mathrm{id}_{\pi_1(X, x_0)}$.

Similarly, for $[h] \in \pi_1(X, x_1)$,

$$\Phi_\alpha(\Psi_\alpha([h])) = \Phi_\alpha([\alpha * h * \overline{\alpha}]) = [\overline{\alpha} * \alpha * h * \overline{\alpha} * \alpha].$$

Since $\overline{\alpha} * \alpha \simeq_p e_{x_1}$, we get

$$[\overline{\alpha} * \alpha * h * \overline{\alpha} * \alpha] = [e_{x_1} * h * e_{x_1}] = [h].$$

Thus $\Phi_\alpha \circ \Psi_\alpha = \mathrm{id}_{\pi_1(X, x_1)}$. Therefore Φ_α is an isomorphism, and hence $\pi_1(X, x_0) \cong \pi_1(X, x_1)$. In particular, if X is path-connected, then for any $x \in X$, all the groups $\pi_1(X, x)$ are isomorphic. □

Proposition 1.4. Let x_0, x_1 be points of a path-connected space X. Then, $\pi_1(X, x_0)$ is abelian if and only if for all paths from x_0 to x_1 denoted by α, β, we have $\widehat{\alpha} = \widehat{\beta}$.

Proof. We first prove the forward direction. Suppose $\pi_1(X, x_0)$ is abelian. Then, for any paths α and β, note that $\alpha * \overline{\beta}$ is a loop at x_0. So,

$$[\beta * \overline{\alpha}] * [\alpha * \overline{\beta}] = e \quad \text{which shows that } \widehat{\alpha} \text{ and } \widehat{\beta} \text{ are inverses of each other.}$$

The reverse direction is easy to see, i.e. need to prove $[f] * [g] = [g] * [f]$. □

Definition 1.11 (simply-connected). A space X is simply-connected if and only if X is a path-connected space and $\pi_1(X, x_0)$ is trivial for all $x \in X$.

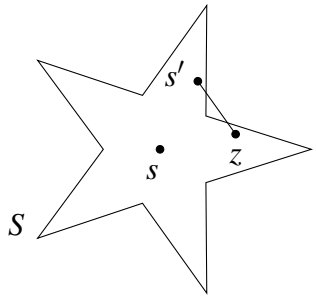
Example 1.5. Any convex subset of $\mathbb{R}^n$ is simply-connected.

Example 1.6. Let $\overline{\mathbb{C}}$ denote the set of extended complex numbers. Then, a connected open subset U of $\mathbb{C} \cong \mathbb{R}^2$ is simply-connected if and only if its complement $\overline{\mathbb{C}} \setminus U$ in $\overline{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ is connected. This requires the Riemann mapping theorem.

Definition 1.12 (star convex). A subset $A \subseteq \mathbb{R}^n$ is star convex if there exists $a_0 \in A$ such that all the line segments joining a_0 to other points of A lie in A.

Example 1.7 (Munkres p. 334 Question 1). Find a star convex set that is not convex.

Solution. We shall consider a star S.



Fix $z \in S$. Then, there exists $s' \in S$ such that the line segment from s' to z does not lie in S, so S is not convex. □

Lemma 1.7 (Munkres p. 334 Question 334). If A is a star convex set, then A is simply-connected.

Proof. Let $p(s)$ be a loop in A, where $0 \leq s \leq 1$. Then, we can construct a straight-line homotopy

$$H(s,t) = ta + (1-t)p(s) \quad \text{with} \quad H(s,0) = p(s) \text{ and } H(s,1) = a.$$

This implies p is homotopic to a point, so A is simply-connected. □

Lemma 1.8. Let X be a simply-connected space. Then, any two paths having the same initial and final points are path homotopic.

Proof. Let α and β be two paths from x_0 to x_1. Then, $\alpha * \overline{\beta}$ is defined and is a loop on X based at x_0. Since X is simply-connected, this loop is path homotopic to the constant loop at x_0. Thus implies

$$[\alpha * \overline{\beta}] * [\beta] = [e_{x_0}] * [\beta]$$

for which it follows that $[\alpha] = [\beta]$. □

Example 1.8 (Munkres p. 335 Question 3). Let x_0 and x_1 be points of the path-connected space X. Show that $\pi_1(X, x_0)$ is abelian if and only if for every pair of paths α, β from x_0 to x_1, we have $\widehat{\alpha} = \widehat{\beta}$.

Solution. Suppose $\pi_1(X, x_0)$ is abelian. Then, for any loops $[f], [g]$ based at x_0, we have $[f] * [g] = [g] * [f]$. Say α and β are paths from x_0 to x_1. Then, $[\alpha * \overline{\beta}]$ is the homotopy class of loops based at x_0. Recall that by Definition 1.10, it suffices to prove that

$$[\overline{\alpha}] * [f] * [\alpha] = [\overline{\beta}] * [f] * [\beta].$$

Replacing g with $\alpha * \overline{\beta}$, we have

$$[f] * [\alpha] * [\overline{\beta}] = [\alpha] * [\overline{\beta}] * [f].$$

The result follows from here. The reverse direction can be proven similarly. □

1.3 Functorial Property

Before proving the functorial property of the fundamental group (Theorem 1.5), let us pause to explain why this result is natural. The fundamental group $\pi_1(X, x_0)$ records information about the loops in a space X based at a point x_0, where two loops are regarded as the same if one can be continuously deformed into the other while keeping the base point fixed (Definition 1.9). Thus, $\pi_1(X, x_0)$ is not merely a set of loops but an algebraic object extracted from the topology of X. A natural question is therefore the following: if we have a continuous map $h : X \to Y$, how does this map transfer loop information from X to Y? Since a loop $f : I \to X$ based at x_0 can be sent to a loop $h \circ f : I \to Y$ based at $h(x_0)$, every continuous map between spaces should induce a corresponding map between their fundamental groups. This induced map is denoted by h_* (Definition 1.13), and it captures the idea that continuous maps preserve homotopy-theoretic information.

The importance of this construction is that it shows π_1 behaves systematically with respect to continuous maps as we would see in Theorem 1.5. In particular, if we first map X into Y and then map Y into Z, the induced homomorphism on fundamental groups is the same as first applying the induced homomorphism from X to Y and then applying the induced homomorphism from Y to Z. Also, the identity map on a space induces the identity homomorphism on its fundamental group. These two properties are what we call the functorial property of π_1. In simpler terms, the fundamental group does not only assign a group to every pointed topological space; it also assigns a group homomorphism to every base point preserving continuous map, and it does so in a way that respects composition and identity maps.

Lemma 1.9. If $k : X \to Y$ is a continuous map and

$$F \text{ is a path homotopy in } X \quad \text{between } f \text{ and } f',$$

then

$$k \circ F \text{ is a path homotopy in } Y \quad \text{between } k \circ f \text{ and } k \circ f'.$$

Example 1.9 (projection map). Let $X = \mathbb{R}^2$ and $Y = \mathbb{R}$. Let $k : X \to Y$ be the continuous map given by $k(x,y) = x$[4]. Consider the two paths

$$f, f' : I \to X \quad \text{defined by} \quad f(s) = (s,0) \text{ and } f'(s) = (s, s(1-s)).$$

These are both paths from $(0,0)$ to $(1,0)$. Define $F : I \times I \to X$ by $F(s,t) = (s, ts(1-s))$. Then, F is continuous, and $F(s,0) = f(s)$, and $F(s,1) = f'(s)$. Also, $F(0,t) = (0,0)$ and $F(1,t) = (1,0)$ for all $t \in I$. As such, F is a path homotopy in X between f and f'.

Composing with k yields

$$k \circ F : I \times I \to Y \quad \text{where} \quad (k \circ F)(s,t) = k(s, ts(1-s)) = s$$

and one can check that $k \circ F$ is indeed a path homotopy in Y between $k \circ f$ and $k \circ f'$.

Geometrically, F deforms the straight line segment f into the curved arc f' in the plane. After applying k, we only remember the x-coordinate, so both paths become the same path $s \mapsto s$ in $\mathbb{R}$.

Lemma 1.10. Let $k : X \to Y$ be a continuous map and f and g be paths in X with $f(1) = g(0)$. Then,

$$k \circ (f * g) = (k \circ f) * (k \circ g).$$

Definition 1.13 (induced homomorphism). Suppose $h : X \to Y$ is a continuous map that carries $x_0 \in X$ to $y_0 \in Y$. We denote this by writing $h : (X, x_0) \to (Y, y_0)$. Then, define the induced homomorphism on fundamental groups, h_*, to be

$$h_* : \pi_1(X, x_0) \to \pi_1(Y, y_0) \quad \text{by} \quad h_*([f]) = [h \circ f].$$

Note that $h_*(\gamma)$ is well-defined, i.e. if F is a path homotopy between the paths f and f', then $h \circ F$ is a path homotopy between the paths $h \circ f = h \circ f'$.

The induced homomorphism h_* depends not only on the map $h : X \to Y$ but also on the choice of the base point x_0. To see why, if x_0 and x_1 are two different points of X, we cannot use the same symbol h_* to denote two different homomorphisms, with one having domain $\pi_1(X, x_0)$ and the other having domain $\pi_1(X, x_1)$. Even if X is path-connected (which implies the groups are isomorphic), they are still not the same group. In such a

[4] This is the projection onto the first coordinate.

case, we use the notation

$$(h_{x_0})_* : \pi_1(X,x_0) \to \pi_1(Y,y_0) \quad \text{for the first homomorphism} \quad \text{and}$$
$$(h_{x_1})_* \quad \text{for the second homomorphism}$$

Lemma 1.11. $h_* : \pi_1(X,x_0) \to \pi_1(Y,y_0)$ is a group homomorphism.

Theorem 1.5 (functorial property of π_1). The following hold:

(i) If $h : (X,x_0) \to (Y,y_0)$ and $k : (Y,y_0) \to (Z,z_0)$ are continuous, then $(k \circ h)_* = k_* \circ h_*$.

(ii) If $i : (X,x_0) \to (X,x_0)$ is the identity map, then i_* is the identity homomorphism.

Proof. **(ii)** is easy to prove — we have $i_*([f]) = [i \circ f] = [f]$. To prove **(i)**, we have

$$(k \circ h)_*([f]) = [(k \circ h) \circ f]$$
$$(k_* \circ h_*)([f]) = k_*(h_*([f])) = k_*([h \circ f]) = [k \circ (h \circ f)]$$

and these two expressions are equivalent by associativity. □

Corollary 1.2. If $h : (X,x_0) \to (Y,y_0)$ is a homeomorphism of X and with Y, then h_* is an isomorphism of $\pi_1(X,x_0)$ with $\pi_1(Y,y_0)$.

Proof. Let

$$k : (Y,y_0) \to (X,x_0) \text{ be the inverse of } h \quad \text{where} \quad h : (X,x_0) \to (Y,y_0).$$

Then, $k_* \circ h_* = (k \circ h)_* = i_*$, where i is the identity map of (X,x_0). Similarly, we have $h_* \circ k_* = j_*$, where j is the identity map of (Y,y_0). Since i_* and j_* are the identity homomorphism of their respective groups, then k_* is the inverse of h_*. □

Theorem 1.6. Let $h : X \to Y$ be continuous with $h(x_0) = y_0$ and $h(x_1) = y_1$. Also, let α be a path in X from x_0 to x_1. Define $\beta = h \circ \alpha$. Then,

$$\widehat{\beta} \circ (h_{x_0})_* = (h_{x_1})_* \circ \widehat{\alpha}, \text{ i.e. the following diagram commutes:}$$

$$\begin{tikzcd}
\pi_1(X,x_0) \arrow[r, "(h_{x_0})_*"] \arrow[d, "\widehat{\alpha}"'] & \pi_1(Y,y_0) \arrow[d, "\widehat{\beta}"] \\
\pi_1(X,x_1) \arrow[r, "(h_{x_1})_*"] & \pi_1(Y,y_1)
\end{tikzcd}$$

Example 1.10 (Munkres p. 335 Question 7). Let G be a topological group $\cdot$ and identity element x_0. Let

$$\Omega(G,x_0) \quad \text{denote the set of all loops in } G \text{ based at } x_0.$$

If $f, g \in \Omega(G, x_0)$, then we define a loop $f \otimes g$ by the formula $(f \otimes g)(s) = f(s) \cdot g(s)$.

(a) Prove that this operation makes the set $\Omega(G, x_0)$ into a group.

(b) Prove that this operation induces a group operation $\otimes$ on $\pi_1(G, x_0)$.

(c) By computing

$$(f * e_{x_0}) \otimes (e_{x_0} * g),$$

prove that the two group operations $*$ and $\otimes$ on $\pi_1(G, x_0)$ are the same.

(d) Prove that $\pi_1(G, x_0)$ is abelian.

Solution.

(a) We first prove that Ω contains the identity element, i.e. there exists $e_{x_0} \in \Omega(G, x_0)$ such that

$$f \otimes e_{x_0} = f = e_{x_0} \otimes f \quad \text{for all } f \in \Omega(G, x_0).$$

We shall define the constant loop $e_{x_0} : I \to G$ such that $e_{x_0}(s) = x_0$ for all $s \in I$. Then,

$$(f \otimes e_{x_0})(s) = f(s) \cdot e_{x_0}(s) = f(s) \cdot x_0 = f(s)$$

and the proof that $e_{x_0} \otimes f = f$ is similar.

We then prove that Ω has an inverse element. That is, for every $f \in \Omega(G, x_0)$, there exists $f^{-1} \in \Omega(G, x_0)$ such that $f \otimes f^{-1} = e_{x_0}$. Define $f^{-1}(s) = (f(s))^{-1}$, where the inverse operation on the RHS is with respect to the group G. Hence,

$$\left(f \otimes f^{-1}\right)(s) = f(s) \cdot f^{-1}(s) = f(s) \cdot (f(s))^{-1} = x_0$$

so it follows that $f \otimes f^{-1} = e_{x_0}$.

Lastly, we prove that Ω is associative under $\otimes$, i.e.

$$((f \otimes g) \otimes h)(s) = (f \otimes (g \otimes h))(s).$$

This is in fact equivalent to

$$(f(s) \cdot g(s)) \cdot h(s) = f(s) \cdot (g(s) \cdot h(s)) \quad \text{which is true since } G \text{ is associative under } \cdot.$$

(b) By definition, the fundamental group $\pi_1(G, x_0)$ denotes the set of homotopy classes of loops based at x_0. We have $[f] \otimes [g] = [f \otimes g]$, so $\otimes$ is a well-defined operation. We need to prove that

$$f \sim f' \text{ and } g \sim g' \quad \text{implies} \quad f \otimes g \sim f' \otimes g'.$$

Let $H_f : I \times I \to G$ be a homotopy between f and f' and $H_g : I \times I \to G$ be a homotopy between g and g'. Define

$$H(s,t) = H_f(s,t) \cdot H_g(s,t).$$

Then,

$$H(s,0) = H_f(s,0) \cdot H_g(s,0) = f(s) \cdot g(s)$$

and

$$H(s,1) = H_f(s,1) \cdot H_g(s,1) = f'(s) \cdot g'(s).$$

So, H is indeed a homotopy between $f \otimes g$ and $f' \otimes g'$. Combining the result in **(a)**, **(b)** follows.

(c) We have

$$((f * e_{x_0}) \otimes (e_{x_0} * g))(s) = (f * e_{x_0})(s) \cdot (e_{x_0} * g)(s)$$

Recall that

$$(f * e_{x_0})(s) = \begin{cases} f(2s) & \text{if } 0 \leq s \leq 1/2; \\ x_0 & \text{if } 1/2 \leq s \leq 1 \end{cases}$$

and

$$(e_{x_0} * g)(s) = \begin{cases} x_0 & \text{if } 0 \leq s \leq 1/2; \\ g(2s-1) & \text{if } 1/2 \leq s \leq 1. \end{cases}$$

By the pasting lemma,

$$(f * e_{x_0})(s) \cdot (e_{x_0} * g)(s) = \begin{cases} f(2s) \cdot x_0 = f(2s) & \text{if } 0 \leq s \leq 1/2; \\ x_0 \cdot g(2s-1) = g(2s-1) & \text{if } 1/2 \leq s \leq 1. \end{cases}$$

This is precisely the definition of $f * g$, so the result follows.

(d) Consider $[f], [g] \in \pi_1(G, x_0)$. Then,

$$\begin{aligned} [f] * [g] &= [f] \otimes [g] \quad \text{by } \textbf{(c)} \\ &= [e_{x_0} * f] \otimes [g * e_{x_0}] \\ &= [g * f] \end{aligned}$$

which is equivalent to $[g] * [f]$, implying that the fundamental group is abelian. □

1.4 Covering Spaces

Definition 1.14 (B-space). A B-space or a space over a topological space of interest B consists of a topological space E and a continuous map $p : E \to B$ which is also known as a structural map.

Definition 1.15 (B-space map). Suppose $p_1 : E_1 \to B$ and $p_2 : E_2 \to B$ are B-spaces. A B-space map from E_1 to E_2 is a continuous map $f : E_1 \to E_2$ such that $p_2 \circ f = p_1$, i.e. the following diagram commutes:

$$\begin{tikzcd} E_1 \arrow[rr, "f"] \arrow[dr, "p_1"'] & & E_2 \arrow[dl, "p_2"] \\ & B & \end{tikzcd}$$

Furthermore, a B-space map $f : E_1 \to E_2$ is an isomorphism of B-spaces if there exists a B-space map $g : E_2 \to E_1$ such that

$$g \circ f = \mathrm{id}_{E_1} \quad \text{and} \quad f \circ g = \mathrm{id}_{E_2}.$$

Example 1.11 (projection map). For any topological space F, the projection map $p : B \times F \to B$, where $(b,a) \mapsto b$ makes $B \times F$ into a B-space (by convention, the projection map is denoted by π instead of p).

Example 1.12 (restriction map). Let $p : E \to B$ be a B-space. For any $B' \subseteq B$, define $E' = p^{-1}(B') \subseteq E$ and the restriction map $p|_{E'} : E' \to B'$. Then, E' becomes a space over B', called the restriction of p over B'.

Example 1.13 (identity map). If $p : E \to B$ is a B-space, the identity map $\mathrm{id}_E : E \to E$ is a B-space map.

Example 1.14. Let E_1 and E_2 be topological spaces. If

$$f : E_1 \to E_2 \text{ and } g : E_2 \to E_3 \quad \text{are } B\text{-spaces maps,}$$

the composite map $g \circ f : E_1 \to E_3$ is a B-space map. This is obvious since E_1 and E_2 are topological spaces and the maps f and g are continuous (since they are B-spaces maps).

Definition 1.16 (even cover). Let $p : E \to B$ be a B-space. An open set $U \subseteq B$ is evenly covered by p if there exists a non-empty discrete space F such that

$$p|_U \cong U \times F \quad \text{as } U\text{-spaces.}$$

That is to say, $p^{-1}(U) \subseteq E$ can be written as a non-empty disjoint union of open sets in E such that $p|_{V_\alpha}$ is a homomorphism of V_α onto U. We say that the collection $\{V_\alpha\}$ is a partition of $p^{-1}(U)$ into slices.

Note that if U is evenly covered by p and $W \subseteq U$ is open, then W is evenly covered by p.

Definition 1.17 (covering map). A covering space of B is a B-space $p : E \to B$ which is locally trivial (p is a covering map), i.e. for all $b \in B$, there exists an open

neighbourhood U of b that is evenly covered by p. If this is the case,

$$p \text{ is a covering map} \quad \text{and} \quad E \text{ is a covering space of } B.$$

If $p : E \to B$ is a covering map, then the following hold:

(i) for all $b \in B$, the fiber $p^{-1}(b) \subseteq E$ given by the subspace topology from E has the discrete topology. To see why, based on the definition of a covering map , each point $e = p^{-1}(b)$ lies in some open set of E that is mapped homeomorphically onto some neighbourhood of B. So, each point in the fiber can be separated using disjoint open sets, giving the fiber the discrete topology.

(ii) $p : E \to B$ is an open map and surjective

(iii) p is a local homomorphism of E with B, i.e. for all $e \in E$, there exists an open neighbourhood V of e such that $p\,|_V : V \to p(V)$ is a homeomorphism

Example 1.15. The identity map $\mathrm{id}_B : B \to B$ is a covering map.

Example 1.16 (Munkres p. 341 Question 1). For any non-empty discrete space F, the projection map $p : B \times F \to B$ such that $(b,a) \mapsto b$ is a covering map. To see why, note that p is continuous and surjective, and

$$p^{-1}(B) = \bigcup_{f \in F} B \times \{f\},$$

where each slice $B \times \{f\}$ is open in $B \times F$ (as F is discrete) and homeomorphic to B.

Example 1.17. If

$$p : E \to B \text{ and } p' : E' \to E \quad \text{are finite covering maps,}$$

then $p \circ p' : E' \to B$ is a finite covering map.

Theorem 1.7. The exponential map $p : \mathbb{R} \to S^1 \subseteq \mathbb{C} \setminus \{0\}$ given by the equation

$$p(x) = (\cos 2\pi x, \sin 2\pi x) \text{ or equivalently } p(x) = \exp(2\pi i x) \quad \text{is a covering map.}$$

Proof. Note that the map p is continuous and surjective. We now prove that p is locally trivial. Consider some point $y_0 = \exp(2\pi x_0) \in S^1$. We can choose an open interval $I \subseteq \mathbb{R}$ around y_0 such that $p\,|_I$ is injective[5].

Now, consider $p^{-1}(V)$, where $V = p(I) \subseteq S^1$. Note that $p(x+1) = p(x)$ since $\cos 2\pi x$ and $\sin 2\pi x$ are periodic with period 1. Hence, $p^{-1}(V)$ consists of a disjoint union of intervals of the form $I_k = I + k$ for all $k \in \mathbb{Z}$. Each I_k is mapped homeomorphically onto V by the map p, so we conclude that p is locally trivial. □

[5]Here is a YouTube video which shows how one can visualise a covering space of S^1 via the exponential map $p : \mathbb{R} \to S^1$.

Example 1.18. The map $p : \mathbb{R}^+ \to S^1$ given by $x \mapsto \exp(2\pi i x)$ is surjective and a local homeomorphism, but it is not a covering map. This is because the point $b = (1,0) \in S^1$ has no neighbourhood U that is not evenly covered by p. To see why, the boundary of $\mathbb{R}^+$ at $x = 0$ disrupts the ability to create disjoint preimages that evenly cover the neighbourhoods of S^1.

Example 1.19 (Munkres p. 341 Question 5). The map $p : S^1 \to S^1$ given by $z \mapsto z^2$ is a covering map. More generally, for any $n \in \mathbb{Z} \setminus \{0\}$, the map $p : S^1 \to S^1$ given by $z \mapsto z^n$ is a covering map.

To see why, for the general case, suppose $p(z) = w$. Consider $w \in \mathbb{C} \setminus \{0\}$, which has a small neighbourhood U around $\mathbb{C} \setminus \{0\}$ which avoids the branch cut of the complex logarithm. So, the pre-image $p^{-1}(w)$ consists of n distinct points, i.e.

$$z_k = |w|^{1/n} \exp\left(\frac{2k\pi}{n} i\right) \quad \text{such that} \quad p(z_k) = w.$$

Proposition 1.5. Let $p : E \to B$ be a covering map. If B_0 is a subspace of B and $E_0 = p^{-1}(B_0)$, then the map $p_0 : E_0 \to B_0$ by restricting p is a covering map.

Proof. Consider the following commutative diagram:

$$\begin{tikzcd} E_0 \arrow[r] \arrow[d, "p|_{E_0}"'] & E \arrow[d, "p"] \\ B_0 \arrow[r] & B \end{tikzcd}$$

Given $b_0 \in B_0$, let U be an open set in B containing b_0 that is evenly covered by p. Also, let $\{V_\alpha\}$ be a partition of $p^{-1}(U)$ into slices. So, $U \cap B_0$ is a neighbourhood of b_0 in B_0, and the sets $V_\alpha \cap E_0$ are disjoint open sets in E_0 such that

$$p^{-1}(U \cap B_0) = \bigcup_\alpha V_\alpha \cap E_0,$$

and $p : V_\alpha \cap E_0 \to U \cap B_0$ is a homeomorphism. □

Proposition 1.6. If $p : E \to B$ and $p' : E' \to B'$ are covering maps, then

$$p \times p' : E \times E' \to B \times B' \quad \text{is a covering map.}$$

Proof. Consider the following commutative diagram:

$$\begin{tikzcd} & E \times E' \arrow[dl] \arrow[d] \arrow[dr] & \\ E \arrow[d, "p"'] & B \times B' \arrow[dl] \arrow[dr] & E' \arrow[d, "p'"] \\ B & & B' \end{tikzcd}$$

Given $b \in B$ and $b' \in B'$, let U and U' be neighbourhoods of b and b' respectively which are evenly covered by p and p'. Let $\{V_\alpha\}$ and $\{V'_\alpha\}$ be partitions of $p^{-1}(U)$ and $(p')^{-1}(U')$ into slices. Then, the inverse image under $p \times p'$ of the open set $U \times U'$ is the union of all the sets $V_\alpha \times V'_\alpha$. These are disjoint open sets of $E \times E'$, and each is mapped homeomorphically onto $U \times U'$ by $p \times p'$. □

Proposition 1.7. Let

$$q : X \to Y \text{ and } r : Y \to Z \quad \text{be covering maps.}$$

Define $p = r \circ q$. If $r^{-1}(z)$ is finite for all $z \in Z$, then p is a covering map.

Example 1.20 (covering map of torus). Let $\mathbb{T} = S^1 \times S^1$ denote the torus. The product map

$$p \times p : \mathbb{R} \times \mathbb{R} \to S^1 \times S^1 \subseteq \mathbb{C}^\times \times \mathbb{C}^\times \quad \text{is a covering of the torus by the plane } \mathbb{R}^2.$$

Here, $\mathbb{C}^\times = \mathbb{C} \setminus \{\mathbf{0}\}$. Also, p maps $(x, y) \in \mathbb{R} \times \mathbb{R}$ to $\left(e^{2\pi i x}, e^{2\pi i y}\right) \in S^1 \times S^1$. Geometrically, think of having an infinite grid in $\mathbb{R}^2$ and first wrap it with respect to the vertical axis to obtain a hollow cylinder. Thereafter, wrap it with respect to the horizontal axis to obtain the torus $\mathbb{T}$.

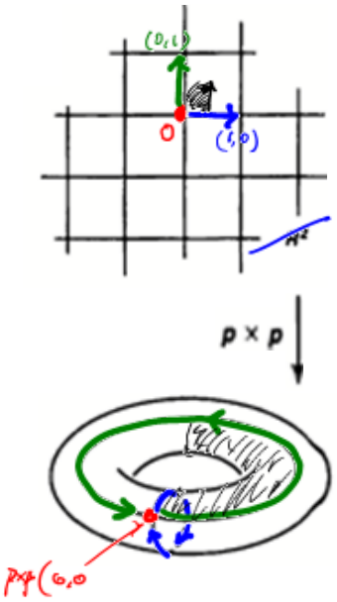


Figure 1.6: Covering space of the torus $\mathbb{T}$

Example 1.21 (figure-eight space). Let b_0 denote the point $p(0) \in S^1$ and B_0 denote the figure-eight space, i.e.

$$B_0 = \left(S^1 \times \{b_0\}\right) \cup \left(\{b_0\} \times S^1\right)$$

which is the union of S^1 and S^1 that have a point in common. This is an example of a space that is not simply-connected; it is the wedge sum of two circles, so B_0 does not contain

any contractible looks that can be continuously shrunk to a point. The space $E_0 = p^{-1}(B)$ is the infinite grid

$$E_0 = (\mathbb{R} \times \mathbb{Z}) \cup (\mathbb{Z} \times \mathbb{R}) \quad \text{for which the grids intersect at } \mathbb{Z} \times \mathbb{Z}.$$

The map $p_0 : E_0 \to B_0$ obtained by restricting $p \times p$ is a covering map.

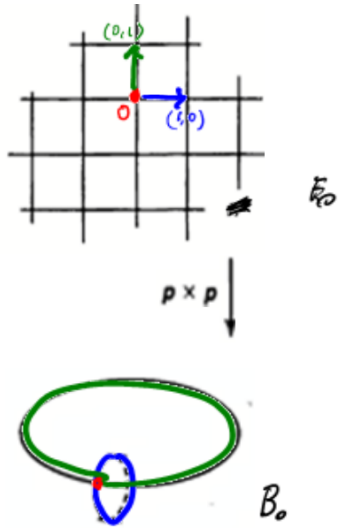


Figure 1.7: Covering space of the figure-eight space B_0

Example 1.22 (punctured plane). Consider the covering map of the punctured plane by the open upper-half plane $\mathcal{H} = \mathbb{R} \times \mathbb{R}^+$ as follows:

$$\mathbb{R} \times \mathbb{R}^+ \to S^1 \times \mathbb{R}^+ \to \mathbb{R}^2 \setminus \{\mathbf{0}\} = \mathbb{C}^\times \quad \text{given by} \quad (x,y) \mapsto \left(e^{2\pi i x}, r\right) \mapsto re^{2\pi i x}.$$

For the map $(x,y) \mapsto \left(e^{2\pi i x}, r\right)$, we see that the first coordinate $x \in \mathbb{R}$ is mapped onto the unit circle S^1 via the complex exponential function $e^{2\pi i x}$, which wraps the real line onto the circle. $r = \sqrt{x^2+y^2}$ measures the distance this point makes with the origin.

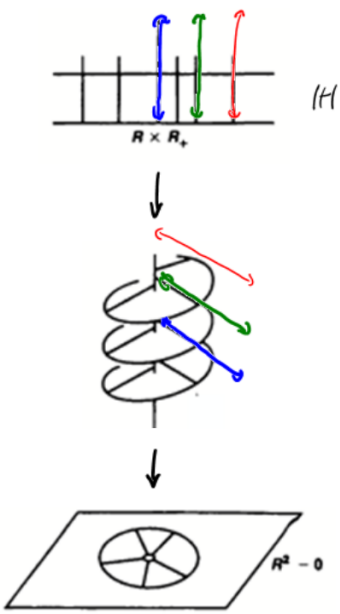


Figure 1.8: Covering space of the punctured plane

Proposition 1.8. Let $p : E \to B$ be a covering map. Suppose U is an open set of B that is evenly covered by p. If U is connected, then the partition of $p^{-1}(U)$ into slices is unique.

Proof. Suppose $\{V_\alpha\}$ and $\{W_\beta\}$ are distinct partitions of $p^{-1}(U)$. Fix an arbitrary α and let $e \in V_\alpha$. Then, there exists a unique β such that $e \in W_\beta$. We will prove that $V_\alpha = W_\beta$. Suppose on the contrary that $V_\alpha \neq W_\beta$. Then, $V_\alpha \setminus W_\beta \neq \emptyset$ and $V_\alpha \cap W_\beta \neq \emptyset$. So, $V_\alpha \cap W_\beta$ is open (intersection of two open sets is open), and

$$V_\alpha = (V_\alpha \cap W_\beta) \cup (V_\alpha \setminus W_\beta).$$

Let $x \in V_\alpha \setminus W_\beta$ (which is open), then there exists $\beta_x \neq \beta$ such that $x \in W_{\beta_x}$. Notice that

$$V_\alpha \setminus W_\beta = V_\alpha \cap \bigcup_{x \in V_\alpha \setminus W_\beta} W_{\beta_x}, \quad \text{which is open.}$$

Since U is connected and $p^{-1}(U)$ is homeomorphic to U through each slice, V_α must be connected as well, i.e. it cannot be decomposed into two disjoint open subsets, as would happen if $V_\alpha \setminus W_\beta$ were non-empty and open. This results in a contradiction. □

Lemma 1.12. Let $p : E \to B$ be a covering map, where B is connected. If

$$\left|p^{-1}(b_0)\right| = k \text{ for some } b_0 \in B, \quad \text{then} \quad \left|p^{-1}(b)\right| = k \text{ for all } b \in B.$$

In such a case, we say that E is a k-fold covering of B.

Proof. By the local triviality property, the set of elements $b \in B$ such that $\left|p^{-1}(b)\right| = \left|p^{-1}(b_0)\right|$ is both open and closed. □

Proposition 1.9. Let $p : E \to B$ be a covering map. If B is Hausdorff, then E is also Hausdorff. The same applies if B is T_3, $T_{3\frac{1}{2}}$, or locally compact Hausdorff. In addition, if B is compact and $p^{-1}(b)$ is finite for all $b \in B$, then E is compact.

1.5 Homotopy Lifting Property of Covering Maps

Definition 1.18. Let $p : E \to B$ be a B-space and $f : X \to B$ be a continuous map. A lifting of f to E over p is a continuous map $\widetilde{f} : X \to E$ such that $p \circ \widetilde{f} = f$.

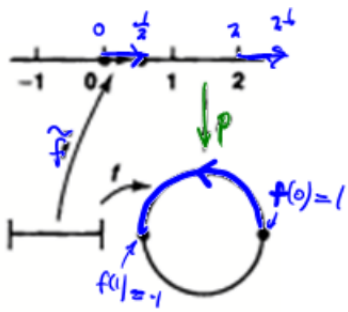
$$\begin{tikzcd} & E \arrow[d, "p"] \\ X \arrow[ur, "\widetilde{f}"] \arrow[r, "f"'] & B \end{tikzcd}$$

Example 1.23. Let $p:\mathbb{R}\to S^1$ be a covering map. The path $f(s)=(\cos\pi s,\sin\pi s)$ lifts to the path $\widetilde{f}(s)=s/2$ beginning at 0 and ending at $1/2$. To verify this statement, consider

$$(p\circ\widetilde{f})(s)=p(s/2)\quad\text{so}\quad p(s)=f(2s)=(\cos 2\pi s,\sin 2\pi s)$$

which wraps around the unit circle anticlockwise once from $(1,0)$ to $(-1,0)$ as s increases from 0 to $1/2$.

$$\begin{tikzcd}
 & \mathbb{R} \arrow[d, "p"] \\
\mathbb{R} \arrow[ur, "\widetilde{f}(s)=s/2"] \arrow[r, "f(s)=(\cos\pi s,\sin\pi s)"'] & S^1
\end{tikzcd}$$

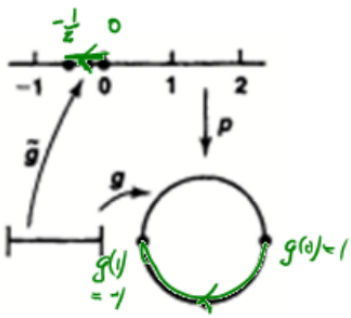


Example 1.24. Let $p:\mathbb{R}\to S^1$ be a covering map. The path $f(s)=(\cos\pi s,-\sin\pi s)$ lifts to the path $\widetilde{g}(s)=-s/2$ beginning at 0 and ending at $-1/2$. Again, we shall verify this statement. Note that

$$(p\circ\widetilde{g})(s)=p(-s/2)\quad\text{so}\quad p(s)=f(-2s)=(\cos 2\pi s,\sin 2\pi s)$$

which wraps around the unit circle from $(1,0)$ to $(-1,0)$ clockwise as s decreases from 0 to $-1/2$.

$$\begin{tikzcd}
 & \mathbb{R} \arrow[d, "p"] \\
\mathbb{R} \arrow[ur, "\widetilde{g}(s)=-s/2"] \arrow[r, "g(s)=(\cos\pi s,-\sin\pi s)"'] & S^1
\end{tikzcd}$$



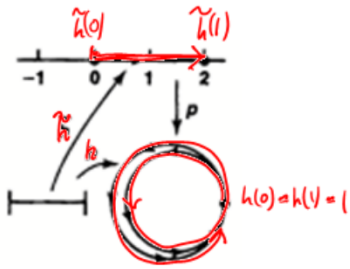
Example 1.25. Let $p:\mathbb{R}\to S^1$ be a covering map. The path $h(s)=(\cos 4\pi s,\sin 4\pi s)$ lifts to the path $\widetilde{h}(s)=2s$ beginning at 0 and ending at 2. Again, we shall verify this statement. Note that

$$\left(p\circ\widetilde{h}\right)(s)=p(2s)\quad\text{so}\quad p(s)=h(s/2)=(\cos 2\pi s,\sin 2\pi s)$$

which wraps around the unit circle from $(1,0)$ to itself in an anticlockwise manner as s

increases from 0 to 2.

$$\begin{tikzcd}
 & \mathbb{R} \arrow[d, "p"] \\
\mathbb{R} \arrow[ur, "\widetilde{h}(s) = 2s"] \arrow[r, "h(s) = (\cos 4\pi s, \sin 4\pi s)"'] & S^1
\end{tikzcd}$$



Lemma 1.13. Let $p : E \to B$ be a covering map and $f : X \to B$ be a continuous map. Suppose

(i) there exists an open subset $U \subseteq B$ with $f(X) \subseteq U$ such that U is evenly covered by p;

(ii) there exists a connected subset $X_0 \subseteq X$ such that there exists a lifting $\widetilde{f_0} : X_0 \to E$ of $f|_{X_0}$ to E

Then, there exists a lifting $\widetilde{f} : X \to E$ of f to E such that $\widetilde{f}|_{X_0} = \widetilde{f_0}$.

Theorem 1.8 (unique path lifting property). Let $p : E \to B$ be a covering map and $f : I \to B$ be any path in B beginning at $b_0 = f(0)$. For any choice $e_0 \in p^{-1}(b_0)$ in the fiber over b_0, there exists

$$\text{a unique lifting } \widetilde{f} : I \to E \quad \text{to a path } E \text{ beginning at } e_0.$$

$$\begin{tikzcd}
 & & E \arrow[d, "p"] \\
\{0\} \arrow[r] \arrow[urr] & I \arrow[r, "f"'] \arrow[ur, dashed, "\widetilde{f}"] & B
\end{tikzcd}$$

Theorem 1.9 (unique path-homotopy lifting property). Let $p : E \to B$ be a covering map and $F : I \times I \to B$ be a continuous map such that $F(0,0) = b_0$. For any choice $e_0 \in p^{-1}(b_0)$ in the fiber over b_0, there exists

$$\text{a unique lifting } \widetilde{F} : I \times I \to E \quad \text{with } \widetilde{F}(0,0) = e_0.$$

Moreover, if F is a path homotopy, then $\widetilde{F}$ is a path homotopy.

Proposition 1.10. Let $p : E \to B$ be a covering map and f and g be two paths in B from b_0 to b_1. Let $e_0 \in p^{-1}(b_0)$ be a point in the fiber over b_0. Let $\widetilde{f}$ and $\widetilde{g}$ be their

respective listings to paths in E beginning at e_0. If f and g are path homotopic in B, then $\widetilde{f}$ and $\widetilde{g}$ end at the same point $e_1 \in p^{-1}(b_1)$ and are path homotopic.

Lemma 1.14. Let $p : E \to B$ be a covering map. Let α and β be paths in B such that $\alpha(1) = \beta(0)$. Suppose $\widetilde{\alpha}$ and $\widetilde{\beta}$ are lifting of them such that $\widetilde{\alpha}(1) = \widetilde{\beta}(0)$. Then, $\widetilde{\alpha} * \widetilde{\beta}$ is a lifting of $\alpha * \beta$.

Definition 1.19 (the lifting correspondence). Let $p : E \to B$ be a covering map. Let $b_0 \in B$ and $e \in p^{-1}(b_0)$. For any path-homotopy class α of paths in B beginning at b_0, there exists a unique path-homotopy class $\widetilde{\alpha}_{e_0}$ of paths in E beginning at e_0 lifting α.

Definition 1.20 (monodromy action). Let $p : E \to B$ be a covering map. Let $b_0 \in B$. For any $\alpha \in \pi_1(B, b_0)$, the lifting correspondence induced by α is a map from $p^{-1}(b_0)$ to itself, i.e.

$$- * \alpha : p^{-1}(b_0) \to p^{-1}(b_0), \quad e_0 \mapsto e_0 * \alpha.$$

The monodromy action of $\pi_1(B, b_0)$ on $p^{-1}(b_0)$ is the map

$$* : p^{-1}(b_0) \times \pi_1(B, b_0) \to p^{-1}(b_0), \quad (e_0, \alpha) \mapsto e_0 * \alpha.$$

This is a well-defined right action of $\pi_1(B, b_0)$ on $p^{-1}(b_0)$.

Example 1.26 (monodromy action for the universal cover of the circle). Consider the covering map

$$p : \mathbb{R} \to S^1 \quad \text{where} \quad p(t) = e^{2\pi i t}.$$

Take the base point $b_0 = 1 \in S^1$. Then the fiber over b_0 is $p^{-1}(1) = \mathbb{Z}$. Also, we would see in Theorem 1.10 that $\pi_1(S^1, 1) \cong \mathbb{Z}$. For each $n \in \mathbb{Z}$, let α_n be the loop in S^1 based at 1 given by

$$\alpha_n(s) = e^{2\pi i n s} \quad \text{where } 0 \leq s \leq 1.$$

This loop winds around the circle n times. Now choose a point $k \in p^{-1}(1) = \mathbb{Z}$. The lift of α_n starting at k is the path $\widetilde{\alpha}_n(s) = k + ns$. Indeed,

$$p(\widetilde{\alpha}_n(s)) = p(k + ns) = e^{2\pi i(k+ns)} = e^{2\pi i n s} = \alpha_n(s)$$

since $e^{2\pi i k} = 1$ for every integer k. Therefore the endpoint of this lifted path is $\widetilde{\alpha}_n(1) = k + n$. Hence the monodromy action is given by $k * [\alpha_n] = k + n$. Under the identifications $p^{-1}(1) = \mathbb{Z}$ and $\pi_1(S^1, 1) \cong \mathbb{Z}$, the monodromy action is simply

$$\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z} \quad \text{where} \quad (k, n) \mapsto k + n.$$

Thus a loop which winds once around S^1 sends each point $k \in p^{-1}(1)$ to $k + 1$, while a loop which winds n times sends k to $k + n$.

Corollary 1.3 (Munkres p. 348 Question 8). Let $p : E \to B$ be a covering map, where E is path-connected. If B is simply-connected, then p is a homeomorphism.

Proof. Since E is path-connected, then the map $H \setminus \pi_1(B, b_0) \to p^{-1}(b_0)$ is bijective for all $b_0 \in B$. Since B is simply-connected, then $\pi_1(B, b_0)$ is trivial, so $p^{-1}(b_0)$ contains exactly one element. Hence, p is a bijection. Lastly, as p is open, then p^{-1} is a continuous map, and it follows that p is a homeomorphism. □

1.6 The Fundamental Group of the Circle

Theorem 1.10 (fundamental group of S^1). The fundamental group of S^1 is isomorphic to $\mathbb{Z}$. More precisely, consider the exponential covering map

$$p : \mathbb{R} \to S^1 \quad \text{given by} \quad p(x) = (\cos 2\pi x, \sin 2\pi x).$$

Since $\mathbb{R}$ is simply-connected, the monodromy action of $\pi_1(S^1, 1)$ on $\pi^{-1}(1) = \mathbb{Z}$ is simply transitive. Hence, we obtain a bijection

$$\phi : \pi_1(S^1, 1) \to \mathbb{Z} \quad \text{where} \quad \alpha \mapsto 0 * \alpha.$$

Proof. We claim that ϕ is a group isomorphism, and hence an isomorphism of groups. Suppose $\alpha, \beta \in \pi_1(S^1, 1)$. Then, we choose loops $f \in \alpha$ and $g \in \beta$ in S^1 based at 1 representing α and β respectively. By the unique path lifting property (Theorem 1.9), we obtain the liftings

$$\widetilde{f} \text{ of } f \text{ and } \widetilde{g} \text{ of } g \quad \text{to paths in } \mathbb{R} \text{ beginning at } 0.$$

Let $n = \widetilde{f}(1)$ and $m = \widetilde{g}(1)$ in $p^{-1}(1) = \mathbb{Z}$. So, $0 * \alpha = n$ and $0 * \beta = m$ by definition.

Let $\widetilde{\widetilde{g}}$ be the path $n + \widetilde{g}(s)$ in $\mathbb{R}$ beginning at n. By the properties of the exponential map $p : \mathbb{R} \to S^1$ (namely a group homomorphism with kernel $\mathbb{Z}$), it follows that $\widetilde{\widetilde{g}}$ is the unique lifting of g beginning at n.

Then, the product path $\widetilde{f} * \widetilde{\widetilde{g}}$ is defined, and it is the unique lifting of $f * g$ beginning at 0. The endpoint of this path is $\widetilde{\widetilde{g}}(1) = n + m$. Since $f * g$ represents $\alpha * \beta$, it follows that $0 * (\alpha * \beta) = n + m$. □

Example 1.27 (Munkres p. 347 Question 3). Let $p : E \to B$ be a covering map. Suppose α and β are paths in B with $\alpha(1) = \beta(0)$; let $\widetilde{\alpha}$ and $\widetilde{\beta}$ be liftings of them such that $\widetilde{\alpha}(1) = \widetilde{\beta}(0)$. Prove that $\widetilde{\alpha} * \widetilde{\beta}$ is a lifting of $\alpha * \beta$.

Solution. Recall the following definition: if $p : E \to B$ is a covering map and $f : X \to B$ be a continuous map. A lifting of f to E over p is a continuous map $\widetilde{f} : X \to E$ such that

$p \circ \widetilde{f} = f$.

$$\begin{tikzcd} & E \arrow[d, "p"] \\ X \arrow[ur, "\widetilde{f}"] \arrow[r, "f"'] & B \end{tikzcd}$$

As such, it suffices to prove that there exists a continuous map $\widetilde{\alpha} * \widetilde{\beta} : X \to E$ such that $p \circ \left(\widetilde{\alpha} * \widetilde{\beta}\right) = \alpha * \beta$. We are given that

$$p \circ \widetilde{\alpha} = \alpha \quad \text{and} \quad p \circ \widetilde{\beta} = \beta.$$

Since $\widetilde{\alpha}(1) = \widetilde{\beta}(0)$, we can concatenate the paths $\widetilde{\alpha}$ and $\widetilde{\beta}$. Since

$$(\alpha * \beta)(t) = \begin{cases} \alpha(2t) & \text{if } 0 \leq t \leq 1/2; \\ \beta(2t-1) & \text{if } 1/2 \leq t \leq 1, \end{cases}$$

we can define

$$\left(\widetilde{\alpha} * \widetilde{\beta}\right)(t) = \begin{cases} \widetilde{\alpha}(2t) & \text{if } 0 \leq t \leq 1/2; \\ \widetilde{\beta}(2t-1) & \text{if } 1/2 \leq t \leq 1. \end{cases}$$

Hence,

$$\begin{aligned} p \circ \left(\widetilde{\alpha} * \widetilde{\beta}\right) = p \circ \widetilde{\alpha}(2t) = (\alpha * \beta)(t) \quad & \text{for } 0 \leq t \leq 1/2 \quad \text{and} \\ p \circ \left(\widetilde{\alpha} * \widetilde{\beta}\right) = p \circ \widetilde{\beta}(2t-1) = (\alpha * \beta)(t) \quad & \text{for } 1/2 \leq t \leq 1. \end{aligned}$$

Hence, $p \circ \left(\widetilde{\alpha} * \widetilde{\beta}\right) = \alpha * \beta$ for all $0 \leq t \leq 1$. □

Example 1.28 (Munkres p. 348 Question 6). Consider the maps

$$g, h : S^1 \to S^1 \quad \text{given by} \quad g(z) = z^n \text{ and } h(z) = 1/z^n.$$

Here, $S^1 = \{z \in \mathbb{C} : |z| = 1\}$. Compute the induced homomorphisms g_*, h_* of the infinite cyclic group $\pi_1(S^1, b_0)$ into itself.

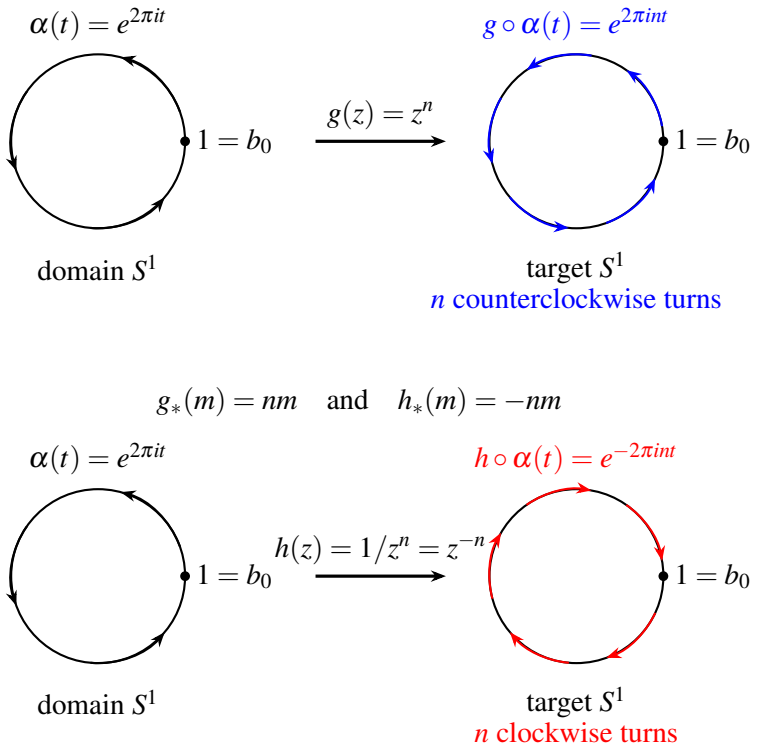


Figure 1.9: The map $g(z) = z^n$ preserves orientation and winds n times, while $h(z) = 1/z^n = z^{-n}$ reverses orientation and winds n times

Solution. We have

$$g_*, h_* : \pi_1(S^1, b_0) \to \pi_1(S^1, b_0).$$

Recall that any loop around S^1 can be represented by $\gamma(t) = e^{2\pi it}$ for $0 \leq t \leq 1$. This corresponds to a generator of $\pi_1(S^1, b_0)$, say $1 \in \mathbb{Z}$. So, $g(\gamma(t)) = e^{2n\pi it}$, for which the loop runs n times around the origin as t goes from 0 to 1. Under the induced homomorphism g_*, the generator 1 of $\pi_1(S^1, b_0)$ is mapped to n, i.e. $g_*(1) = n$. Since $\pi_1(S^1, b_0) \cong \mathbb{Z}$, then $g_* : \mathbb{Z} \to \mathbb{Z}$ is multiplication by n, i.e. for any $k \in \mathbb{Z}$, we have $g_*(k) = nk$.

In a similar fashion, $h_* : \mathbb{Z} \to \mathbb{Z}$ is multiplication by $-n$, i.e. for any $k \in \mathbb{Z}$, we have $h_*(k) = -nk$. □

1.7 Retractions

We now establish some classical topological results that follow from our knowledge of $\pi_1(S^1, 1) \cong \mathbb{Z}$.

Definition 1.21 (retraction and retract). Let X be a topological space and $A \subseteq X$ be a subspace. We have the canonical continuous inclusion map $j : A \to X$.

- A retraction of X onto A is a continuous map

$$r : X \to A \quad \text{such that} \quad r \circ j = \mathrm{id}_A$$

- A is a retract of X if and only if there exists a retraction of X onto A

Example 1.29. A singleton $\{x_0\}$ is a retract of the unit disc B^2. In particular, we can consider the retraction $p(x_0) = x_0$ for all $x_0 \in B^2$.

Example 1.30. The unit circle S^1 is a retract of the annulus. To see why, consider the following retraction: the inner ring of the annulus being *pushed outwards* such that it coincides with the outer ring. Alternatively, the outer ring can be *pushed inwards* such that it coincides with the inner ring. In either case, we would obtain S^1.

Example 1.31. The unit circle S^1 is a retract of the punctured disc.

Example 1.32. The unit circle S^1 is a retract of the punctured plane $\mathbb{R}^2 \setminus \{\mathbf{0}\}$[6] via the retraction $r : \mathbb{R}^2 \setminus \{\mathbf{0}\} \to S^1$ via $r(x) = x/\|x\|$.

Lemma 1.15. If A is a retract of X, then $j_* : \pi_1(A,a) \to \pi_1(X,a)$ is injective.

Proof. We have

$$\begin{aligned} r_* \circ j_* &= (r \circ j)_* && \text{by functoriality} \\ &= (\mathrm{id}_A)_* && \text{since there exists a retraction of } X \text{ onto } A \\ &= \mathrm{id}_{\pi_1(A,a)} && \text{by functoriality} \end{aligned}$$

This means that for any $[\alpha] \in \pi_1(A,a)$ (recall that $[\alpha]$ is a homotopy class of loops based at a in A), applying j_* to $[\alpha]$ first and then applying r_* returns $[\alpha]$. Also, note that $_* : X \to A$ is a retraction. Hence, given any $[\alpha], [\beta] \in \pi_1(A,a)$, if $j_*([a]) = j_*([\beta])$, then applying r_* to both sides yields $[\alpha] = [\beta]$. □

Theorem 1.11 (no-retraction theorem). There is no retraction of B^2 onto S^1.

Proof. Suppose on the contrary that the unit circle S^1 were a retract of the unit disc B^2. Then, there exists $j_* : \pi_1(S^1,1) \to \pi_1(B^2,1)$ is injective. However,

$$\text{the fundamental group of } S^1 \text{ is } \mathbb{Z} \quad \text{but} \quad \text{the fundamental group of } B^2 \text{ is trivial.}$$

By considering group isomorphism, we see that there is no injection j_* from $\mathbb{Z}$ to the trivial group, which is a contradiction. □

Corollary 1.4. The following hold:

$$\begin{aligned} \text{the identity map } i : S^1 \to S^1 \quad &\text{is not null-homotopic in } S^1 \quad \text{and} \\ \text{the inclusion map } j : S^1 \to \mathbb{R}^2 \setminus \{\mathbf{0}\} \quad &\text{is not null-homotopic in } \mathbb{R}^2 \setminus \{\mathbf{0}\} \end{aligned}$$

Lemma 1.16. Let $h : S^1 \to X$ be a continuous map. Then, the following are equivalent:

[6] Here is an animation in Desmos

(i) h is null-homotopic in X

(ii) h extends to a continuous map $k : B^2 \to X$, i.e. $k \circ j = h$ where $j : S^1 \hookrightarrow B^2$

(iii) h_* is the trivial homomorphism of fundamental groups, i.e. $\pi_1(S^1, 1) \to \pi_1(X, h(1))$

Proof. We first prove **(i)** implies **(ii)**. Let $H : S^1 \times I \to X$ be a homotopy between h and a constant map. Let $\pi : S^1 \times I \to B^2$ be the map $\pi(x,t) = (1-t)x$. Then, π is a quotient map as it collapses $S^1 \times 1$ to the point $\mathbf{0}$ and is otherwise injective. Since H is constant on $S^1 \times 1$, it induces a continuous map $k : B^2 \to X$ that is an extension of h via the quotient map π.

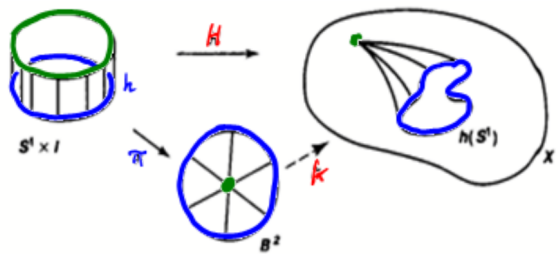


Figure 1.10: Visual proof that **(i)** implies **(ii)**

We then prove **(ii)** implies **(iii)**. Suppose $j : S^1 \hookrightarrow B^2$ is the inclusion map. Then, $h = k \circ j$. However,

$$j_* : \pi_1(S^1, b_0) \to \pi_1(B^2, b_0) = 1 \quad \text{is trivial.}$$

This is because the fundamental group of S^1 is $\mathbb{Z}$ whereas the fundamental group of B^2 is trivial. Hence, $h_* = k_* \circ j_*$ is trivial by functoriality.

Lastly, we prove **(iii)** implies **(i)**. We refer to the following figure:

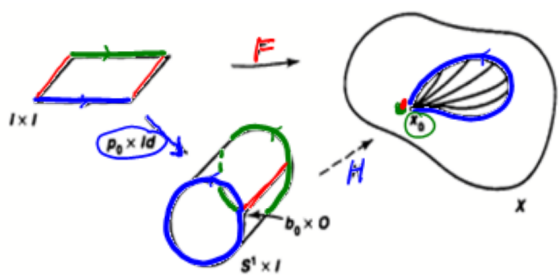


Figure 1.11: Visual proof that **(iii)** implies **(i)**

Let $p_0 : I \to S^1$ be a loop in S^1 such that $[p_0]$ generates $\pi_1(S^1, 1)$. Since h_* is trivial, the loop $f = h \circ p_0$ in X represents 1 in $\pi_1(X, x_0)$. Hence, there exists a path homotopy F in X between f and the constant path at x_0. The map

$$p_0 \times \mathrm{id} : I \times I \to S^1 \times I \quad \text{is a quotient map for each } i \in I.$$

It maps $0 \times t$ and $1 \times t$ to $b_0 \times t$ but is otherwise injective. The path homotopy F maps $0 \times I$ and $1 \times I$ and $I \times 1$ to the point $x_0 \in X$, which induces a continuous map $H : S^1 \times I \to X$ that is a homotopy between h and a constant map at x_0. □

Example 1.33 (Munkres p. 366 Question 6). Show that a retract of a contractible space is contractible.

Solution. Intuitively, this is obvious. Let X be a contractible space and $f : X \times I \to X$ be the homotopy that continuously deforms X to a point, say $x_0 \in X$. Suppose $r : X \to Y$ is a retract of X, i.e. $r \circ i = \mathrm{id}_Y$, where $i : Y \to X$ is the inclusion map. Hence, Y is a subspace of X and r restricts to the identity on Y. We wish to prove that Y is also contractible, so we shall construct a homotopy on Y using the given homotopy F on X.

Define the map $g : Y \times I \to Y$ by setting

$$g(y,t) = r(F(i(y),t)) \quad \text{for all } y \in Y \text{ and } t \in I.$$

This map is well-defined. We shall also check the boundary conditions. At $t = 0$, we see that $g(y,0) = r(F(i(y),0)) = r(i(y)) = y$ so $g(y,0) = y$ for all $y \in Y$. Also, at $t = 1$, we have $g(y,1) = r(F(i(y),1)) = r(x_0)$, where we recall that $x_0 \in X$ is the point that X contracts to. Hence,the image of $r(x_0)$ is some fixed point in Y, say $Y \ni y_0 = r(x_0)$.

Hence, g defines a homotopy between the identity map on Y and the constant map that sends every point in Y to y_0, showing that Y is contractible. □

Theorem 1.12 (Munkres p. 366 Question 7). Let A be a subspace of X; let $j : A \to X$ be the inclusion map, and let $f : X \to A$ be a continuous map. Suppose there is a homotopy $H : X \times I \to X$ between the map $j \circ f$ and the identity map of X. Then, the following hold:

(a) If f is a retraction, then j_* is an isomorphism.

(b) If H maps $A \times I$ into A, then j_* is an isomorphism.

1.8 Fixed Points

Definition 1.22. A vector field on B^2 is a continuous map $v : B^2 \to \mathbb{R}^2$. v is said to be non-vanishing if and only if $v(x) \neq \mathbf{0}$ for all x. In such a case, v is a continuous map $v : B^2 \to \mathbb{R}^2 \setminus \{\mathbf{0}\}$.

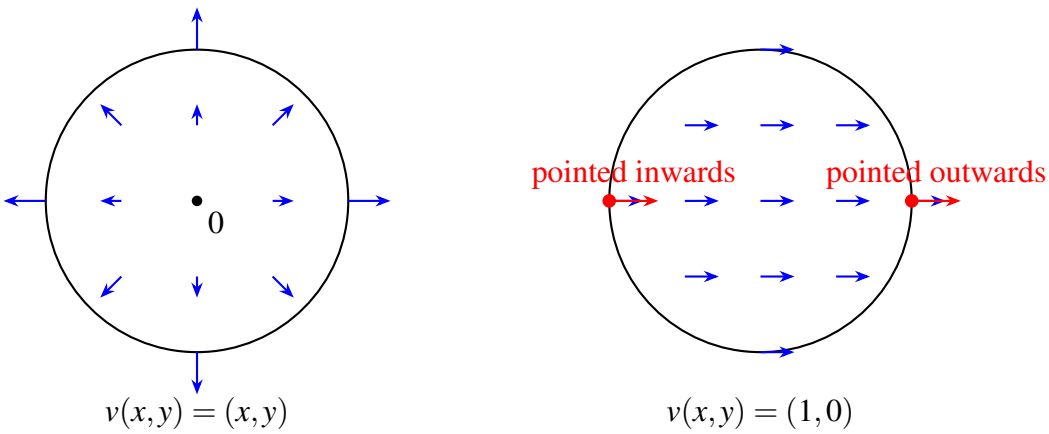


Figure 1.12: The radial vector field $v : B^2 \to \mathbb{R}^2$ given by $v(\mathbf{x}) = \mathbf{x}$, which vanishes at $(0,0)$, is shown on the left; the non-vanishing vector field $v : B^2 \to \mathbb{R}^2$ given by $v(x,y) = (1,0)$ is shown on the right

Theorem 1.13 (inward outward pointing for the disc). Given a non-vanishing vector field on B^2, i.e. $v : B^2 \to \mathbb{R}^2 \setminus \{\mathbf{0}\}$, there exists

$$\begin{aligned} \text{a point of } S^1 \quad & \text{where the vector field points directly inward} \\ \text{and a point of } S^1 \quad & \text{where the vector field points directly outward} \end{aligned}$$

That is to say,

$$\begin{aligned} \text{there exists } p \in S^1 \quad & \text{such that } v(p) \in -\mathbb{R}_{>0} \cdot p \\ \text{and there exists } q \in S^1 \quad & \text{such that } v(q) \in \mathbb{R}_{>0} \cdot q \end{aligned}$$

Proof. Let $w : S^1 \to \mathbb{R}^2 \setminus \{\mathbf{0}\}$ denote the restriction of v to S^1. Since v is defined on the whole closed disc B^2, the map w extends to B^2. Hence, w is null-homotopic in $\mathbb{R}^2 \setminus \{\mathbf{0}\}$. Indeed, a homotopy is given by

$$H : S^1 \times I \to \mathbb{R}^2 \setminus \{\mathbf{0}\} \quad \text{where} \quad H(x,t) = v(tx).$$

When $t = 1$, we have $H(x,1) = v(x) = w(x)$; when $t = 0$, we have $H(x,0) = v(\mathbf{0})$ which is a constant map. Thus, w is null-homotopic.

We first prove that there exists $p \in S^1$ such that $v(p) \in -\mathbb{R}_{>0} \cdot p$. Suppose on the contrary that $v(p) \notin -\mathbb{R}_{>0} \cdot p$ for every $p \in S^1$. Equivalently, $w(x) \notin -\mathbb{R}_{>0} \cdot x$ for every $x \in S^1$. We consider the map

$$F : S^1 \times I \to \mathbb{R}^2 \quad \text{where} \quad F(x,t) = tx + (1-t)\,w(x).$$

We claim that $F(x,t) \neq \mathbf{0}$ for all $x \in S^1$ and $t \in I$. This is clear when $t = 0$ since $F(x,0) = w(x) \neq \mathbf{0}$ and when $t = 1$, we have $F(x,1) = x \neq \mathbf{0}$. Suppose $0 < t < 1$. If $F(x,t) = \mathbf{0}$, then $tx + (1-t)\,w(x) = \mathbf{0}$ so $w(x) = -\frac{t}{1-t}x$ so $w(x) \in -\mathbb{R}_{>0} \cdot x$ which contradicts our assumption.

Hence, $F(x,t) \neq \mathbf{0}$ for all $x \in S^1$ and $t \in I$. So, F is a homotopy in $\mathbb{R}^2 \setminus \{\mathbf{0}\}$ from w to the inclusion map $j : S^1 \hookrightarrow \mathbb{R}^2 \setminus \{\mathbf{0}\}$ where $j(x) = x$. Since w is null-homotopic, then j is also null-homotopic in $\mathbb{R}^2 \setminus \{\mathbf{0}\}$. However, this is impossible since the map j represents the generator of $\pi_1\left(\mathbb{R}^2 \setminus \{\mathbf{0}\}\right) \cong \mathbb{Z}$.[7] The other result can be proven similarly. □

Theorem 1.14 (Brouwer fixed-point theorem for the disc). If $f : B^2 \to B^2$ is continuous, then there exists $\mathbf{x} \in B^2$ such that $f(\mathbf{x}) = \mathbf{x}$.

Proof. Suppose non the contrary that for all $\mathbf{x} \in B^2$, we have $f(\mathbf{x}) \neq \mathbf{x}$. Equivalently, there exists a non-vanishing vector field on B^2 given by the map

$$v : B^2 \to \mathbb{R}^2 \setminus \{0\} \quad \text{where} \quad v(\mathbf{x}) = f(\mathbf{x}) - \mathbf{x}.$$

So, there exists $\mathbf{x} \in S^1$ such that $v(\mathbf{x}) \in \mathbb{R}_{>0} \cdot \mathbf{x}$ points directly outward, i.e. there exists $a > 0$ such that $f(\mathbf{x}) - \mathbf{x} = a\mathbf{x}$ so $f(\mathbf{x}) = (1+a)\mathbf{x}$ lies outside B^2. This contradicts Theorem 1.13. □

Note that $B^n \subseteq \mathbb{R}^n$. Here is a funny remark by Prof. Chin. If you place a map of Singapore on a table in our room such that the map is a scaled-down version of the area it represents, there exists a point on the map that corresponds exactly to the point in the room directly below it.

Theorem 1.15 (Brouwer fixed-point theorem for B^n). Let $f : B^n \to B^n$ be a continuous map. Then, there exists $\mathbf{x} \in B^n$ such that $f(\mathbf{x}) = \mathbf{x}$.

Corollary 1.5 (Perron-Frobenius theorem). Any $n \times n$ matrix $\mathbf{A}$ with positive real entries has a positive real eigenvalue. Moreover, $\mathbf{A}$ has an eigenvector whose entries are all positive.

Proof. Let

$$\Delta^{n-1} = \left\{ \mathbf{x} \in \mathbb{R}^n : x_i \geq 0 \text{ for all } i \text{ and } \sum_{i=1}^{n} x_i = 1 \right\}$$

denote the standard $(n-1)$-simplex. It is compact, convex and homeomorphic to the closed ball B^{n-1}. Hence, the Brouwer fixed point theorem (Theorem 1.15) applies to continuous maps $\Delta^{n-1} \to \Delta^{n-1}$. For example, Figure 1.13 shows a 3-simplex.

[7] Equivalently, j winds once around the origin and hence cannot be null-homotopic in the punctured plane $\mathbb{R}^2 \setminus \{\mathbf{0}\}$.

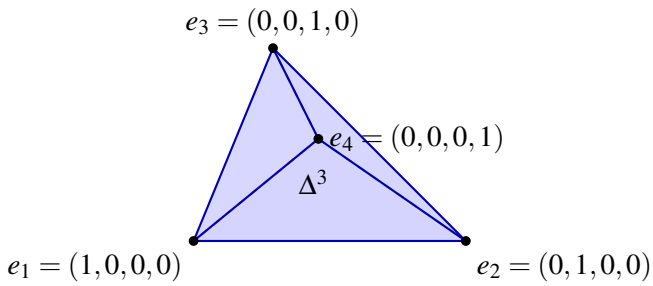


Figure 1.13: The standard 3-simplex Δ^3 drawn as a tetrahedron

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be the linear transformation given by multiplication by $\mathbf{A}$, so $T(\mathbf{x}) = \mathbf{A}\mathbf{x}$. Since every entry of $\mathbf{A}$ is positive, if $\mathbf{x} \in \Delta^{n-1}$, then $\mathbf{x} \neq \mathbf{0}$, and at least one component of $\mathbf{x}$ is > 0. Hence, every component of $\mathbf{A}\mathbf{x}$ is positive since the ith entry of $\mathbf{A}\mathbf{x}$ is $\sum_{j=1}^n a_{ij}x_j$.

Now, define

$$F : \Delta^{n-1} \to \Delta^{n-1} \quad \text{where} \quad F(\mathbf{x}) = \frac{\mathbf{A}\mathbf{x}}{\sum_{i=1}^n (\mathbf{A}\mathbf{x})_i}.$$

This is well-defined because every component of $\mathbf{A}\mathbf{x}$ is > 0. Moreover, we have $F(\mathbf{x}) \in \Delta^{n-1}$ because every component of $F(\mathbf{x})$ is ≥ 0 and

$$\sum_{i=1}^n (F(\mathbf{x}))_i = \sum_{i=1}^n \frac{(\mathbf{A}\mathbf{x})_i}{\sum_{k=1}^n (\mathbf{A}\mathbf{x})_k} = 1.$$

Also, F is continuous because it is obtained from polynomial functions by dividing by a strictly positive continuous function. By the Brouwer fixed-point theorem (Theorem 1.15), F has a fixed point $\mathbf{x}_0 \in \Delta^{n-1}$. Hence $F(\mathbf{x}_0) = \mathbf{x}_0$, which implies

$$\frac{\mathbf{A}\mathbf{x}_0}{\sum_{i=1}^n (\mathbf{A}\mathbf{x}_0)_i} = \mathbf{x}_0.$$

Let $\lambda = \sum_{i=1}^n (\mathbf{A}\mathbf{x}_0)_i$. Since every component of $\mathbf{A}\mathbf{x}_0$ is > 0, we have $\lambda > 0$. As such, $\mathbf{A}\mathbf{x}_0 = \lambda\mathbf{x}_0$ so λ is a positive eigenvalue of $\mathbf{A}$. Finally, since $F(\mathbf{x})$ has all components positive for every $\mathbf{x} \in \Delta^{n-1}$, the fixed point $\mathbf{x}_0 = F(\mathbf{x}_0)$ also has all components positive. Hence $\mathbf{x}_0$ is a positive eigenvector corresponding to the positive eigenvalue λ. □

1.9 Deformation Retracts and Homotopy Type

Let X be a topological space and $A \subseteq X$ be a subspace. Then, we have the canonical continuous inclusion map $j : A \to X$. Recall from Definition 1.21 that a retraction of X onto A is a continuous map

$$r : X \to A \quad \text{such that} \quad r \circ j = \mathrm{id}_A.$$

In other words, $r|_A$ is the identity map of A. We say that A is a retract of X if and only if there exists a retraction of X onto A.

Definition 1.23 (deformation retraction). A deformation retraction of X onto A is a continuous map $H : X \times I \to X$ such that

$$\text{for all } x \in X \quad \text{we have } H(x,0) = x \text{ and } H(x,1) \in A$$

and for all $t \in I$ and $a \in A$, one has $H(a,t) = a$, i.e. H fixes a throughout all time $t \in I$.

Equivalently, a deformation retraction is such that the map

$$\text{the map } r : X \to A \text{ defined by } r(x) = H(x,1) \quad \text{is a retraction of } X \text{ onto } A$$

and H is a homotopy between the identity map of X and the map $j \circ r$ such that each point of A remains fixed during the homotopy.

From Definition 1.23, we see that A is a deformation retract of X if and only if there exists a deformation retraction of X onto A. From Figure 1.14, we see that a deformation retraction moves every point of X into A, while every point of A stays fixed.

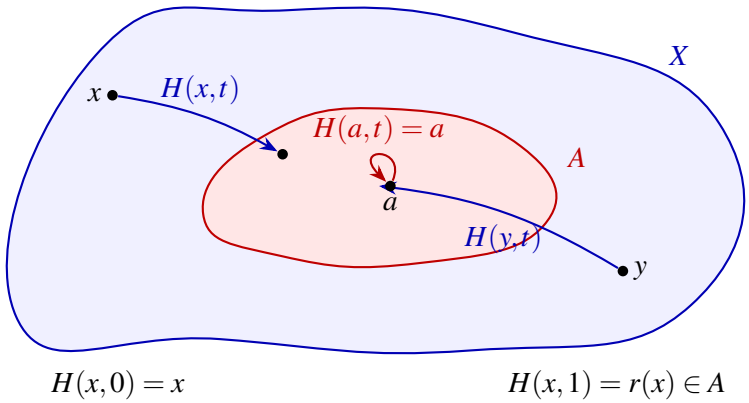


Figure 1.14: A deformation retraction of X onto A

Example 1.34. A singleton is a deformation retract of $I = [0,1]$, the unit disc B^2 and $\mathbb{R}^2$.

Example 1.35. S^n is a deformation retract of $X = \mathbb{R}^{n+1} \setminus \{\mathbf{0}\}$. To see why, we have a continuous map

$$H : X \times I \to X \quad \text{such that} \quad H(x,t) = (1-t)x + t\frac{x}{\|x\|}.$$

Example 1.36. A singleton is a retract, but not a deformation retract, of two distinct singletons. To see why, use connectedness.

Example 1.37. A singleton is a retract, but not a deformation retract, of a circle.

Example 1.38 (Munkres p. 366 Question 1). Show that if A is a deformation retract of X and B is a deformation retract of A, then B is a deformation retract of X.

Solution. By definition, there exist continuous maps $G : X \times I \to X$ and $H : A \times I \to A$ such that

$$\begin{aligned} &\text{for all } x \in X \quad \text{we have } G(x,0) = x \text{ and } G(x,1) \in A \quad \text{and} \\ &\text{for all } a \in A \quad \text{we have } H(a,0) = a \text{ and } H(a,1) \in B \end{aligned}$$

Moreover, for all $t \in I$, $a \in A$, and $b \in B$, we have

$$G(a,t) = a \quad \text{and} \quad H(a,t) = b.$$

We need to construct a continuous map $F : X \times I \to X$ such that

$$\text{for all } x \in X \quad \text{we have } F(x,0) = x \text{ and } F(x,1) \in B.$$

Also, for all $t \in I$ and $b \in B$, we must have $F(b,t) = b$. The required continuous map is

$$F(x,t) = \begin{cases} G(x,2t) & \text{if } 0 \le t \le 1/2; \\ H(G(x,1),2t-1) & \text{if } 1/2 \le t \le 1. \end{cases}$$

We proceed with a verification — when $t = 1/2$, the first piece-wise component implies $F(x,1/2) = G(x,1) \in A$ whereas the second piece-wise component implies $F(x,1/2) = H(G(x,1),0)$. By the hypothesis, it was mentioned that $G(x,1) \in A$, so there exists $a_0 \in A$ such that $G(x,1) = a_0$. Hence, $H(G(x,1),0) = H(a_0,0) = a_0 \in A$, which corresponds with what we had earlier.

Moreover, when $t = 1$, we have $F(x,1) = H(G(x,1),1) = H(a_0,1) \in B$, which corresponds to the hypothesis as we require $F(x,1) \in B$. □

Lemma 1.17. Let $h,k : (X,x_0) \to (Y,y_0)$ be continuous maps. If h and k are homotopic and the image of the base point x_0 of X remains fixed at y_0 during the homotopy, then the homomorphism

$$h_*, k_* : \pi_1(X,x_0) \to \pi_1(Y,y_0) \quad \text{where} \quad [f] \mapsto [h \circ f], [k \circ f] \text{ are equal.}$$

Proof. By assumption, there is a homotopy $H : X \times I \to Y$ between h and k such that $H(x_0,t) = y_0$ for all t. If f is a loop in X based at x_0, the composite

$$I \times I \xrightarrow{f \times \mathrm{id}} X \times I \xrightarrow{H} Y \quad \text{is a path homotopy between } h \circ f \text{ and } k \circ f.$$

This is because H maps $x_0 \times I$ to y_0. □

Theorem 1.16. If A is a deformation retract of X and $x_0 \in A$, the inclusion map $j : (A,x_0) \to (X,x_0)$ induces

$$\text{an isomorphism} \quad j_* : \pi_1(A,x_0) \cong \pi_1(X,x_0) \text{ of fundamental groups.}$$

Corollary 1.6. The inclusion map

$$j : S^n \to \mathbb{R}^{n+1} \setminus \{\mathbf{0}\} \quad \text{induces an isomorphism} \quad j_* : \pi_1(S^1, b_0) \to \pi_1(\mathbb{R}^{n+1} \setminus \{\mathbf{0}\}, b_0)$$

of fundamental groups.

Example 1.39. The punctured xy-plane $(\mathbb{R}^2 \setminus \{\mathbf{0}\}) \times 0$ is a deformation retract of the space $\mathbb{R}^3 \setminus B$, where B is the z-axis in $\mathbb{R}^3$ via

$$H : (\mathbb{R}^3 \setminus B) \times I \to (\mathbb{R}^3 \setminus B), \quad \text{where} \quad H(x,y,z,t) = (x,y,(1-t)z).$$

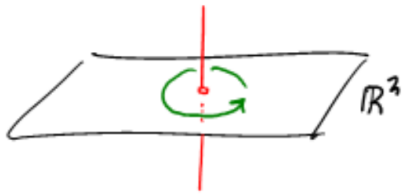


Figure 1.15: The punctured xy-plane is a deformation retract of $\mathbb{R}^3 \setminus B$

As shown in the figure above, we have $\pi_1(\mathbb{R}^2 \setminus \{\mathbf{0}\}, b_0) \cong \pi_1(\mathbb{R}^3 \setminus B, b_0)$. When $t = 0$, we have $H(x,y,z,0) = (x,y,z)$, which means H starts with the identity map on $\mathbb{R}^3 \setminus B$; when $t = 1$, we have $H(x,y,z,1) = (x,y,0)$, which collapses every point $(x,y,z) \in \mathbb{R}^3 \setminus B$ onto the plane $z = 0$.

Example 1.40 (figure-eight space). The figure-eight space is a deformation retract of the doubly punctured plane $\mathbb{R}^2 \setminus \{p,q\}$ via a three-stage deformation as shown in Figure 1.16.

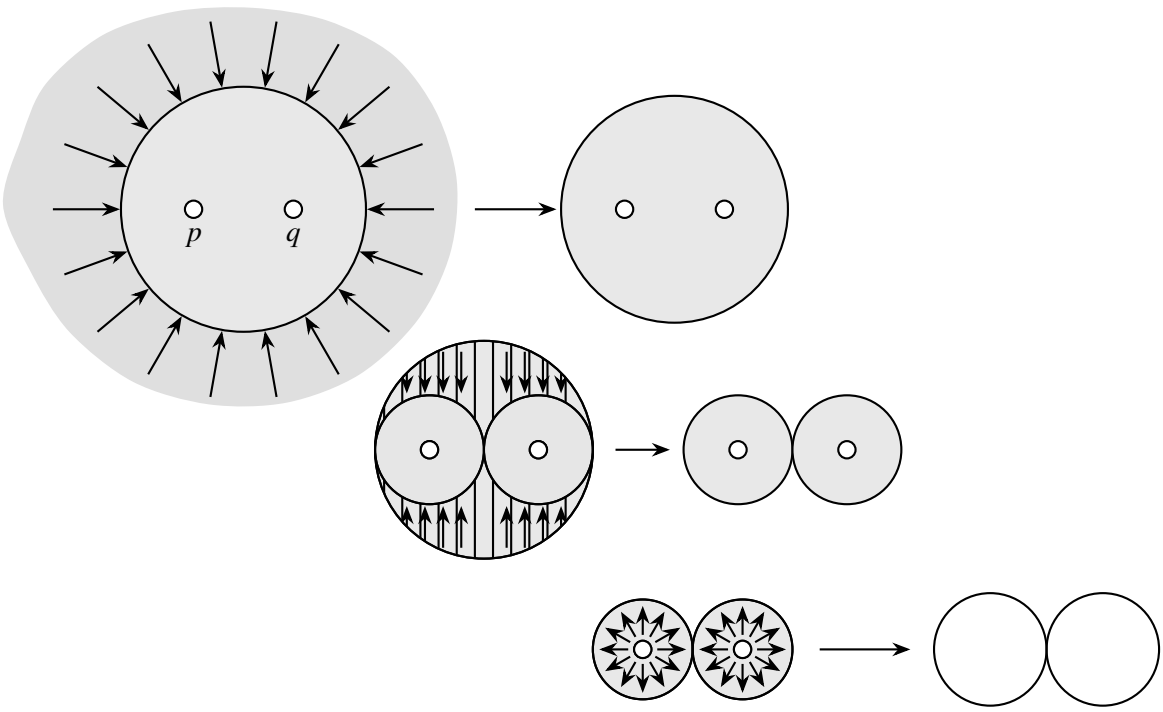


Figure 1.16: The figure-eight space is a deformation retract of the doubly punctured plane $\mathbb{R}^2 \setminus \{p,q\}$

Intuitively, this means that $\mathbb{R}^2 \setminus \{p,q\}$ can be continuously shrunk onto a space homeomorphic to two circles meeting at one point, without ever moving the points of the final figure-eight space. We describe the deformation in three stages as shown in Figure 1.16.

First, choose a closed disc $D \subseteq \mathbb{R}^2$ containing the two missing points p and q in its interior. The region outside D may be pushed radially inward onto the boundary ∂D. Since the points p and q lie inside D, this deformation does not touch the punctures. Thus $\mathbb{R}^2 \setminus \{p,q\}$ deformation retracts onto $D \setminus \{p,q\}$.

Next, inside the punctured disc $D \setminus \{p,q\}$, choose two smaller closed disks D_p and D_q, centred at p and q, respectively. We choose them so that they are tangent to each other at exactly one point and also tangent to the boundary of D. The remaining part of $D \setminus \big(D_p \cup D_q\big)$ can be pushed continuously onto $D_p \cup D_q$. Since p and q are missing, what remains is

$$(D_p \setminus \{p\}) \cup \big(D_q \setminus \{q\}\big)$$

where the two punctured disks touch at one point.

Third, each punctured disk deformation retracts onto a circle surrounding its missing point. Indeed, $D_p \setminus \{p\}$ retracts onto ∂D_p, and $D_q \setminus \{q\}$ retracts onto ∂D_q. Since the two disks are tangent, their boundary circles meet at one common point. Hence the final deformation retract is $\partial D_p \cup \partial D_q$, which is precisely a figure-eight space, namely two copies of S^1 joined at one point.

Example 1.41 (theta space). The theta space $\theta = S^1 \cup (\{0\} \times [-1,1])$ is a deformation retract of the doubly punctured plane $\mathbb{R}^2 \setminus \{p,q\}$. Refer to Figure 1.17.

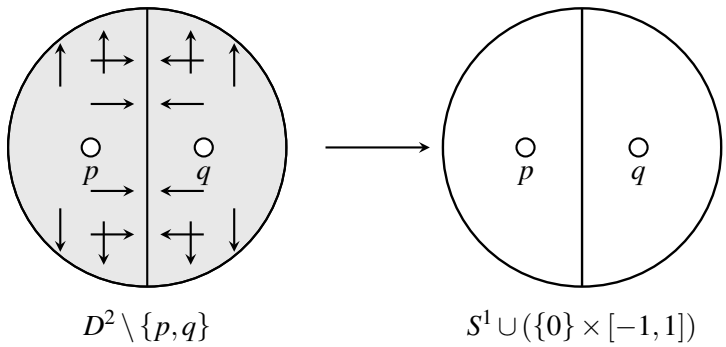


Figure 1.17: The theta space as a deformation retract of the doubly punctured plane

Intuitively, we first choose a closed disc $D \subseteq \mathbb{R}^2$ containing the two missing points p and q. The part of the plane outside D may be pushed radially inward onto the boundary ∂D. Hence $\mathbb{R}^2 \setminus \{p,q\}$ deformation retracts onto $D \setminus \{p,q\}$.

Now place the punctures p and q on opposite sides of a vertical diameter of D. The vertical diameter separates the two punctures. Inside the disc, every point may be pushed continuously either toward the boundary circle ∂D or toward this vertical diameter. The

boundary circle is kept fixed, and the vertical diameter is also kept fixed throughout the deformation. Thus the punctured disc $D \setminus \{p,q\}$ deformation retracts onto the union of the boundary circle and the vertical diameter.

As such, the resulting deformation retract is $\partial D \cup (\{0\} \times [-1,1])$ which is homeomorphic to $S^1 \cup (\{0\} \times [-1,1])$. Intuitively, this space is called the theta space because it looks like the Greek letter θ.

From Examples 1.40 and 1.41, we infer that the figure-eight space and the theta space have isomorphic fundamental groups even though neither is a deformation retract of the other.

Definition 1.24 (homotopy inverse). A homotopy inverse of a continuous map $f : X \to Y$ is a continuous map $g : Y \to X$ such that

$$\begin{aligned} &g \circ f : X \to X \text{ is homotopic to the identity map of } X \quad \text{and} \\ &f \circ g : Y \to Y \text{ is homotopic to the identity map of } Y. \end{aligned}$$

Then, f is a homotopy inverse of g.

Definition 1.25 (homotopy equivalence). A homotopy equivalence from X to Y is a continuous map $f : X \to Y$ for which there exists a homotopy inverse $g : Y \to X$ (Definition 1.24). X and Y are homotopy equivalent if and only if there exists a homotopy equivalence from X to Y.

Example 1.42 (a point and a contractible space). Let $X = \mathbb{R}^n$ and let $Y = \{0\}$. Define $f : \mathbb{R}^n \to \{0\}$ by $f(x) = 0$ and $g : \{0\} \to \mathbb{R}^n$ by $g(0) = 0$. Then, $f \circ g$ is the identity map on $\{0\}$, whereas $g \circ f$ is the constant map $x \mapsto 0$ which is homotopic to the identity map on $\mathbb{R}^n$. Indeed, if we define

$$H : \mathbb{R}^n \times I \to \mathbb{R}^n \quad \text{where} \quad H(x,t) = (1-t)x,$$

then $H(x,0) = x$ and $H(x,1) = 0$. So, $g \circ f \simeq \mathrm{id}_{\mathbb{R}^n}$, which implies $\mathbb{R}^n$ is homotopy equivalent to a point.

Recall Definition 1.25. Homotopy equivalence is an equivalence relation among topological spaces. Reflexivity and symmetry are obvious. To see why homotopy equivalence is transitive, suppose

$$\begin{aligned} &f : X \to Y \text{ is a homotopy equivalence from } X \text{ to } Y \quad \text{and} \\ &h : Y \to Z \text{ is a homotopy equivalence from } Y \text{ to } Z, \end{aligned}$$

then $h \circ f : X \to Z$ is a homotopy equivalence from X to Z. An equivalence class for this equivalence relation is a homotopy type.

Example 1.43. Let X be a topological space and $A \subseteq X$ be a deformation retract of X. Then, the canonical inclusion $j : A \to X$ is a homotopy equivalence from A to X.

Example 1.44. As seen in Examples 1.40 and 1.41, the figure-eight space and the theta space are deformation retracts of the doubly punctured plane $\mathbb{R}^2 \setminus \{p,q\}$ so they have the same homotopy type (or that they are homotopy equivalent spaces) even though neither is a deformation retract of the other.

Theorem 1.17. Let $f : X \to Y$ be a homotopy equivalence with $f(x_0) = y_0$. Then,

$$f_* : \pi_1(X,x_0) \to \pi_1(Y,y_0) \quad \text{is an isomorphism.}$$

In other words, $\pi_1(X,x_0)$ is a homotopy type invariant of (X,x_0).

Proof. Let $g : Y \to X$ be a homotopy inverse for f. Consider the maps and the induced homomorphism.

$$\begin{tikzcd}
(X,x_0) \arrow[r, "f"] \arrow[d, "g\circ f"'] & (Y,y_0) \arrow[d, "f\circ g"] \arrow[dl, "g"'] \\
(X,x_1) \arrow[r, "f"'] & (Y,y_1)
\end{tikzcd}
\qquad
\begin{tikzcd}
\pi_1(X,x_0) \arrow[r, "(f_{x_0})_*"] \arrow[d, "(g\circ f)_*"'] & \pi_1(Y,y_0) \arrow[d, "(f\circ g)_*"] \arrow[dl, "g_*"'] \\
\pi_1(X,x_1) \arrow[r, "(f_{x_1})_*"'] & \pi_1(Y,y_1)
\end{tikzcd}$$

By the hypothesis,

$$g \circ f : (X,x_0) \to (X,x_1) \quad \text{is isomorphic to } \mathrm{id}_X : (X,x_0) \to (X,x_0).$$

So, there exists a path α in X such that

$$(g \circ f)_* = \widehat{\alpha} \circ (\mathrm{id}_X)_* = \widehat{\alpha}$$

so $g_* \circ (f_{x_0})_* = (g \circ f)_*$ is an isomorphism. Similarly, $(f_{x_1})_* \circ g_* = (f \circ g)_*$ is an isomorphism. Therefore, g_* is an isomorphism, and so is $(f_{x_0})_* = (g_*)^{-1} \circ \widehat{\alpha}$. □

Theorem 1.18 (Fuch's theorem). Two spaces X and Y have the same homotopy type if and only if there exists a single space Z such that

$$X \text{ and } Y \text{ are homeomorphic to deformation retracts of } Z.$$

Definition 1.26 (degree of continuous map). Let $h : S^1 \to S^1$ be a continuous map. We define the degree of h as follows: let $b_0 = (1,0) \in S^1$. Choose a generator γ for the infinite cyclic group $\pi_1(S^1,b_0)$. If x_0 is any point of S^1, choose a path α in S^1 from b_0 to x_0 and define $\gamma(x_0) = \widehat{\alpha}(\gamma)$. Then, $\gamma(x_0)$ generates $\pi_1(S^1,x_0)$. The element $\gamma(x_0)$ is independent of the choice of the path α since the fundamental group of S^1 is abelian.

Choose $x_0 \in S^1$ and let $h(x_0) = x_1$. Consider the homomorphism

$$h_* : \pi_1(S^1, x_0) \to \pi_1(S^1, x_0)$$

Since both groups are infinite cyclic, we have

$$h_*(\gamma(x_0)) = d\gamma(x_1) \quad \text{for some } d \in \mathbb{Z}.$$

We say that d is the degree of h and it is denoted by $\deg h$.

Note that the degree of h is independent of the choice of the generator γ; choosing the other generator would merely change the sign of both sides of the equation $h_*(\gamma(x_0)) = d\gamma(x_1)$.

Example 1.45 (Munkres p. 366 Question 9). The following properties hold:

(a) The degree d is independent of the choice of x_0.

(b) If

$$h, k : S^1 \to S^1 \text{ are homotopic}, \quad \text{then} \quad \text{they have the same degree.}$$

(c) We have $\deg(h \circ k) = (\deg h) \cdot (\deg k)$.

(d) The degree of the constant map is 0, the degree of the identity map is 1, the degree of the reflection map $\rho(x_1, x_2) = (x_1, -x_2)$ is -1, and the degree of the nth power map $h(z) = z^n$, where $z \in \mathbb{C}$, is n.

(e) If $h, k : S^1 \to S^1$ have the same degree, they are homotopic.

1.10 Fundamental Group of a Union

Theorem 1.19 (Seifert-van Kampen theorem). Suppose $X = U \cup V$, where U and V are open and path-connected subsets of X, and suppose $U \cap V$ is non-empty and path-connected. Let $x_0 \in U \cap V$, and let

$$i : U \hookrightarrow X \text{ and } j : V \hookrightarrow X \quad \text{be the inclusion maps.}$$

Then, $\pi_1(X, x_0)$ is generated by the images of the induced homomorphisms

$$i_* : \pi_1(U, x_0) \to \pi_1(X, x_0) \quad \text{and} \quad j_* : \pi_1(V, x_0) \to \pi_1(X, x_0).$$

Equivalently, for every loop $f : I \to X$ based at x_0, there exist loops $g_1, \ldots, g_n$, each lying entirely in U or entirely in V, such that

$$[f] = [g_1] * \cdots [g_n] \quad \text{in } \pi_1(X, x_0).$$

$$\begin{tikzcd}
\pi_1(U \cap V, x_0) \arrow[r, "(k_U)_*"] \arrow[d, "(k_V)_*"'] & \pi_1(U, x_0) \arrow[d, "i_*"] \\
\pi_1(V, x_0) \arrow[r, "j_*"'] & \pi_1(X, x_0)
\end{tikzcd}$$

Proof. Let $f : I \to X$ be a loop based at x_0, so $f(0) = f(1) = x_0$. Since I is compact and $\{f^{-1}(U), f^{-1}(V)\}$ is an open cover of I, the Lebesgue number lemma allows us to choose a subdivision $0 = a_0 < a_1 < \cdots < a_n = 1$ such that, for each $i = 1, \ldots, n$, the image $f([a_{i-1}, a_i])$ is contained entirely in either U or V.

By refining the subdivision if necessary, we may assume that each subdivision point $f(a_i)$ lies in $U \cap V$. Indeed, if $f(a_i) \notin U \cap V$ for some $1 \leq i \leq n-1$, then $f(a_i) \in U \setminus V$ or $f(a_i) \in V \setminus U$. Suppose, for example, that $f(a_i) \in U \setminus V$. Since $f([a_{i-1}, a_i])$ and $f([a_i, a_{i+1}])$ both contain $f(a_i)$, neither of these two subpaths can lie in V. Hence both lie in U, so their union $f([a_{i-1}, a_{i+1}])$ lies in U. Thus the point a_i is unnecessary and may be removed from the subdivision. Repeating this process, we obtain a subdivision such that

$$f(a_i) \in U \cap V \quad \text{for all } i = 0, 1, \ldots, n.$$

For each $i = 1, \ldots, n$, let $f_i : I \to X$ be the reparametrised path

$$f_i(t) = f((1-t)a_{i-1} + ta_i).$$

Then f_i is a path from $f(a_{i-1})$ to $f(a_i)$, and its image lies entirely in either U or V. Since $U \cap V$ is path-connected and $x_0, f(a_i) \in U \cap V$, for each $i = 0, \ldots, n$, choose a path $\alpha_i : I \to U \cap V$ from x_0 to $f(a_i)$. We choose α_0 and α_n to be the constant path at x_0, since $f(a_0) = f(a_n) = x_0$.

Now define, for each $i = 1, \ldots, n$, $g_i = \alpha_{i-1} * f_i * \overline{\alpha_i}$. Then g_i is a loop based at x_0. Moreover, if f_i lies in U, then g_i lies in U, because α_{i-1} and α_i lie in $U \cap V \subseteq U$. Similarly, if f_i lies in V, then g_i lies in V. Hence each g_i is a loop based at x_0 lying entirely in U or entirely in V.

We claim that

$$[f] = [g_1] * [g_2] * \cdots * [g_n]. \tag{1.1}$$

Indeed, we have

$$[g_1] * \cdots * [g_n] = [\alpha_0 * f_1 * \overline{\alpha_1}] * [\alpha_1 * f_2 * \overline{\alpha_2}] * \cdots * [\alpha_{n-1} * f_n * \overline{\alpha_n}].$$

Since $\overline{\alpha_i} * \alpha_i$ is path homotopic to the constant path at $f(a_i)$, the consecutive factors cancel up to path homotopy. Also, α_0 and α_n are constant paths at x_0. Therefore,

$$[g_1] * \cdots * [g_n] = [f_1] * [f_2] * \cdots * [f_n].$$

But the concatenation $f_1 * f_2 * \cdots * f_n$ is just a reparametrisation of the original loop f. Hence, $[f_1] * [f_2] * \cdots * [f_n] = [f]$ so (1.1 holds. Thus every element of $\pi_1(X, x_0)$ can be written as a product of elements coming from $\pi_1(U, x_0)$ and $\pi_1(V, x_0)$. Hence $\pi_1(X, x_0)$ is generated by the images of i_* and j_*. □

Corollary 1.7. Suppose $X = U \cup V$, where U and V are open sets of X with $U \cap V \neq \emptyset$ and path-connected. If U and V are simply-connected, then X is simply-connected.

Proof. Suppose $X = U \cup V$, where U and V are open subsets of X, and suppose $U \cap V \neq \emptyset$ is path-connected. Since U and V are simply-connected, they are path-connected. Choose a base point $x_0 \in U \cap V$. By the Seifert-van Kampen theorem (Theorem 1.19), $\pi_1(X, x_0)$ is generated by the images of

$$i_* : \pi_1(U, x_0) \to \pi_1(X, x_0) \quad \text{and} \quad j_* : \pi_1(V, x_0) \to \pi_1(X, x_0),$$

where $i : U \hookrightarrow X$ and $j : V \hookrightarrow X$ are the inclusion maps. However, as U and V are simply-connected, then $\pi_1(U, x_0) = 0$ and $\pi_1(V, x_0) = 0$. Hence, the images of i_* and j_* are both trivial subgroups of $\pi_1(X, x_0)$. Since $\pi_1(X, x_0)$ is generated by these two trivial images, it follows that $\pi_1(X, x_0) = 0$ so X is simply-connected. □

Theorem 1.20. If $n \geq 2$, the n-sphere S^n is simply-connected.

Proof. Let $n \geq 2$. We prove that S^n is simply-connected using Corollary 1.7. Let $N = (0, \ldots, 0, 1)$ be the north pole and $S = (0, \ldots, 0, -1)$ be the south pole of S^n. Define

$$U = S^n \setminus \{N\} \quad \text{and} \quad V = S^n \setminus \{S\}.$$

Then U and V are open subsets of S^n, and $S^n = U \cup V$. Moreover, by stereographic projection, both U and V are homeomorphic to $\mathbb{R}^n$. Hence both U and V are simply-connected since $\mathbb{R}^n$ is simply-connected.

It remains to check that $U \cap V$ is path-connected. We have $U \cap V = S^n \setminus \{N, S\}$. Geometrically, this is the sphere with the north and south poles removed. This space is homeomorphic to $S^{n-1} \times (-1, 1)$. Indeed, each point of $S^n \setminus \{N, S\}$ lies on a unique horizontal sphere S^{n-1} at height $t \in (-1, 1)$. Since $n \geq 2$, we have $n - 1 \geq 1$, so S^{n-1} is path-connected. Also, $(-1, 1)$ is path-connected. Therefore $S^{n-1} \times (-1, 1)$ is path-connected, and hence $U \cap V$ is path-connected.

Thus $S^n = U \cup V$, where U and V are open and simply-connected, and $U \cap V$ is non-empty and path-connected. By Corollary 1.7, S^n is simply-connected. □

1.11 Fundamental Groups of Some Surfaces

We now turn to the computation of fundamental groups of several important surfaces and surface-like spaces. The guiding idea is that the fundamental group records the different

ways in which loops in a space can wind around holes. Thus spaces with different types of holes often have different fundamental groups. For example, a circle has fundamental group $\mathbb{Z}$, since a loop may wind around the circle any integer number of times. A torus has two independent circular directions, and its fundamental group is therefore $\mathbb{Z} \times \mathbb{Z}$. On the other hand, the figure-eight space has two loops joined at a point, but with no commutativity relation between them, so its fundamental group is the free group on two generators.

Theorem 1.21 (fundamental group of a product). Let X and Y be topological spaces, and $x_0 \in X$ and $y_0 \in Y$. Then,

$$\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0).$$

More precisely, let $p : X \times Y \times X$ where $p(x,y) = x$ and $q : X \times Y \to Y$ where $q(x,y) = y$ be the projection maps. Then, the induced homomorphisms

$$p_* : \pi_1(X \times Y, (x_0, y_0)) \to \pi_1(X, x_0) \quad \text{and} \quad q_* : \pi_1(X \times Y, (x_0, y_0)) \to \pi_1(Y, y_0)$$

define a homomorphism

$$\Phi : \pi_1(X \times Y, (x_0, y_0)) \to \pi_1(X, x_0) \times \pi_1(Y, y_0)$$

by

$$\Phi([f]) = (p_*([f]), q_*([f])) = ([p \circ f], [q \circ f]).$$

Then Φ is an isomorphism.

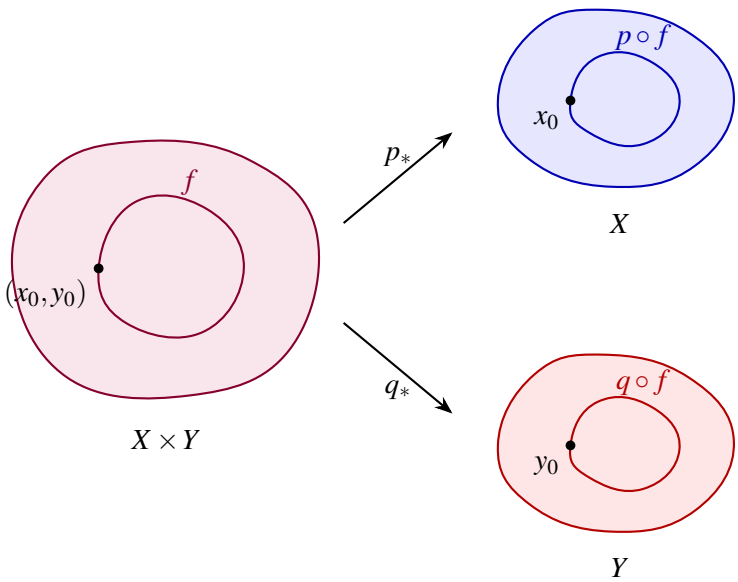


Figure 1.18: A loop f in $X \times Y$ projects to loops $p \circ f$ in X and $q \circ f$ in Y

Proof. Let

$$\Phi : \pi_1(X \times Y, (x_0, y_0)) \to \pi_1(X, x_0) \times \pi_1(Y, y_0)$$

be defined by

$$\Phi([f]) = ([p \circ f], [q \circ f]).$$

We first check that this is well-defined. Suppose $[f] = [g]$ in $\pi_1(X \times Y, (x_0, y_0))$. Then $f \simeq_p g$, so there exists a path homotopy $H : I \times I \to X \times Y$ such that $H(s,0) = f(s)$, $H(s,1) = g(s)$, and $H(0,t) = H(1,t) = (x_0, y_0)$ for all $s,t \in I$. Composing with the projection maps gives path homotopies

$$p \circ H : I \times I \to X \quad \text{and} \quad q \circ H : I \times I \to Y.$$

Thus $p \circ f \simeq_p p \circ g$ in X, and $q \circ f \simeq_p q \circ g$ in Y. Hence

$$[p \circ f] = [p \circ g] \quad \text{and} \quad [q \circ f] = [q \circ g].$$

Therefore $\Phi([f]) = \Phi([g])$, so Φ is well-defined.

Next, we show that Φ is a homomorphism. Let $[f], [g] \in \pi_1(X \times Y, (x_0, y_0))$. Then

$$\Phi([f] * [g]) = \Phi([f * g]) = ([p \circ (f * g)], [q \circ (f * g)]).$$

Since projections commute with concatenation, we have

$$p \circ (f * g) = (p \circ f) * (p \circ g) \quad \text{and} \quad q \circ (f * g) = (q \circ f) * (q \circ g).$$

Therefore,

$$\Phi([f] * [g]) = ([p \circ f] * [p \circ g], [q \circ f] * [q \circ g]) = \Phi([f]) * \Phi([g]).$$

Hence Φ is a homomorphism. We now construct an inverse map. Define

$$\Psi : \pi_1(X, x_0) \times \pi_1(Y, y_0) \to \pi_1(X \times Y, (x_0, y_0))$$

by

$$\Psi([\alpha], [\beta]) = [(\alpha, \beta)],$$

where $(\alpha, \beta) : I \to X \times Y$ is the loop given by $(\alpha, \beta)(s) = (\alpha(s), \beta(s))$. This is a loop based at (x_0, y_0), since

$$(\alpha, \beta)(0) = (\alpha(0), \beta(0)) = (x_0, y_0) \quad \text{and} \quad (\alpha, \beta)(1) = (\alpha(1), \beta(1)) = (x_0, y_0).$$

We check that Ψ is well-defined. Suppose $[\alpha] = [\alpha']$ in $\pi_1(X, x_0)$, and $[\beta] = [\beta']$ in $\pi_1(Y, y_0)$. Then there are path homotopies $F : I \times I \to X$ and $G : I \times I \to Y$ from from α to α', and from β to β', respectively. Define

$$K : I \times I \to X \times Y \quad \text{where} \quad K(s,t) = (F(s,t), G(s,t)).$$

Then K is a path homotopy from (α, β) to (α', β') because

$$\begin{aligned} K(s,0) &= (F(s,0), G(s,0)) = (\alpha(s), \beta(s)) \quad \text{and} \\ K(s,1) &= (F(s,1), G(s,1)) = \big(\alpha'(s), \beta'(s)\big). \end{aligned}$$

Also, for every $t \in I$, $K(0,t) = K(1,t) = (x_0, y_0)$. Hence $(\alpha, \beta) \simeq_p (\alpha', \beta')$, so $[(\alpha, \beta)] = [(\alpha', \beta')]$. Thus Ψ is well-defined. It remains to show that Φ and Ψ are inverses. We omit the proof. $\square$

Corollary 1.8 (fundamental group of torus). The fundamental group of the torus $\mathbb{T} = S^1 \times S^1$ is isomorphic to $\mathbb{Z} \times \mathbb{Z}$.

Corollary 1.9. The fundamental group of the n-dimensional torus is isomorphic to $\mathbb{Z}^{\oplus n}$.

Proof. The torus is defined by $\mathbb{T} = S^1 \times S^1$. Choose a base point $x_0 \in S^1$. Then the base point of $\mathbb{T}$ is $(x_0, x_0) \in S^1 \times S^1$. By the theorem on the fundamental group of a product (Theorem 1.21), we have

$$\pi_1\left(S^1 \times S^1, (x_0, x_0)\right) \cong \pi_1(S^1, x_0) \times \pi_1(S^1, x_0) \cong \mathbb{Z} \times \mathbb{Z}.$$

The result follows. □

Example 1.46 (Munkres p. 375 Question 1). Compute the fundamental groups of the solid torus $S^1 \times B^2$ and the product space $S^1 \times S^2$.

Solution. By Theorem 1.21,

$$\pi_1\left(S^1 \times B^2, (x_0, y_0)\right) \cong \pi_1\left(S^1, x_0\right) \times \pi_1\left(B^2, y_0\right).$$

We then use Theorem 1.10 and Figure 1.4 respectively to conclude that the fundamental group is isomorphic to $\mathbb{Z} \times \{e\} \cong \mathbb{Z}$.

Next, using Theorem 1.21, we compute

$$\pi_1\left(S^1 \times S^2, (x_0, y_0)\right) \cong \pi_1\left(S^1, x_0\right) \times \pi_1\left(S^2, y_0\right)$$

which is $\cong \mathbb{Z} \times \{e\} \cong \mathbb{Z}$ by Theorem 1.10 and Figure 1.4 respectively. □

Example 1.47 (Munkres p. 366 Question 2). For each of the following spaces, the fundamental group is either trivial, infinite cyclic, or isomorphic to the fundamental group of the figure eight. Determine for each space which of the three alternatives holds.

(a) The solid torus $B^2 \times S^1$.

(b) The torus $\mathbb{T}$ with a point removed

(c) The cylinder $S^1 \times I$

(d) The infinite cylinder $S^1 \times \mathbb{R}$

(e) $\mathbb{R}^3$ with non-negative x, y, and z axes deleted

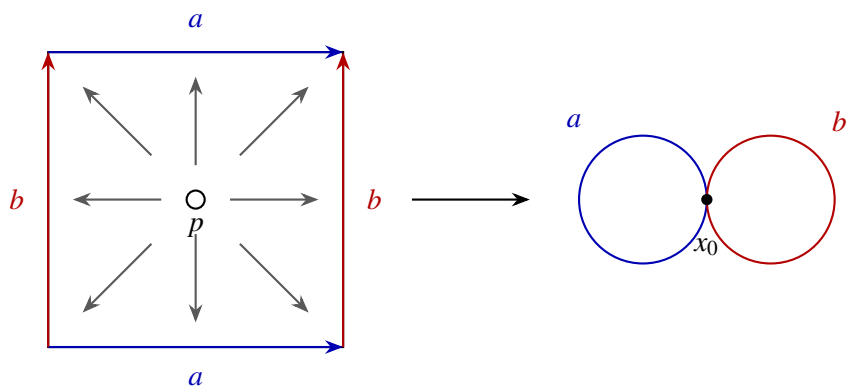
The following subsets of $\mathbb{R}^2$:

(i) $\{x : \|x\| > 1\}$ **(ii)** $\{x : \|x\| \geq 1\}$ **(iii)** $\{x : \|x\| < 1\}$ **(iv)** $S^1 \cup (\mathbb{R}_+ \times 0)$ **(v)** $S^1 \cup (\mathbb{R}_+ \times \mathbb{R})$ **(vi)** $S^1 \cup (\mathbb{R} \times 0)$ **(vii)** $\mathbb{R}^2 \setminus (\mathbb{R}_+ \times 0)$

Solution. We classify each space up to homotopy equivalence. The possible answers are the trivial group, $\mathbb{Z}$ or $\pi_1\left(S^1 \vee S^1\right)$ which is isomorphic to the fundamental group of the figure eight. We first discuss **(a)** to **(e)**.

(a) The fundamental group is $\mathbb{Z}$, which is infinite cyclic (see Example 1.46).

(b) The torus $\mathbb{T}$ with one point removed deformation retracts onto a figure-eight space. To see this, regard the torus as a square with opposite sides identified. Removing one point from the interior of the square allows the remaining surface to retract onto the boundary of the square. Under the edge identifications, this boundary becomes a wedge of two circles.



Punctured square model of $\mathbb{T} \setminus \{p\}$ Boundary after edge identifications

Figure 1.19: The punctured torus deformation retracts onto the figure-eight space

Hence,

$$\pi_1\left(\mathbb{T} \setminus \{p\}\right) \cong \pi_1\left(S^1 \vee S^1\right)$$

whose fundamental group is isomorphic to that of the figure eight[8].

(c) The cylinder $S^1 \times I$ deformation retracts onto $S^1 \times \{0\} \cong S^1$. Therefore, the fundamental group is just $\pi_1\left(S^1\right) \cong \mathbb{Z}$, which is infinite cyclic.

(d) Use the same idea as **(c)** to show that the infinite cylinder $S^1 \times \mathbb{R}$ deformation retracts onto $S^1 \times \{0\} \cong S^1$. So, the fundamental group is infinite cyclic[9].

(e) Let X denote the $\mathbb{R}^3$ with the non-negative x, y, z axes deleted. Radial projection gives a deformation retraction of X onto $S^2 \setminus \{p, q, r\}$, where p, q, r are the three points where the positive coordinate axes meet S^2. Removing one of these three points identifies this space with $\mathbb{R}^2 \setminus \{s, t\}$[10]. By Example 1.40, the doubly punctured plane deformation retracts onto the figure-eight space. Thus the fundamental group is isomorphic to that of the figure eight.

We then discuss **(i)** to **(vii)**.

[8] See this video for a geometric interpretation.
[9] See this video for a geometric interpretation.
[10] See this video for a geometric interpretation.

(i) The space is the exterior of the closed unit disc. It deformation retracts onto the unit circle S^1 by radial projection. Hence its fundamental group is $\mathbb{Z}$.

(ii) This also deformation retracts onto S^1. Hence its fundamental group is $\mathbb{Z}$.

(iii) This is the open unit disc, which is contractible. Hence its fundamental group is trivial.

(iv) The space $S^1 \cup (\mathbb{R}_+ \times \{0\})$ is obtained by attaching a ray to the circle at the point $(1,0)$ (Figure 1.20). The ray is contractible and retracts to its endpoint $(1,0)$. Hence the whole space deformation retracts onto S^1. Therefore its fundamental group is $\mathbb{Z}$.

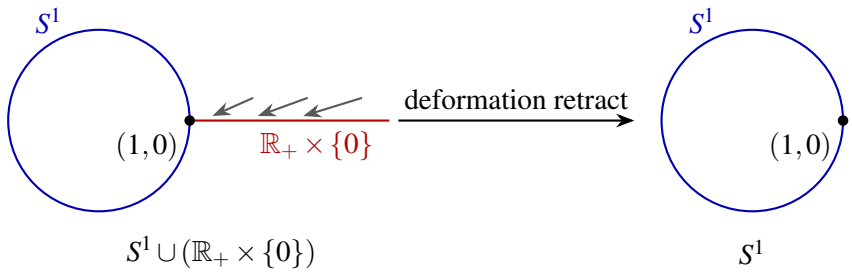


Figure 1.20

(v) The space $S^1 \cup (\mathbb{R}_+ \times \mathbb{R})$ is the union of the unit circle with the right half-plane. The right half-plane is contractible, and the remaining left semicircle together with a path in the half-plane forms one essential loop (Figure 1.21). Thus the space deformation retracts onto a circle. Hence its fundamental group is $\mathbb{Z}$.

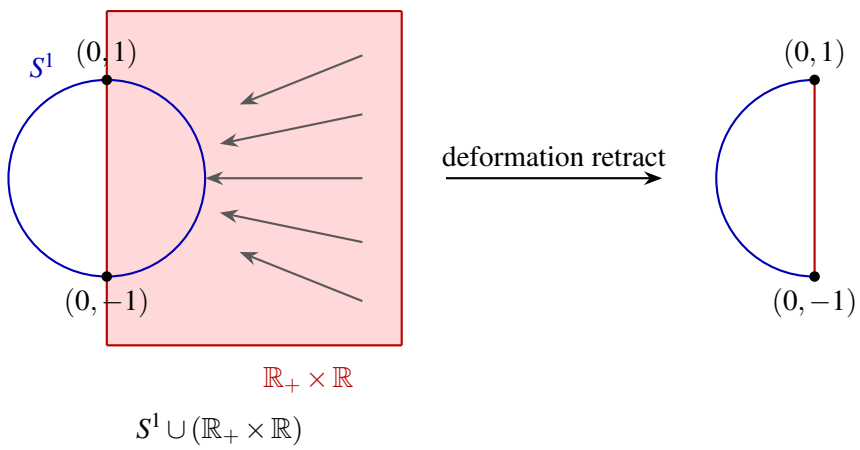
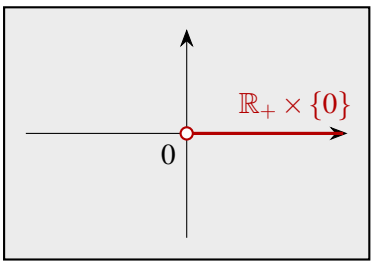


Figure 1.21

(vi) The space $S^1 \cup (\mathbb{R} \times \{0\})$ is the unit circle together with the horizontal diameter extended to an entire line. The two unbounded parts of the line retract onto the endpoints $(-1,0)$ and $(1,0)$, so the space deformation retracts onto $S^1 \cup ([-1,1] \times \{0\})$ which is the theta space. The theta space is homotopy equivalent to the figure-eight space (Example 1.40).

(vii) The space $\mathbb{R}^2 \setminus (\mathbb{R}_+ \times \{0\})$ is the plane with a slit removed (Figure 1.22). This space is simply-connected.



$\mathbb{R}^2 \setminus (\mathbb{R}_+ \times \{0\})$

Figure 1.22

Hence its fundamental group is trivial. □

Definition 1.27 (real projective space). The real projective n-space $\mathbb{RP}^n$ is the quotient space obtained from S^n by identifying each point $x \in S^n$ with its antipode $-x \in S^n$. That is,

$$\mathbb{RP}^n = S^n / \{\pm 1\}.$$

Definition 1.28 (topological n-manifold). Let X be a topological space. We say that X is an n-manifold if

(i) X is locally homeomorphic to $\mathbb{R}^n$, i.e. for any $x \in X$, there exists an open neighbourhood $U \subseteq X$ of x such that U is homeomorphic to $\mathbb{R}^n$

(ii) X is Hausdorff

(iii) X is second-countable (countable basis for its topology)

Theorem 1.22 (covering map from sphere to real projective space). $\mathbb{RP}^n$ is a compact topological n-manifold. The quotient map $p : S^n \to \mathbb{RP}^n$ is a covering map.

Example 1.48. A surface is a connected topological 2-manifold.

Example 1.49 (fundamental group of projective space). $\pi_1(\mathbb{RP}^2, y)$ is a group of order 2.

To see why, consider the projection $p : S^2 \to \mathbb{RP}^2$ which is a covering map. Since S^2 is simply-connected, then there exists a bijective correspondence between $\pi_1(\mathbb{RP}^2, y)$ and $p^{-1}(y)$. Since this is a two-element set, then $\pi_1(\mathbb{RP}^2, y)$ is a group of order 2. Any group of integers is isomorphic to $\mathbb{Z}/2$, i.e. the integers modulo 2.

Example 1.50 (Munkres p. 375 Question 2). Let X be the quotient space obtained from B^2 by identifying each point $x \in S^1$ with its antipode $-x$. Show that X is homeomorphic to the projective plane $\mathbb{RP}^2$.

Solution. Consider X as constructed from B^2 identified with the closed upper hemisphere of S^2. Let

$$p : S^2 \to \mathbb{RP}^2 \text{ and } q : B^2 \to X \quad \text{be quotient maps}$$

and $\pi : S^2 \to B^2$ be such that

$$\pi(x) = \begin{cases} x & \text{if } x \in B^2; \\ -x & \text{if } x \notin B^2. \end{cases}$$

This is a quotient map as $U \subseteq S^2$ implies $\pi(U)$ is open and $V \subseteq B^2$ is open implies $\pi^{-1}(V) = V \cup (-V)$ is open. We construct the following commutative diagram:

$$\begin{tikzcd} S^2 \arrow[r, "p"] \arrow[d, "\pi"'] & \mathbb{RP}^2 \arrow[d, "r"] \\ B^2 \arrow[r, "q"] & X \end{tikzcd}$$

As $q \circ \pi$ is a quotient map, then $r : \mathbb{RP}^2 \to X$ is also a quotient map. Since for every $x \in X$, we have $(q \circ \pi)^{-1}(x) = \{x, -x\} \in \mathbb{RP}^2$ and every equivalence class $\{x, -x\}$ can be regarded as the inverse image of this type, then r is a bijection. This implies $\mathbb{RP}^2 = \left\{(q \circ \pi)^{-1}(x) : x \in X\right\}$. We conclude that r is a homeomorphism. □

Example 1.51 (Munkres p. 375 Question 4). The space $\mathbb{RP}^1$ and the covering map $p : S^1 \to \mathbb{RP}^1$ are familiar ones. What are they?

Solution. Think of the space $\mathbb{RP}^1$ as the set of lines through the origin in $\mathbb{R}^2$. In other words, this is the space of equivalence classes of points on the circle S^1, where two points on S^1 are *equivalent* if and only if they are antipodal (will formally define this when discussing the Borsuk-Ulam theorem in Theorem 1.28). As such, the covering map p gives a natural way to relate points on the circle S^1 to lines through the origin in $\mathbb{R}^2$. □

Lemma 1.18 (fundamental group of the figure eight). Let $X = S^1 \vee S^1$ be the figure eight space. Then $\pi_1(X, x_0)$ is not abelian.

Proof. Let $X = A \vee B$ be the union of two circles A and B meeting only at the common point x_0. We shall prove that $\pi_1(X, x_0)$ is not abelian by constructing a covering space $p : E \to X$ and comparing the lifts of two loops.

The covering space E may be described as follows (refer to Figure 1.23). Start with the two coordinate axes in $\mathbb{R}^2$. At every non-zero integer point on the x-axis, attach a small circle tangent to the x-axis at that point. Similarly, at every non-zero integer point on the y-axis, attach a small circle tangent to the y-axis at that point. The covering map $p : E \to X$ is defined so that the x-axis wraps around the circle A, the y-axis wraps around the circle B, every integer point on either axis maps to the wedge point x_0, every small circle attached to the x-axis maps homeomorphically onto B, and every small circle attached to the y-axis maps homeomorphically onto A.

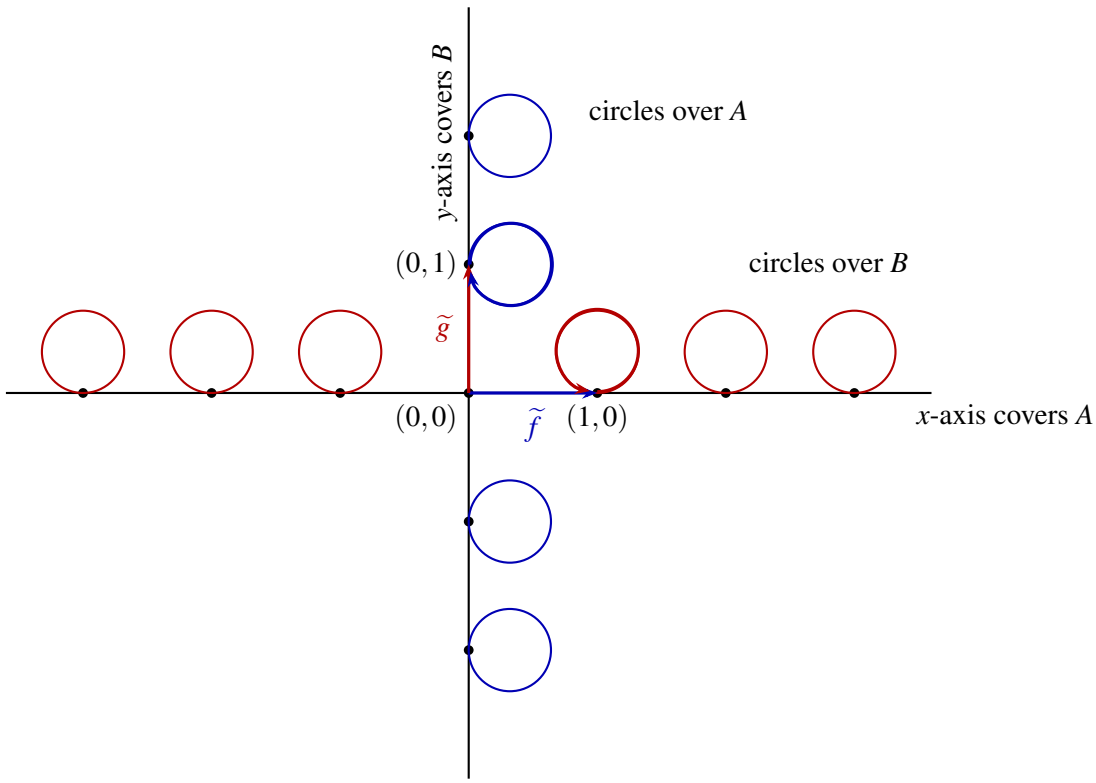


Figure 1.23: A covering space of the figure eight. The x-axis covers A, the y-axis covers B, and the small tangent circles cover the other circle.

Let $\widetilde{f} : I \to E$ be the path $\widetilde{f}(s) = (s,0)$, going from $(0,0)$ to $(1,0)$ along the x-axis. Let $\widetilde{g} : I \to E$ be the path $\widetilde{g}(s) = (0,s)$, going from $(0,0)$ to $(0,1)$ along the y-axis. Define $f = p \circ \widetilde{f}$ and $g = p \circ \widetilde{g}$. Then f and g are loops in X based at x_0, with f going once around A and g going once around B.

We claim that the loop $f * g$ is not path homotopic to the loop $g * f$. By the uniqueness of path lifting (Theorem 1.9), it is enough to compare their lifts beginning at $(0,0)$. The lift of $f * g$ beginning at $(0,0)$ first follows the x-axis from $(0,0)$ to $(1,0)$, and then travels once around the small circle tangent to the x-axis at $(1,0)$. Hence the lift of $f * g$ ends at $(1,0)$. On the other hand, the lift of $g * f$ beginning at $(0,0)$ first follows the y-axis from $(0,0)$ to $(0,1)$, and then travels once around the small circle tangent to the y-axis at $(0,1)$. Hence the lift of $g * f$ ends at $(0,1)$.

Thus the two lifted paths beginning at the same point $(0,0)$ have different endpoints. If $f * g$ and $g * f$ were path homotopic relative to endpoints, then their lifts starting at $(0,0)$ would also end at the same point. This is impossible since $(1,0) \neq (0,1)$. Therefore $f * g$ and $g * f$ are not path homotopic. Consequently, $[f][g] \neq [g][f]$ in $\pi_1(X,x_0)$. Hence $\pi_1(X,x_0)$ is not abelian. $\square$

Example 1.52 (fundamental group of figure eight). We shall compute the fundamental group of the figure eight space X, which can be regarded as the wedge sum of two circles $S^1 \wedge S^1$. Let $U, V = S^1$ which intersect at a single point x_0. So, $X = U \cup V = S^1 \cup S^1$ and $U \cap V = \{x_0\}$.

Since it is a well-known fact that the fundamental group of a circle based at x_0 is the integers $\mathbb{Z}$, then $\pi_1(U,x_0) = \pi_1(V,x_0) \cong \mathbb{Z}$. Also, the fundamental group of a singleton is the trivial group $\{e\}$. So, $\pi_1(U \cap V, x_0) = \{e\}$. By the Seifert-Van Kampen theorem, we have the isomorphism

$$\pi_1(X,x_0) \cong \pi_1(U,x_0) * \pi_1(V,x_0)/N.$$

Recall that N is the least normal subgroup of the free product that contains all elements represented by words of the form $(i_1(g))^{-1} i_2(g)$, where $g \in \pi_1(U \cap V, x_0)$. Since $\pi(U \cap V, x_0) = \{e\}$, then any normal subgroup of this fundamental group is also trivial, for which it follows that $\pi_1(X,x_0) \cong \mathbb{Z} * \mathbb{Z} \cong \langle a,b \rangle = F_2$. Here,

$\langle a,b \rangle$ denotes the free group generated by a and b, or equivalently, the free group on two letters $= F_2$

So, the fundamental group of the figure eight space X is F_2.

Example 1.53 (Munkres p. 375 Question 3). Let X be the union of two circles A and B in $\mathbb{R}^2$ (figure eight space). Define $p : E \to X$ to be the map which wraps the x-axis around circle A and wraps the y-axis around the other circle B. Let E' be the subspace of E that is the union of the x-axis and the y-axis. Show that $p|_{E'}$ is not a covering map.

Solution. Consider the base point x_0, which is the centre of the figure eight space. A neighbourhood $x_0 \in U$ contains the union V of open intervals in A and B whose intersection is exactly $\{x_0\}$.

So, $(p|_{E'})^{-1}(V)$ is equal to the union of open intervals around integers on the x-axis and the union of open intervals around integers on the y-axis. However, none of these intervals is homeomorphic to V since removing an integer in an interval gives two connected components, while removing its image in V gives four connected components. It follows that $p|_{E'}$ is not a covering map. □

Definition 1.29 (double torus). The double torus $\mathbb{T}\#\mathbb{T}$ is the surface obtained by taking two copies of the torus $\mathbb{T}$, deleting a small open disc from each of them, and pasting the remaining pieces together along their edges.

Theorem 1.23. The fundamental group of the double torus $\mathbb{T}\#\mathbb{T}$ is non-abelian.

Proof. The figure eight is a retract of $\mathbb{T}\#\mathbb{T}$.

Figure 1.24: Fundamental group of double torus

□

1.12 The Borsuk-Ulam Theorem

One of the most familiar consequences of continuity is the intermediate value theorem. In its simplest form, it says that if a continuous function changes sign on an interval, then it must vanish somewhere. For example, if $f : [a,b] \to \mathbb{R}$ is continuous and $f(a)$ and $f(b)$ have opposite signs, then there exists $c \in [a,b]$ such that $f(c) = 0$. Geometrically, this means that a continuous path from one side of the real line to the other must cross the origin.

As we would see, the Borsuk-Ulam theorem (Theorem 1.27) may be viewed as a powerful higher-dimensional analogue of this idea. Instead of comparing the values of a function at the two endpoints of an interval, we compare the values of a function at antipodal points on a sphere. The theorem says, roughly speaking, that any continuous map from a sphere to Euclidean space of the same dimension must identify at least one pair of opposite points. Thus, just as the intermediate value theorem forces a continuous real-valued function to take a prescribed value somewhere, the Borsuk-Ulam theorem forces a continuous map on a sphere to take the same value at some pair of antipodal points.

Definition 1.30 (antipode-preserving map). Let $n, m \in \mathbb{Z}_{\geq 0}$. A map $h : S^n \to S^m$ is said to be antipode-preserving if it sends antipodal points in S^n to antipodal points in S^m. That is, for every $x \in S^n$, one has $h(-x) = -h(x)$.

Theorem 1.24. Let $n \in \mathbb{Z}^+$, and let $h : S^n \to S^n$ be a continuous antipode-preserving map (Definition 1.30). Then h has odd degree. In particular, h is not null-homotopic.

Theorem 1.25. Let $h : S^1 \to S^1$ be a continuous antipode-preserving map (Definition 1.30), i.e. $h(-z) = -h(z)$. Then, h is not null-homotopic by Theorem 1.24. More precisely, if $h(1) = 1$, then the induced homomorphism

$$h_* : \pi_1\left(S^1, 1\right) \to \pi_1\left(S^1, 1\right) \quad \text{is non-trivial.}$$

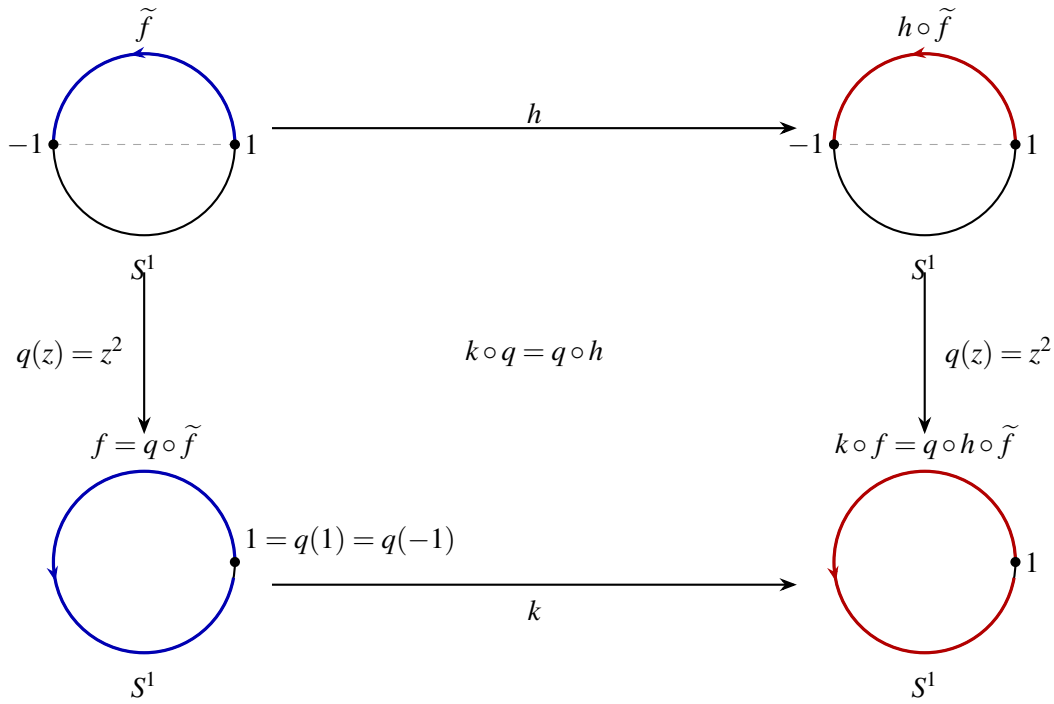
Proof. The key idea is that an antipode-preserving map $h : S^1 \to S^1$ cannot collapse the circle to a point because it must always send opposite points to opposite points. In particular, once we rotate the target circle so that $h(1) = 1$, the antipode-preserving condition (Definition 1.30) forces $h(-1) = -1$.

Thus any path in S^1 going from 1 to -1 is sent by h to another path going from 1 to -1. So h preserves the basic half-turn structure of the circle.

The covering map

$$q : S^1 \to S^1 \quad \text{where} \quad q(z) = z^2$$

identifies antipodal points. Geometrically, it folds the circle by declaring z and $-z$ to be the same point. Hence a path upstairs from 1 to -1 becomes, after applying q, a closed loop downstairs. For example, the path $\widetilde{f}(s) = e^{\pi i s}$ goes from 1 to -1, but $q(\widetilde{f}(s)) = e^{2\pi i s}$ goes once around the circle and returns to 1. Thus $q \circ \widetilde{f}$ represents a non-trivial element of $\pi_1\left(S^1, 1\right)$.



Since h sends antipodal points to antipodal points, the map $q \circ h$ cannot distinguish between z and $-z$. Therefore it factors through the quotient map q, giving a continuous map $k : S^1 \to S^1$ such that

$$k \circ q = q \circ h.$$

The commutative diagram says that applying h upstairs and then folding by q gives the same result as first folding by q and then applying k downstairs:

$$\begin{tikzcd} S^1 \arrow[r, "h"] \arrow[d, "q"'] & S^1 \arrow[d, "q"] \\ S^1 \arrow[r, "k"'] & S^1. \end{tikzcd}$$

Now take the path $\widetilde{f}$ from 1 to -1. Since $h(1) = 1$ and $h(-1) = -1$, the path $h \circ \widetilde{f}$ also goes from 1 to -1. Therefore $q \circ h \circ \widetilde{f}$ is again a loop downstairs whose lift starts at 1 and ends at -1. Such a loop cannot be null-homotopic, because a null-homotopic loop would lift to a loop when lifted from 1. Hence k sends a non-trivial loop to a non-trivial loop, so k_* is non-trivial.

Finally, the induced diagram on fundamental groups forces h_* to be non-trivial as well. If h_* were trivial, then $q_* \circ h_*$ would be trivial. But by commutativity,

$$q_* \circ h_* = k_* \circ q_*,$$

and the right-hand side is non-trivial. This contradiction shows that h_* is non-trivial. Therefore h is not null-homotopic. □

Theorem 1.26. For any positive integers m and n such that $m > n$, there does not exist a continuous and antipode-preserving map

$$g : S^m \to S^n \quad \text{from the } m\text{-sphere to the } n\text{-sphere.}$$

Corollary 1.10. There does not exist a continuous and antipode-preserving map

$$g : S^2 \to S^1 \quad \text{from the 2-sphere to the 1-sphere.}$$

Proof. Suppose on the contrary that $g : S^2 \to S^1$ is a continuous and antipode-preserving map. Let $j : S^1 \to S^2$ be the inclusion map of the equator, which is also continuous and antipode-preserving.
Then, $h = g \circ j$ is a continuous and antipode-preserving map $S^1 \to S^1$. So, h is not null-homotopic. Define E to be the upper hemisphere of S^2. Then, $g|_E$ is a continuous extension of h to $E \cong B^2$, so h is null-homotopic, which is a contradiction. As such, such a g cannot exist. □

Theorem 1.27 (Borsuk-Ulam theorem). For any $n \in \mathbb{Z}_{\geq 0}$ and any continuous map $f : S^{n+1} \to \mathbb{R}^{n+1}$, there exists $x \in S^n$ such that $f(x) = f(-x)$ in $\mathbb{R}^{n+1}$.

Corollary 1.11. Any continuous map $f : S^{n+1} \to \mathbb{R}^{n+1}$ is not injective.

Theorem 1.28 (Borsuk-Ulam theorem for S^2). For any continuous map $f : S^2 \to \mathbb{R}^2$, there exists $x \in S^2$ such that $f(x) = f(-x)$ in $\mathbb{R}^2$.

Proof. Suppose on the contrary that $f : S^2 \to \mathbb{R}^2$ is a continuous map such that $f(x) = f(-x)$ for all $x \in S^2$. Then, the trick is to define $S^2 \to S^1$ given by

$$g(x) = \frac{f(x) - f(-x)}{\|f(x) - f(-x)\|}$$

which is a continuous and antipode-preserving map, which contradicts Corollary 1.10 that there does not exist a continuous and antipode-preserving map from the 2-sphere to the 1-sphere. □

We then introduce the Lusternik-Schnirelmann theorem[11].

Theorem 1.29 (Lusternik-Schnirelmann theorem). For $n \geq 1$, suppose S^n is covered by $n+1$ many closed subsets. Then, at least one of them contains a pair of antipodal

[11] When I audited this class back in AY24/25 Semester 1, Prof. Chin Chee Whye mentioned that *it is kind of cute — this is the nature of topological results* when relating the Borsuk-Ulam theorem to the following theorem by Lusternik and Schnirelmann.

points.

Theorem 1.30 (bisection theorem in $\mathbb{R}^n$). For any $n \in \mathbb{Z}_{\geq 0}$, given n many bounded Jordan-measurable sets (boundary has Lebesgue measure) in $\mathbb{R}^n$, there exists a hyperplane of dimension $n-1$ in $\mathbb{R}^n$ that bisects them all.

Theorem 1.31 (bisection theorem in $\mathbb{R}^2$). For any two bounded polygonal regions in $\mathbb{R}^2$, there exists a line that bisects each of them.

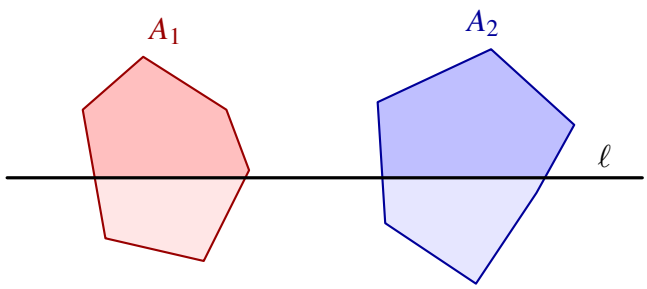


Figure 1.25: Bisection theorem in $\mathbb{R}^2$

Proof. We take two bounded polygonal regions A_1 and A_2 in the plane and show that there exists a line L in this plane that bisects each of them. Given a point $u \in S^2$, we shall consider the plane P in $\mathbb{R}^3$ passing through the origin that has u has its unit normal vector. This plane divides $\mathbb{R}^3$ into two half-spaces. Let $f_i(u)$ be the area of the portion of A_i that lies on the same side of P as does the vector u.

If u is the unit vector $\mathbf{k}$, then $f_i(u)$ is the area of A_i; if $u = -\mathbf{k}$, then $f_i(u) = 0$. Otherwise, the plane P intersects the plane $\mathbb{R}^2 \times 1$ in a line L that splits $\mathbb{R}^2 \times 1$ into two half-planes, and $f_i(u)$ is the area of that part of A_i that lies on one side of this line. Replacing u by $-u$ gives us the same plane, but the other half-space, so that $f_i(-u)$ is the area of that part of A_i that lies on the other side of P from u. It follows that

$$f_i(u) + f_i(-u) = \text{area of } A_i.$$

Now, consider the map

$$F : S^2 \to \mathbb{R}^2 \quad \text{defined by} \quad F(u) = (f_1(u), f_2(u)).$$

The Borsak-Ulam theorem (Theorem 1.28) gives us a point $u \in S^2$ for which $F(u) = F(-u)$. Then, $f_i(u) = f_i(-u)$ for $i = 1, 2$ so that $f_i(u)$ is $\frac{1}{2} \cdot \text{area } A_i$ as desired. $\square$

Chapter 2

The Cauchy Integral Formula

2.1 The Winding Number of a Simple Closed Curve

Recall the definition of an induced homomorphism on fundamental groups from Definition 1.13, i.e. $h_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$. This is defined to be the map that takes a loop in X based at x_0 and applies h to produce a loop in Y based at y_0.

Definition 2.1 (winding number). Let $h : S^1 \to \mathbb{R}^2 \setminus \{(0,0)\}$ be a continuous map. Suppose the induced homomorphism h_* carries a generator of the fundamental group of S^1 to some integral power of a generator of the fundamental group of $\mathbb{R}^2 \setminus \{(0,0)\}$. Then, the integral power n is called the winding number of h with respect to the origin $(0,0)$.

Geometrically, one can think of the winding number of a continuous map h as the number of times h wraps the circle S^1 around the origin $(0,0)$; its sign depends on the choice of generators.

Example 2.1 (winding numbers of some closed curves). Consider the following curves in $\mathbb{R}^2$ as shown in Figure 2.1. Recall some basic topology that these curves are simple as they loop back onto themselves, forming continuous shapes with no open ends. However, they are not simple because they intersect themselves at multiple points.

One can easily check that the winding numbers of these closed curves in Figure 2.1 are ± 2 and 0 respectively.

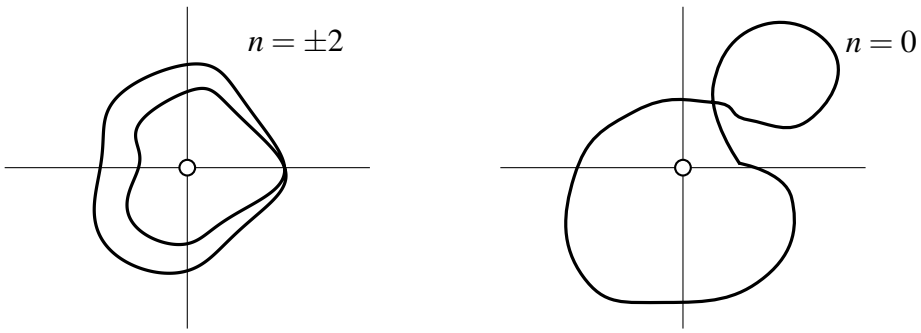


Figure 2.1: Winding numbers of some closed curves in $\mathbb{R}^2$

Example 2.2 (winding numbers of some simple closed curves). Now, what can one say about the winding number of h if h is injective, i.e. h is a homeomorphism of S^1 with a simple closed curve C in $\mathbb{R}^2 \setminus \{(0,0)\}$?

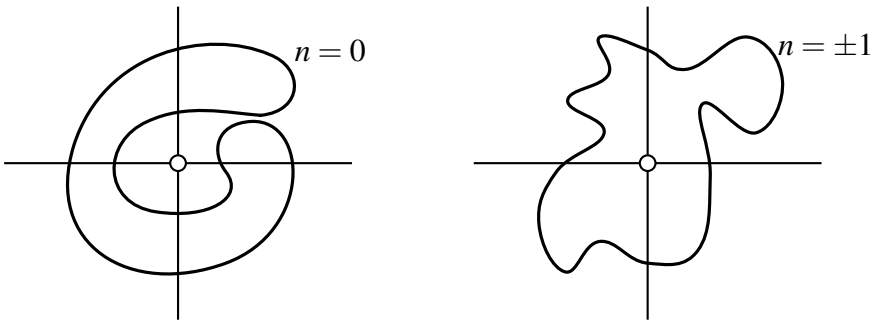


Figure 2.2: Winding numbers of some simple closed curves in $\mathbb{R}^2$

Figure 2.2 suggests that if $(0,0)$ belongs to the unbounded component of $\mathbb{R}^2 \setminus C$, then $n = 0$; if $(0,0)$ belongs to the bounded component of $\mathbb{R}^2 \setminus C$, then $n = \pm 1$. The first property is quite easy to prove but the second is deep and difficult. However, we will prove neither result here.

The main result needed to prove it is as follows.

Theorem 2.1. Let C be a simple closed curve in S^2 and suppose p, q lie in different components of $S^2 \setminus C$. Then, the inclusion map

$$j : C \hookrightarrow S^2 \setminus \{p, q\} \quad \text{induces an isomorphism of fundamental groups.}$$

Theorem 2.1 is a special case of a rather deep theorem of algebraic topology, concerning with what is called the linking number of two disjoint subspaces of S^{m+n+1}, one homeomorphic to an m-sphere and the other homeomorphic to an n-sphere. This is related to the Alexander duality theorem. The special case of our theorem is that of a 0-sphere (a two-point space) and a 1-sphere (a simple closed curve) in S^2.

2.2 The Cauchy Integral Formula

One of the central theorems in the study of functions of a complex variable is the one concerning the Cauchy integral formula for analytic functions. For the classical version

of this theorem, we need to assume two things, which are the Jordan curve theorem and the winding number theorem (Theorem 2.1). We begin our discussion with a more formal introduction of winding numbers.

Definition 2.2 (winding number). Let f be a loop in $\mathbb{R}^2$ and a be a point not in the image of f. Set

$$g(s) = \frac{f(s)-a}{\|f(s)-a\|} \quad \text{so } g \text{ is a loop in } S^1.$$

Let $p : \mathbb{R} \to S^1$ be the standard covering map and $\widetilde{g}$ be the lifting of g to S^1. Since g is a loop, the difference $\widetilde{g}(1)-\widetilde{g}(0)$, which is an integer, is known as the winding number of f with respect to a, denoted by $n(f,a)$.

$$\begin{tikzcd} & \mathbb{R} \arrow[d, "p"] \\ S^1 \arrow[ur, "\widetilde{g}"] \arrow[r, "g"'] & S^1 \end{tikzcd}$$

The lift $\widetilde{g}$ tracks how g winds around the circle. Since g is a loop, it makes sense that $\widetilde{g}(1)-\widetilde{g}(0)$ is an integer, which tells one how many times g winds around S^1. Also, note that $n(f,a)$ is independent of the choice of the lifting of g, i.e. if $\widetilde{g}$ is one lifting of g, then uniqueness of lifting implies that any other lifting of g has the form $\widetilde{g}(s)+m$ for some $m \in \mathbb{Z}$.

Definition 2.3 (free homotopy). Let $F : I \times I \to X$ be a continuous map such that

$$F(0,t) = F(1,t) \quad \text{for all } t.$$

Then, for each t, the map $f_t(s) = F(s,t)$ is a loop in X. The map F is called a free homotopy between the loops f_0 and f_1. It is a homotopy of loops in which the base point of the loop is allowed to move during the homotopy (Definition 2.3).

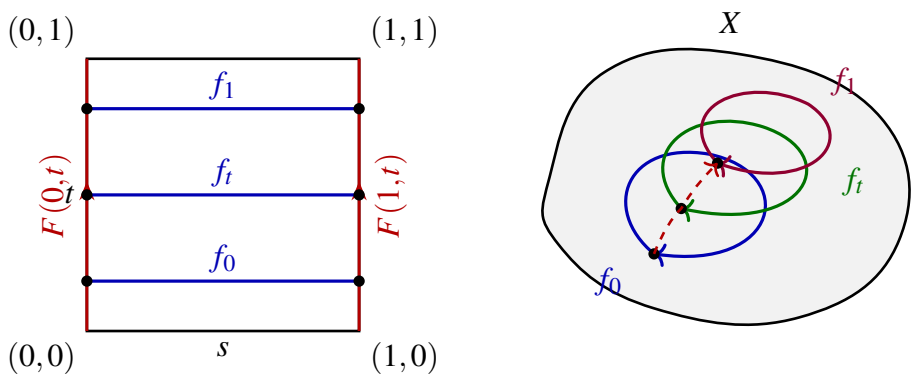


Figure 2.3: A free homotopy between loops f_0 and f_1

Lemma 2.1. Let f be a loop in $\mathbb{R}^2 \setminus \{a\}$. Then, the following hold:

(a) If $\overline{f}$ is the reverse of f, then $n\left(\overline{f}, a\right) = -n(f,a)$.

(b) If f is freely homotopic to f' through loops lying in $\mathbb{R}^2 \setminus \{a\}$, then $n(f,a) = n(f',a)$.

(c) If a and b lie in the same component of $\mathbb{R}^2 \setminus f(I)$, then $n(f,a) = n(f,b)$.

Proof. We first prove **(a)**. Recall that

$$n(f,a) = \widetilde{g}(1) - \widetilde{g}(0) \quad \text{where} \quad \widetilde{g}(s) \text{ is the lifting of } g(s) = \frac{f(s)-a}{\|f(s)-a\|} \text{ to } S^1.$$

By replacing s with $1-s$, we can compute $n\left(\overline{f}, a\right)$. Consequently, this changes $\widetilde{g}(1) - \widetilde{g}(0)$ by a sign.

To prove **(b)**, suppose F is a free homotopy between f and f'. Define

$$G : I \times I \to S^1 \quad \text{by the equation} \quad G(s,t) = \frac{F(s,t)-a}{\|F(s,t)-a\|}.$$

Let $\widetilde{G}$ be a lifting of G to $\mathbb{R}$, so $\widetilde{G}(1,t) - \widetilde{G}(0,t)$ is an integer for all t. Since $\widetilde{G}$ is continuous, then it is a constant.

Lastly, we prove **(c)**. Let α be a path in $\mathbb{R}^2 \setminus f(I)$ from a to b. By the definition of winding number (Definition 2.2), we have

$$n(f,a) = n(f-a,(0,0)).$$

This is obvious as n is invariant under translations, so the winding number of f around a is the same as that of the translated map $f-a$ around the origin. Since $f(s) - \alpha(t)$ is a free homotopy in $\mathbb{R}^2 \setminus \{(0,0)\}$ between $f-a$ and $f-b$, the result follows. □

Definition 2.4 (simple loop). Let f be a loop in X. We say that f is a simple loop if f is injective, i.e. $f(s) = f(s')$ implies $s = s'$, or if one of the points s, s' is 0 and the other is 1.

Proposition 2.1. If $f : X \to X$ is a simple loop, then its image set is a simple closed curve in X.

Theorem 2.2. Let f be a simple loop in $\mathbb{R}^2$. We have

$$n(f,a) = \begin{cases} 0 & \text{if } a \text{ lies in the unbounded component of } \mathbb{R}^2 \setminus f(I); \\ \pm 1 & \text{otherwise.} \end{cases}$$

Definition 2.5 (counterclockwise and clockwise loops). Let f be a simple loop in $\mathbb{R}^2$. f is a counterclockwise loop if $n(f,a) = 1$ for some a (hence, for every a) in the bounded component of $\mathbb{R}^2 \setminus f(I)$. On the other hand, f is a clockwise loop if $n(f,a) = -1$.

Example 2.3. The standard loop $p(s) = (\cos 2\pi s, \sin 2\pi s)$ is a counterclockwise loop.

Theorem 2.3 (Cauchy integral formula). Let C be a simple closed piecewise differentiable curve in the complex plane. Let B be the bounded component of $\mathbb{R}^2 \setminus C$. If $f(z)$ is analytic (can be expressed as a Taylor series) in an open set Ω that contains B and C, then for every $a \in B$, we have

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-a}\, dz.$$

The sign is positive if C is oriented counterclockwise, and negative otherwise.

Chapter 3

Groups, Rings and Modules

3.1 Generation of Modules

Recall from MA3201 Algebra II the following: if we have an R-module M and let $\{N_\lambda\}_{\lambda\in\Lambda}$ be any collection of R-submodules of M, then the arbitrary intersection of R-submodules of M is also an R-submodule of M. In fact, the intersection is the largest R-submodule contained in every N_λ.

Every abelian group can be regarded as a $\mathbb{Z}$-module. So, we have the following fact:

$$\{\text{abelian groups}\} \subseteq \{R\text{-modules}\}.$$

Definition 3.1 (submodule generated by a subset). Let M be an R-module. For every $A \subseteq M$, let

$$RA = \left\{ \sum_{i=1}^{m} r_i a_i : r_i \in R, a_i \in A \text{ for all } 1 \leq i \leq m, m \in \mathbb{Z}^+ \right\}$$

be the submodule of M generated by A.

Alternatively, Definition 3.1 can be interpreted as follows. For any $A \subseteq M$, consider the collection of all R-submodules N of M containing A, i.e. $A \subseteq N$. Then, the intersection over this collection is thus also an R-submodule of M. It is called the R-submodule of M generated by A, denoted by RA or (A) (this shares a similar notion with ideals). As mentioned, this intersection is the smallest submodule of M which contains A.

For any $A \subseteq M$, we can also interpret RA as follows:

Proposition 3.1. RA can be interpreted as the set of finite sums in M of the form $r_1\alpha_1 + \ldots + r_k\alpha_k$, where $r_1, \ldots, r_k \in R$ and $\alpha_1, \ldots, \alpha_k \in A$.

If $A = \{a_1, \ldots, a_n\} \subseteq M$ is a finite set, we shall write $Ra_1 + \ldots + Ra_n$ to denote RA, which is equivalent to $\mathbb{Z}a_1 + \ldots + \mathbb{Z}a_n$.

Theorem 3.1 (universal property of free modules). The subset $A \subseteq M$ corresponds to the set-map $\varphi : A \hookrightarrow M$. By the universal property of the free R-module $R^{\oplus A}$, there exists a unique R-module homomorphism such that the following diagram commutes:

$$\begin{tikzcd} R^{\oplus A} \arrow[r, "\Phi"] & M \\ A \arrow[u] \arrow[ur, "\varphi"'] & \end{tikzcd}$$

Then, $RA = \Phi\left(R^{\oplus A}\right)$ is the image submodule of this homomorphism.

Definition 3.2 (generating set of a module). If N is a submodule of M (possibly $N = M$) and $N = RA$ for some $A \subseteq M$, we say that A is a set of generators or generating set for N, and we say N is generated by A.

This is equivalent to saying that N is generated by the subset $A \subseteq M$ if and only if the image module $\Phi\left(R^{\oplus A}\right)$ in M is equal to N.

Definition 3.3 (finitely generated module). A submodule N of M (possibly $N = M$) is finitely generated if there exists some finite subset $A \subseteq M$ such that $N = RA$, i.e. if N is generated by some finite subset.

Equivalently, there exists a finite subset $A \subseteq M$ such that N is equal to the image module $\Phi\left(R^{\oplus A}\right)$ in M if and only if there exists $n \in \mathbb{N}$ such that N is isomorphic to a quotient of the free module R^n of rank n.

Definition 3.4 (cyclic submodule). A submoule N of M (possibly $N = M$) is cyclic if there exists an element $a \in M$ such that $N = Ra$, that is, if N is generated by one element, i.e.

$$N = Ra = \{ra : r \in R\}.$$

Equivalently, there exists $a \in M$ such that N is equal to the image module $\Phi(R) = \{ra \in M : r \in R\}$ in M if and only if $N \cong R/I$ for some ideal I of R.

Definition 3.5 (sum of submodules). Let M be an R-module and let $\{N_\lambda\}_{\lambda \in \Lambda}$ be any collection of R-submodules of M. Take

$$A = \bigcup_{\lambda \in \Lambda} N_\lambda.$$

The sum of the R-submodules N_λ is equal to

$$\sum_{\lambda \in \Lambda} N_\lambda = \left\{\text{finite sums } x_{\lambda_1} + \ldots + x_{\lambda_k} : \lambda_1, \ldots, \lambda_k \in \Lambda \text{ and } x_{\lambda_i} \in N_{\lambda_i}\right\}.$$

It is the smallest submodule of M containing every N_λ.

Definition 3.6 (direct product). Let M, N be R-modules. The direct product of M and N is the R-module $M \times N$ which can be regarded as the product of the additive groups $(M, +, 0_M, -)$ and $(N, +, 0_N, -)$ and the properties of scalar multiplication are satisfied.

In general, let $\{M_\alpha\}_{\alpha \in A}$ be a family of R-modules (indexed by A). The direct product of the family $\{M_\alpha\}_{\alpha \in A}$ is the R-module

$$\prod_{\alpha \in A} M_\alpha$$

with an underlying additive group and the properties of scalar multiplication are satisfied.

Lemma 3.1. The canonical projection maps

$$\pi_M : M \times N \to M \text{ where } (m,n) \mapsto m \quad \text{and} \quad \pi_N : M \times N \to N \text{ where } (m,n) \mapsto n$$

are surjective R-module homomorphisms with

$$\ker(\pi_M) = \{0_M\} \times N \quad \text{and} \quad \ker(\pi_N) = M \times \{0_N\}.$$

Definition 3.7 (direct sum). Let M, N be R-modules. The direct sum of M and N is the R-module $M \times N$, denoted by $M \oplus N$.

In general, if $\{M_\alpha\}_{\alpha \in A}$ is a family of R-modules (indexed by A), the direct sum of the family $\{M_\alpha\}_{\alpha \in A}$ is the R-module

$$\bigoplus_{\alpha \in A} M_\alpha$$

which is defined to be the R-submodule of

$$\prod_{\alpha \in A} M_\alpha \quad \text{consisting of elements which are zero in almost all components.}$$

Example 3.1 (Munkres p. 411 Question 1). Suppose that $G = \sum G_\alpha$. Show this sum is direct if and only if the equation

$$x_{\alpha_1} + \cdots + x_{\alpha_n} = 0$$

implies that each x_{α_i} equals 0. Here $x_{\alpha_i} \in G_{\alpha_i}$ and the indices α_i are distinct.

Solution. We first prove the forward direction. Suppose the sum is direct. Given the uniqueness of the representation and that $0 = 0 + \ldots + 0$ regardless of which groups are

considered in the sum, it follows that $x_{\alpha_i} = 0$ for all $1 \leq i \leq n$.

We then prove the reverse direction. Suppose for some $x \in G$, we have two decompositions as follows:

$$x = x_{\alpha_1} + \ldots + x_{\alpha_n} = y_{\beta_1} + \ldots + y_{\beta_m}.$$

Then, $x_{\alpha_1} + \ldots + x_{\alpha_n} - y_{\beta_1} - \ldots - y_{\beta_m} = 0$. If there exist i and j such that $\alpha_i = \beta_j$, then $x_{\alpha_i} = y_{\beta_j}$, and for all other indices α_i and β_j, we must have $x_{\alpha_i} = y_{\beta_j} = 0$, implying that the sum is direct. □

Example 3.2 (Munkres p. 411 Question 2). Show that if G_1 is a subgroup of G, there may be no subgroup G_2 of G such that $G = G_1 \oplus G_2$.
Hint: Set $G = \mathbb{Z}$ and $G_1 = 2\mathbb{Z}$.

Solution. Take $G = \mathbb{Z}$ and $G_1 = 2\mathbb{Z}$. One checks that $G_1 \leq G$. Suppose on the contrary that there exists $G_2 \leq G$ such that $G = G_1 \oplus G_2$. Then, G_2 must be the set of odd integers (since the sum $G_1 + G_2$ is direct and they must generate the group of integers $\mathbb{Z}$). However, the closure property in the set of odd integers $2\mathbb{Z} + 1$ fails since the sum of two odd integers is not odd. □

Lemma 3.2. The canonical inclusion maps

$$i_M : M \to M \oplus N \text{ where } m \mapsto (m, 0_N) \quad \text{and} \quad i_N : N \to M \oplus N \text{ where } n \mapsto (0_M, n)$$

are injective R-module homomorphism with

$$i_M(M) = M \times \{0_N\} \quad \text{and} \quad i_N(N) = \{0_M\} \times N.$$

3.2 Free Modules

Definition 3.8 (free module). For any set A, the free R-module on the set A is the R-module

$$F(A) = R^{\oplus A} = R^{(A)} = \bigoplus_{\alpha \in A}.$$

In other words, the free R-module is the direct sum of the family of R-modules indexed by A, with each member being a copy of the regular R-module R.

Definition 3.9 (free module). An R-module F is said to be free if and only if there exists a set A such that $F \cong F(A)$ as R-modules, i.e. there exists an R-module isomorphism $f : F \to F(A)$.

Lemma 3.3. If two sets A and B have the same cardinality, then $F(A) \cong F(B)$ as R-modules.

Proposition 3.2. Suppose $R \neq 0$. If two sets A and B are such that $F(A) \cong F(B)$ as R-modules, then they have the same cardinality.

Corollary 3.1 (rank of free module). Suppose $R \neq 0$. If F is a free R-module, then the cardinality of A is well-defined and it is called the rank of F as a free R-module.

Example 3.3. For each $n \in \mathbb{N}$, the set $R^n = \{(r_1, \ldots, r_n) : r_i \in R \text{ for all } i\}$ is a free module of rank n over R.

3.3 Free Products of Groups

Definition 3.10 (free product and its universal property). Let $\{G_\alpha\}_{\alpha \in J}$ be an indexed family of groups. The free product of the groups G_α is a group G given with a family $i_\alpha : G_\alpha \to G$ of homomorphism satisfying the following universal property: for any group and any family of homomorphism $h_\alpha : G_\alpha \to H$, there exists a unique homomorphism $h : G \to H$ such that for each $\alpha \in J$, the following diagram commutes:

$$\begin{tikzcd} G_\alpha \arrow[r, "i_\alpha"] \arrow[dr, "h_\alpha"'] & G \arrow[d, "h"] \\ & H \end{tikzcd}$$

Theorem 3.2 (existence of free products). Let $\{G_\alpha\}_{\alpha \in J}$ be a family of groups. Then, there exists a free product of the groups G_α, i.e. there exists a group G with a family $i_\alpha : G_\alpha \to G$ of homomorphism satisfying the universal property (Definition 3.10).

Theorem 3.3 (uniqueness of free products). Let $\{G_\alpha\}_{\alpha \in J}$ be a family of groups. Suppose

$$G \text{ with } i_\alpha : G_\alpha \to G \quad \text{and} \quad G' \text{ with } i'_\alpha : G_\alpha \to G'$$

are both free products of the groups G_α in the sense of having the universal property (Definition 3.10). Then, there exist canonical isomorphism

$$\phi : G \to G' \quad \text{and} \quad \psi : G' \to G$$

such that for each $\alpha \in J$, we have

$$\phi \circ i_\alpha = i'_\alpha \quad \text{and} \quad \psi \circ i'_\alpha = i_\alpha.$$

In other words, the following diagram commutes:

$$\begin{tikzcd} & G \arrow[d, shift right, "\phi"'] \\ G_\alpha \arrow[ur, "i_\alpha"] \arrow[r, "i'_\alpha"'] & G' \arrow[u, bend right, "\psi"'] \end{tikzcd}$$

Thus, the free product of groups G_α exists, and it is unique up to a canonical isomorphism. It is denoted by

$$G = \prod_{\alpha \in J} G_\alpha = ** \bigoplus_{\alpha \in J} G_\alpha \quad \text{or in the finite case,} \quad G = G_1 * \ldots * G_n.$$

Lemma 3.4. Let $\{G_\alpha\}_{\alpha \in J}$ be a family of groups. Suppose G with $i_\alpha : G_\alpha \to G$ is a free product of the groups G_α. Then, the following hold:

(a) each $i_\alpha : G_\alpha \to G$ is injective

(b) the family $\{i_\alpha(G_\alpha)\}_{\alpha \in J}$ of subgroups of G generate G.

Corollary 3.2. Let $G = G_1 * G_2$ be a free product of groups, where

$$G_1 \text{ is the free product of } \{H_\alpha\}_{\alpha \in J} \quad \text{and} \quad G_2 \text{ is the free product of } \{H_\beta\}_{\beta \in K}.$$

If $J \cap K = \emptyset$, then G is the free product of the subgroups $\{H_\gamma\}_{\gamma \in J \cup K}$. In particular,

$$G_1 * (G_2 * G_3) = G_1 * G_2 * G_3 = (G_1 * G_2) * G_3.$$

Definition 3.11 (least normal subgroup). If S is a subset of a group G, the normal subgroup of G generated by S is the intersection of all normal subgroups of G that contain S. It is the least normal subgroup of G that contains S. Moreover, it is the subgroup of G generated by all conjugates of elements of S.

Theorem 3.4. Let $N_i \trianglelefteq G_i$ for $i = 1, 2$ and $N \trianglelefteq (G_1 * G_2)$ is generated by N_1 and N_2. Then,

$$(G_1 * G_2)/N \cong (G_1/N_1) * (G_2/N_2) \quad \text{canonically.}$$

3.4 Free Groups

Definition 3.12 (free group). Let $\{a_\alpha\}$ be a family of elements of a group G. Suppose each a_α generates an infinite cyclic subgroup G_α of G. If G is the free product of the groups $\{G_\alpha\}$, then G is said to be a free group and the family $\{a_\alpha\}$ is a system of free generators for G.

Proposition 3.3. Let G be a group; let $\{a_\alpha\}_{\alpha\in J}$ be a family of elements of G. If G is a free group with system of free generators $\{a_\alpha\}$, then G satisfies the following condition: given any group H and any family $\{y_\alpha\}$ of elements of H,

$$\text{there exists a homomorphism } h : G \to H \quad \text{such that} \quad h(a_\alpha) = y_\alpha \text{ for each } \alpha.$$

Corollary 3.3. Let

$$G = G_1 * G_2 \quad \text{be the free product of two free groups.}$$

Here, G_1 and G_2 are free groups with $\{a_\alpha\}_{\alpha\in J}$ and $\{a_\alpha\}_{\alpha\in K}$ as respective systems of free generators. If $J \cap K = \emptyset$, then

$$G \text{ is a free group} \quad \text{with} \quad \{a_\alpha\}_{\alpha\in J\cup K} \text{ as a system of free generators.}$$

Example 3.4 (Munkres p. 412 Question 3). If G is free abelian with basis $\{x, y\}$, show that $\{2x+3y, x-y\}$ is also a basis for G.

Solution. G being free Abelian simply means that it is Abelian with a basis. In particular, any element in G can be uniquely expressed as a linear combination of x and y with integer coefficients. Analogous to concepts in MA2001, it suffices to show that the matrix

$$\begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix} \quad \text{is invertible,}$$

which is easy to verify. □

Example 3.5 (Munkres p. 412 Question 4). The *order* of an element a of an abelian group G is the smallest positive integer m such that $ma = 0$, if such exists; otherwise, the order of a is said to be infinite. The order of a thus equals the order of the subgroup generated by a.

(a) Show the elements of finite order in G form a subgroup of G, called its *torsion subgroup*.

(b) Show that if G is free abelian, it has no elements of finite order.

(c) Show the additive group of rationals has no elements of finite order, but is not free abelian.
Hint: If $\{a_\alpha\}$ is a basis, express $\frac{1}{2}a_\alpha$ in terms of this basis.

Solution.

(a) Let G be an Abelian group and $a_1, a_2 \in G$ be of finite order, i.e. there exist smallest $m_1, m_2 \in \mathbb{Z}^+$ such that $m_1 a_1 = 0$ and $m_2 a_2 = 0$. Then, $\operatorname{lcm}(m_1, m_2)(a_1 + a_2) = 0$, satisfying the closure property. The other three group axioms are easy to verify.

(b) Suppose G is free Abelian. Then, G is Abelian with a basis. Let

$$a = \sum_{\alpha \in J} m_\alpha a_\alpha \quad \text{so given that} \quad ma = \sum_{\alpha \in J} m m_\alpha a_\alpha = 0,$$

we must have $mm_\alpha = 0$. This forces $mm_\alpha a = 0$ for all α. As such, $a = 0$. Since we are only interested in elements which are non-zero, it follows that G has no non-zero element of finite order.

(c) Consider $G = (\mathbb{Q}, +)$. Suppose on the contrary that there exists a non-zero element $x \in G$ of finite order, i.e. there exists $m \in Z^+$ such that $mx = 0$. Then, $m = 0$ or $x = 0$, which yields a contradiction.

Given that

$$\frac{1}{2} a_\beta = \sum_{\alpha \in J} m_\alpha a_\alpha \quad \text{this implies} \quad a_\beta = \sum_{\alpha \in J} 2 m_\alpha a_\alpha.$$

As such, $2m_\beta = 1$, which is a contradiction. □

Example 3.6 (Munkres p. 412 Question 5). Give an example of a free abelian group G of rank n having a subgroup H of rank n for which $H \neq G$.

Solution. Note that $\mathbb{Z}$ is free Abelian of rank 1 since it is generated by the element 1, i.e. $G = \langle 1 \rangle$. Consider $G = \mathbb{Z}$ and we note that $H = 2\mathbb{Z} \leq \mathbb{Z} = G$, where H is of rank 1 since H is generated by the element 2, i.e. $2\mathbb{Z} = \langle 2 \rangle$. However, $H \neq G$. □

Recall the following fact (Definition 3.13) from MA2202 Algebra I.

Definition 3.13 (commutator). For any $x, y \in G$, the commutator of x and y is the element $[x, y] = xyx^{-1}y^{-1}$ of G.

Definition 3.14 (commutator subgroup). The commutator subgroup of G is the subgroup of G generated by all commutators in G, i.e.

$$[G, G] = \langle [x, y] : x, y \in G \rangle .$$

Proposition 3.4 (commutator subgroup is normal subgroup). $[G, G] \trianglelefteq G$

Lemma 3.5. $G/[G, G]$ is the maximal abelian quotient of G with the following universal property: for any homomorphism

$$h : G \to H \quad \text{from } G \text{ to an abelian group } H,$$

there exists a unique homomorphism $k : G/[G, G] \to H$ such that the following

diagram commutes:

$$\begin{tikzcd} G \arrow[r, "h"] \arrow[d] & H \\ G/[G,G] \arrow[ur, "k"'] & \end{tikzcd}$$

Theorem 3.5. If G is the free group on the set J, then $G/[G,G]$ is the free Abelian group on the set J.

Definition 3.15 (presentation of a group). A presentation of a group G, denoted by $G = \langle S \mid R \rangle$, consists of

$$\begin{aligned} &\text{a subset } S \subseteq G \text{ comprising the generators} \quad \text{and} \\ &\text{a subset } R \subseteq \mathcal{F}(S) \text{ comprising a complete set of relations} \end{aligned}$$

such that the canonical homomorphism $\mathcal{F}(S) \xrightarrow{h} G$ which maps $a \in S$ to $a \in G$ is surjective.

Definition 3.16 (finitely generated and finitely presented group). The group G is finitely generated if and only if

$$\text{there exists a presentation } G = \langle S \mid R \rangle \quad \text{with } S \text{ finite.}$$

The group G is finitely presented if and only if

$$\text{there exists a presentation } G = \langle S \mid R \rangle \quad \text{with both } S \text{ and } R \text{ finite.}$$

Say G is finitely presented with

$$S = \{s_1, \ldots, s_n\} \quad \text{and} \quad R = \{w_1, \ldots, w_k\}.$$

Then, we write

$$G = \langle s_1, \ldots, s_n : w_1 = \ldots = w_k = 1 \rangle$$

and if w is the word $w_1 w_2^{-1}$, we write $w_1 = w_2$ instead of $w = 1$.

Example 3.7 (finite group has finite presentation). Every finite group $G = \{g_1, \ldots, g_n\}$ is finitely presented. To see why, take $S = G$ and R to be the set of words $g_i g_j g_k^{-1}$ where $g_i g_j = g_k$ in G.

Example 3.8 (presentation of $\mathbb{Z}$). We have $\mathbb{Z} \cong F(\{a\}) = \langle a \rangle$. In fact, this is a trivial result that was covered in MA2202 Algebra I.

To see why this is true, no relations are imposed on a so a can be interpreted as the generator of $\mathbb{Z}$, where powers of a correspond to integers. Specifically, a^n corresponds to

the integer n and a^{-n} corresponds to the integer $-n$. Hence, $\mathbb{Z}$ is isomorphic to a group generated by a single element with no further relations.

Example 3.9 (presentation of $\mathbb{Z} \times \mathbb{Z}$). We have

$$\mathbb{Z} \times \mathbb{Z} \cong \langle a, b \mid [a,b] = 1 \rangle .$$

The only relation here is that a and b commute, i.e. $[a,b] = 1$. This means that $ab = ba$. Note that $\mathbb{Z} \times \mathbb{Z}$ consists of ordered pairs of integers, and in terms of generators, each generator a and b corresponds to a copy of $\mathbb{Z}$, with the two copies commuting.

Example 3.10 (presentation of $\mathbb{Z}^n \times \mathbb{Z}^m$). We have

$$\mathbb{Z}_n \times \mathbb{Z}_m \cong \langle a^n = b^m = [a,b] = 1 \rangle .$$

Example 3.11 (presentation of quaternion group Q_8). Let $\mathbb{Q}_8$ be the quaternion group, which is a non-Abelian group of order 8, consisting of elements that are closely related to the quaternions. The quaternion group Q_8 is defined by the following set of elements:

$$Q_8 = \{1, -1, i, -i, j, -j, k, -k\}$$

with multiplication rules given as follows:

1. $i^2 = j^2 = k^2 = -1$
2. $ij = k$ and $ji = -k$
3. $jk = i$ and $kj = -i$
4. $ki = j$ and $ik = -j$

We note that

$$Q_8 = \left\langle i, j \mid i^4 = 1, j^2 = i^2, j^{-1}ij = i^{-1} \right\rangle .$$

Example 3.12 (presentation of dihedral group D_{2n}). Recall that the group D_{2n} is the dihedral group of order $2n$ and it represents the symmetries of a regular n-gon, including both rotations r and reflections s. Here, r represents a rotation by $2\pi/n$ in the anticlockwise direction; s represents a reflection symmetry.

We have

$$D_{2n} = \left\langle r, s \mid r^n = s^2 = 1, s^{-1}rs = r^{-1} \right\rangle .$$

Example 3.13 (presentation of symmetric group S_n). We have

$$S_n \cong \left\langle t_1, \ldots, t_{n-1} \mid t_i^2 = 1, (t_i, t_{i+1})^3 = 1, [t_i, t_j] = 1 \text{ whenever } |i-j| \geq 2, 1 \leq i, j \leq n-1 \right\rangle .$$

There is an interesting problem in group theory known as the group isomorphism problem, which states that given two finite presentations of groups, determine whether or not the groups are isomorphic. There is no effective procedure for solving this problem.

Chapter 4

Classification of Covering Spaces

4.1 Equivalence of Covering Spaces

We now turn to the problem of comparing covering spaces over the same base space. In general, two covering maps $p : E \to B$ and $p' : E' \to B$ may have very different total spaces, but they may still represent the same covering structure over B. The correct notion of sameness is not merely a homeomorphism $E \cong E'$, but a homeomorphism which respects the projections down to B. In other words, we want a homeomorphism $h : E \to E'$ such that the diagram

$$\begin{tikzcd} E \arrow[rr, "h"] \arrow[dr, "p"'] & & E' \arrow[dl, "p'"] \\ & B & \end{tikzcd}$$

commutes. Such a map says that E and E' cover B in essentially the same way.

The central idea of this section is that covering spaces can be studied through their effect on fundamental groups. Given a covering map $p : E \to B$ and a point $e_0 \in p^{-1}(b_0)$, the subgroup $p_*(\pi_1(E, e_0)) \leq \pi_1(B, b_0)$ records precisely which loops in B lift to loops in E starting at e_0. Thus, the algebraic information contained in this subgroup reflects the geometry of the covering space. In particular, equality of such subgroups corresponds to equivalence of covering spaces preserving a chosen point in the fiber, while conjugacy of such subgroups corresponds to equivalence of covering spaces without fixing a particular point in the fiber.

This relationship is one of the most important bridges between covering space theory and group theory. It leads to a classification of path-connected covering spaces of B in terms of conjugacy classes of subgroups of $\pi_1(B, b_0)$. In especially simple cases, such as when $\pi_1(B, b_0)$ is abelian, conjugacy becomes irrelevant, so covering spaces are classified directly by subgroups. The examples that follow illustrate how this principle works for spaces such as S^1 and the torus.

Unless stated otherwise, all topological spaces are assumed to be path-connected and locally path-connected. Let B be a topological space, i.e. a base space of interest. Recall from Definition 1.14 that a B-space is a topological space E equipped with a continuous map $p : E \to B$.

Definition 4.1 (equivalence of covering maps/covering spaces). Let

$$p : E \to B \text{ and } p' : E' \to B \quad \text{be covering spaces of } B.$$

An isomorphism of B-spaces $h : E \to E'$ is an equivalence of covering maps/covering spaces. If there exists such an equivalence, then p and p' are equivalent as covering maps of B (alternatively, E and E' are equivalent as covering spaces of B).

Theorem 4.1. Let $p : E \to B$ and $p' : E' \to B$ be covering spaces of B and let $b_0 \in B$. Let $e_0 \in p^{-1}(b_0)$ and $e_0' \in (p')^{-1}(b_0)$ be in the fiber over b_0. Then, the following hold:

(i) There exists an equivalence $h : E \to E'$ of covering spaces with $h(e_0) = e_0'$ if and only if

$$p_*(\pi_1(E, e_0)) \text{ and } p'_*(\pi_1(E', e_0')) \quad \text{are equal as subgroups of } \pi_1(B, b_0).$$

If such an equivalence h exists, it is unique.

(ii) There exists an equivalence $h : E \to E'$ of covering spaces if and only if

$$p_*(\pi_1(E, e_0)) \text{ and } p'_*(\pi_1(E', e_0')) \quad \text{are conjugate as subgroups of } \pi_1(B, b_0).$$

Lemma 4.1 (the general lifting lemma). Let $p : E \to B$ be a covering map and let $b_0 \in B$. Let $e_0 \in p^{-1}(b_0)$ be in the fiber over b_0. Suppose Y is path-connected and locally path-connected. Let $f : Y \to B$ be a continuous map and let $y_0 \in Y$ be such that $f(y_0) = b_0$. Then, there exists a lifting $\widetilde{f} : Y \to E$ of f with $\widetilde{f}(y_0) = e_0$ if and only if $f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(E, e_0))$ as subgroups of $\pi_1(B, b_0)$. If such a lifting exists, it is unique.

$$\begin{tikzcd} & E \arrow[d, "p"] \\ X \arrow[ur, "\widetilde{f}"] \arrow[r, "f"'] & B \end{tikzcd}$$

Corollary 4.1. The map

$$\begin{aligned} &\{\text{equivalence classes of path-connected covering spaces of } B\} \\ \to &\{\text{conjugacy classes of subgroups of } \pi_1(B, b_0)\} \end{aligned}$$

given by $[p : E \to B] \mapsto [p_*(\pi_1(E, e_0))$ for any choice of $e_0 \in p^{-1}(b_0)]$ is well-defined and injective.

Remark 4.1. If $\pi_1(B, b_0)$ is abelian, then

$$\{\text{conjugacy classes of subgroups of } \pi_1(B, b_0)\} = \{\text{subgroups of } \pi_1(B, b_0)\}.$$

Example 4.1 (Munkres p. 483 Question 1). Use the lifting lemma to show that for $n > 1$,

$$\text{every continuous map } f : S^n \to S^1 \quad \text{is null-homotopic.}$$

Solution. Recall that if f and f' are paths such that $f \sim f'$ and f' is the constant map, then f is said to be null-homotopic. Consider the following commutative diagram:

$$\begin{tikzcd} & \mathbb{R} \arrow[d, "p"] \\ S^n \arrow[ur, "\widetilde{f}"] \arrow[r, "f"'] & S^1 \end{tikzcd}$$

Suppose

$$p : \mathbb{R} \to S^1 \quad \text{defined by} \quad p(x) = e^{2\pi i x} \text{ is a covering map.}$$

As a side remark, one should already perceive the above-mentioned *trick* as second nature at this point. Since $\pi_1(S^n)$ is the trivial group for $n > 2$, then for any continuous map $f : S^n \to S^1$, we see that $f_* : \pi_1(S^n) \to \pi_1(S^1)$ is trivial, so it is a subgroup of $\pi_1(S^1)$.

As such, the lifting $\widetilde{f} : S^n \to \mathbb{R}$ is a map from a compact space to $\mathbb{R}$, so $\widetilde{f}$ is homotopic to a constant map. Since $f = p \circ \widetilde{f}$, it follows that f is also homotopic to a constant map. □

Example 4.2. Recall the monodromy action of the exponential covering map, which refers to how the fibers of the covering space transform as we go around loops in the base space. We have

$$\pi_1(S^1, 1) \to \mathbb{Z} \text{ (in fact it is an isomorphism)} \quad \text{where} \quad \alpha \mapsto 0 * \alpha$$

via the monodromy action of the exponential covering map $p : \mathbb{R} \to S^1$, where $x \mapsto e^{2\pi i x}$. Since $\mathbb{R}$ is simply-connected, then $p_*(\pi_1(\mathbb{R}, 0)) = 0$ in $\pi_1(S^1, 1) \cong \mathbb{Z}$.

By Corollary 4.1, the map

$$\{\text{equivalence classes of path-connected covering spaces of } S^1\} \to \{\text{subgroups of } \pi_1(S^1, 1) \cong \mathbb{Z}\}$$

given by

$$[p' : E' \to S^1] \mapsto [p'_*(\pi_1(E', e'_0)) \quad \text{for any choice of } e'_0 \in (p')^{-1}(1)] \quad \text{is well-defined and injective.}$$

Example 4.3. Also, as another example, for any non-zero integer n,

$$\text{the } n^{\text{th}} \text{ power map} \quad p_n : S^1 \to S^1 \text{ with } z \mapsto z^n \quad \text{is a covering map.}$$

Then, $(p_n)_*(\pi_1(S^1,1)) = n \cdot \pi_1(S^1,1) \cong n\mathbb{Z}$. Recall the following basic fact from group theory, which states that the map $\mathbb{Z}_{\geq 0} \to \{\text{subgroups of } \mathbb{Z}\}$ is bijective, where $n \mapsto n\mathbb{Z}$. Applying Corollary 4.1, we see that the map

$$\left\{\text{equivalence classes of path-connected covering spaces of } S^1\right\} \to \left\{\text{subgroups of } \pi_1(S^1,1) \cong \mathbb{Z}\right\}$$

is bijective. Moreover, the equivalence classes of path-connected covering spaces of S^1 is precisely the exponential map $p : \mathbb{R} \to S^1$, which is also equal to the set of nth power maps $p_n : S^1 \to S^1$ with $n \in \mathbb{Z}_{>0}$.

In particular, for any $n \in \mathbb{Z}_{>0}$, p_n and p_{-n} are equivalent as covering maps (this is closely related to Example 1.28 from one of the exercises in Munkres' textbook), i.e. the following diagram commutes:

$$\begin{tikzcd}
z \arrow[rr, mapsto, "\cong"] & & z^{-1} \\
S^1 \arrow[rr] \arrow[dr, "p_n"'] & & S^1 \arrow[dl, "p_{-n}"] \\
& S^1 &
\end{tikzcd}$$

Example 4.4 (Munkres p. 483 Question 2).

(a) Show that every continuous map $f : \mathbb{P}^2 \to S^1$ is null-homotopic.

(b) Find a continuous map of the torus into S^1 that is not null-homotopic.

Solution.

(a) The idea here is to use the fundamental groups of $\mathbb{P}^2$ and S^1. The fundamental group of the real projective plane $\mathbb{P}^2$ is isomorphic to $\mathbb{Z}/2\mathbb{Z}$, whereas the fundamental group of the circle S^1 is isomorphic to $\mathbb{Z}$. So, $f : \mathbb{P}^2 \to S^1$, it induces a homomorphism on fundamental groups

$$f_* : \pi_1\left(\mathbb{P}^2\right) \to \pi_1\left(S^1\right).$$

So, any homomorphism from $\mathbb{Z}/2\mathbb{Z}$ to $\mathbb{Z}$ must map the entire group $\mathbb{Z}/2\mathbb{Z}$ to the identity in $\mathbb{Z}$ as there are no non-trivial homomorphisms from a finite group to an infinite group. As such, the induced map on the fundamental group is trivial, so any map f can be continuously deformed to a constant map.

(b) We wish to find

$$\text{a continuous map } f : \mathbb{T} \to S^1 \quad \text{which is not null-homotopic.}$$

Note that $\mathbb{T} = S^1 \times S^1$. Say we define a projection map

$$p : \mathbb{T} \to S^1 \quad \text{where} \quad p(x,y) = x.$$

By definition of the projection map, p is continuous. However, the homomorphism from the fundamental group of the torus to the fundamental group of the circle is surjective, and hence, non-trivial. As such, π_1 is not null-homotopic. □

Example 4.5 (Munkres p. 483 Question 3). Let $p : E \to B$ be a covering map; let $p(e_0) = b_0$. Show that $H_0 = p_*(\pi_1(E, e_0))$ is a normal subgroup of $\pi_1(B, b_0)$ if and only if for every pair of points e_1, e_2 of $p^{-1}(b_0)$, there is an equivalence $h : E \to E$ with $h(e_1) = e_2$.

Solution. We first prove the forward direction. Suppose

$$p_*(\pi_1(E, e_0)) \trianglelefteq \pi_1(B, b_0).$$

Take

$$\text{any loop } \gamma \in \pi_1(B, b_0) \quad \text{and} \quad \text{any element } \alpha \in \pi_1(E, e_0).$$

Then, the conjugate $\gamma\alpha\gamma^{-1}$ is also contained in $p_*(\pi_1(E, e_0))$. As such, any loop in B can be lifted to a path in E that maps one point in $p^{-1}(b_0)$ to another.

Take $e_1, e_2 \in p^{-1}(b_0)$. By the covering space property, for any path γ in B from b_0 to itself (essentially a loop), each point in the fiber $p^{-1}(b_0)$ lifts this path uniquely to a path in E starting at that point in $p^{-1}(b_0)$. Since p_* is normal in $\pi_1(B, b_0)$, for any $e_1, e_2 \in p^{-1}(b_0)$, we can construct a homeomorphism $h : E \to E$ by lifting appropriate loops in B. This homeomorphism can be chosen to map e_1 to e_2.

Proof of reverse direction is omitted. □

Example 4.6 (Munkres p. 483 Question 4). Let $\mathbb{T} = S^1 \times S^1$, the torus. There is an isomorphism of $\pi_1(T, b_0 \times b_0)$ with $\mathbb{Z} \times \mathbb{Z}$ induced by projections of $\mathbb{T}$ onto its two factors.

(a) Find a covering space of $\mathbb{T}$ corresponding to the subgroup of $\mathbb{Z} \times \mathbb{Z}$ generated by the element $m \times 0$, where m is a positive integer.

(b) Find a covering space of $\mathbb{T}$ corresponding to the trivial subgroup of $\mathbb{Z} \times \mathbb{Z}$.

(c) Find a covering space of $\mathbb{T}$ corresponding to the subgroup of $\mathbb{Z} \times \mathbb{Z}$ generated by $m \times 0$ and $0 \times n$, where m and n are positive integers.

Solution.

(a) The subgroup generated by $(m, 0)$ is

$$\Gamma = \{(km, 0) : k \in \mathbb{Z}\} \cong m\mathbb{Z} \times \{0\}.$$

Since Γ acts by translations along the x-axis with period m, the quotient space $\mathbb{R}^2/\Gamma$ identifies points that differ by multiples of $(m, 0)$, i.e. along the x-direction, the space is wrapped every m units; along the y-direction, the space remains unwrapped. So,

$$\mathbb{R}^2/\Gamma \quad \text{is homeomorphic to a cylinder.}$$

This implies that the covering map $p : \mathbb{R}^2/\Gamma \to \mathbb{T}$ is induced by the inclusion $\Gamma \subseteq \mathbb{Z}^2$. Explicitly, what the covering action does is that the cylinder wraps around the torus m times in the x-direction and infinitely in the y-direction. Hence,

$$p : \mathbb{R}^2/\Gamma \quad \text{where} \quad p([x,y]) = ([x \pmod 1], [y \pmod 1]).$$

(b) Since the subgroup Γ is trivial, then $\mathbb{R}^2/\Gamma = \mathbb{R}^2$. So, the covering space is simply the universal cover of $\mathbb{T}$, which is $\mathbb{R}^2$.

As such, the covering map is given as follows:

$$p : \mathbb{R}^2 \to \mathbb{T} \quad \text{where} \quad p([x,y]) = ([x \pmod 1], [y \pmod 1]).$$

(c) Let Γ denote the subgroup. Then, Γ is generated by $(m,0)$ and $(0,n)$, so

$$\Gamma = m\mathbb{Z} \times n\mathbb{Z}.$$

Since Γ is a lattice generated by $(m,0)$ and $(0,n)$, the covering space $\mathbb{R}^2/\Gamma$ is a torus, where the respective periods along the x- and y-directions are m and y.

Again, the covering map is the same as the preceding subparts, but in order to visualise the covering space, it is in fact a torus that covers T m times in one direction and n times in the other. □

4.2 The Universal Covering Space

By convention, throughout unless otherwise stated, all topological spaces are assumed to be path-connected and locally path-connected[1].

Definition 4.2 (universal covering space). Let B be a topological space, i.e. a base space of interest. A universal covering space of B is a covering space $p : E \to B$ such that E is simply-connected.

There are examples of topological spaces B that have no universal covering space. Take the infinite earring/Hawaiian earring (Figure 4.1) for instance.

Theorem 4.2 (universal property of a universal covering space). Let $p : E \to B$ be a universal covering space of B. Suppose $b_0 \in B$ and $e_0 \in p^{-1}(b_0)$ are in the fiber over b_0. Then, for any covering space $p' : E' \to B$ and any $e_0' \in (p')^{-1}(b_0)$,

$$\text{there exists a unique } B\text{-space map } h : E \to E' \quad \text{such that} \quad h(e_0) = e_0'.$$

[1]Recall from MA3209 Metric and Topological Spaces that a locally path-connected topological space X is defined as follows: for all $x \in X$ and for all open neighbourhoods $V \subseteq X$ of x, there exists an open neighbourhood $U \subseteq V$ of x such that U is path-connected.

Proof. Since E is simply-connected, its fundamental group is trivial, so

$$p_*(\pi_1(E,e_0)) = 1 \subseteq p'_*(\pi_1(E',e'_0)) \quad \text{as subgroups of } \pi_1(B,b_0).$$

We then apply the general lifting lemma to the covering map $p' : E' \to B$ and the continuous map $p : E \to B$ to obtain this universal property. □

Lemma 4.2. Let

$$p : E \to B \text{ and } p' : E' \to B \quad \text{be covering maps of } B.$$

Then, any B-space map $h : E \to E'$ is also a covering map.

We will not discuss the proof in detail. Briefly, construct a commutative diagram which involves $p' \circ h = p$. Then, one needs to show that for every point $x \in E$, there exists an open neighbourhood $V \subseteq E'$ of x which is evenly covered by h. In other words, one needs to show that

$$h \text{ is surjective} \quad \text{and} \quad E' \text{ is locally evenly covered by } h.$$

Lemma 4.3 (Munkres p. 487 Question 1). Let

$$q : X \to Y \text{ and } r : Y \to Z \text{ be maps} \quad \text{and} \quad p = r \circ q.$$

If q and r are covering maps and Z has a universal covering space, then p is a covering map.

Proof. Since Z has a universal covering space, say $\widetilde{Z}$, then there exists a covering map $\pi : \widetilde{Z} \to Z$ such that $\widetilde{Z}$ is simply-connected. We wish to prove that $p = r \circ q$ is a covering map, i.e. for every $z \in Z$, there exists an open neighbourhood $V \subseteq Z$ such that $p^{-1}(V)$ is a disjoint union of open sets in X, each of which is mapped homeomorphically onto V.

Since $r : Y \to Z$ is a covering map, then for every $z \in Z$, there exists an open set $V_z \subseteq Z$ such that

$$r^{-1}(V_z) = \bigsqcup_{i \in I_z} U_i \quad \text{such that} \quad r|_{U_i} : U_i \to V_z \text{ is homeomorphic.}$$

Similarly, for every $y \in r^{-1}(V_z)$, as $q : X \to Y$ is a covering map, then there exists an open set $W_y \subseteq Y$ such that

$$q^{-1}(W_y) = \bigsqcup_{j \in J_y} U_j \quad \text{such that} \quad q|_{U_j} : U_j \to W_y \text{ is homeomorphic.}$$

As $p^{-1}(V_z) = q^{-1}\left(r^{-1}(V_z)\right)$, we see that this is a disjoint union of open sets in X, each of which maps homeomorphically onto V_z under p. □

However, the composition of covering maps is not necessarily a covering map. As pointed out in Lemma 4.3, we need Z to be a universal covering space for $p : X \to Z$ to be a covering map. As an example of a topological space that does not have a universal covering space, consider the Hawaiian earring (will be discussed in due course).

Definition 4.3 (semi-locally simply-connected space). A space B is semi-locally simply-connected if and only if for any $b \in B$, there exists an open neighbourhood $U \subseteq B$ of b such that

the homomorphism $i_* : \pi_1(U,b) \to \pi_1(B,b)$ induced by the inclusion $i : U \hookrightarrow B$ is trivial.

In other words, any loop in U at b is null-homotopic in B.

Remark 4.2. If $U \subseteq B$ is a semi-locally simply-connected space, then so is any smaller neighbourhood of b, i.e. we can shrink U if necessary.

Example 4.7 (simply-connected implies semi-locally simply-connected). Let B be simply-connected. Then, it is semi-locally simply-connected.

We provide some examples of the aforementioned result.

Example 4.8. Note that convex subsets of $\mathbb{R}^n$ are simply-connected. Hence, they are semi-locally simply-connected. To see why, consider a point $b \in B$, where $B \subseteq \mathbb{R}^n$. Then, there exists an open neighbourhood U of b such that any loop in U based at b is null-homotopic in B. Here, we have exploited the convexity of subsets of $\mathbb{R}^n$.

Hence, $i_* : \pi_1(U,b) \to \pi_1(B,b)$ induced by the inclusion map $i : U \hookrightarrow B$ is trivial.

Example 4.9. Spheres S^n, where $n > 1$ are simply-connected. Hence, they are semi-locally simply-connected.

Example 4.10. Let B be a locally simply-connected space, i.e. for all $b_0 \in B$ and open neighbourhoods $V \subseteq B$ of b_0, there exists an open neighbourhood $U \subseteq V$ of b_0 such that U is simply-connected. Take for example,

$$B = \text{an arbitrary topological manifold.}$$

Then, B is semi-locally simply-connected.

Theorem 4.3. Let B be a path-connected and locally-path connected space, with $b_0 \in B$. Then, B is semi-locally simply-connected if and only if for any $H \leq \pi_1(B,b_0)$, there exists a covering space $p : E \to B$ and $e_0 \in p^{-1}(b_0)$ such that $H = p_*(\pi_1(E,e_0))$ as subgroups of $\pi_1(B,b_0)$.

Corollary 4.2. Let B be a path-connected and locally path-connected space. Then, there exists a universal covering space of B if and only if B is semi-locally simply-connected.

Example 4.11 (infinite earring/Hawaiian earring). Let C_n be the circle of radius $1/n$ in $\mathbb{R}^2$ with centre at the point $(1/n, 0)$. Let X be the union of these circles as a subspace of $\mathbb{R}^2$ (Figure 4.1), i.e.

$$X = \bigcup_{n=1}^{\infty} \left\{ (x,y) \in \mathbb{R}^2 : \left(x - \frac{1}{n}\right)^2 + y^2 = \left(\frac{1}{n}\right)^2 \right\}.$$

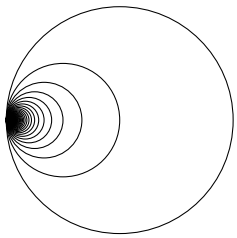


Figure 4.1: Hawaiian earring

Then, X is path-connected and locally path-connected, but not semi-locally simply-connected. As such, X does not have a universal covering space. More precisely, if U is any neighbourhood of $b_0 = 0$, then

$$i_* : \pi_1(U, b_0) \to \pi_1(X, b_0) \quad \text{is non-trivial.}$$

Given n, there exists a retraction $r : X \to C_n$ obtained by letting r map each circle C_i to the point b_0, where $i \neq n$. Then, for n sufficiently large such that C_n lies in U, we see that in the following diagram homomorphism induced by inclusion, j_* is injective. Hence, i_* is non-trivial.

$$\begin{tikzcd}
\pi_1(C_n, b_0) \arrow[rr, "j_*"] \arrow[dr, "k_*"'] & & \pi_1(X, b_0) \arrow[dl, "i_*"] \\
& \pi_1(U, b_0) &
\end{tikzcd}$$

Theorem 4.4 (Galois correspondence for covering spaces). Let B be a path-connected and locally path-connected semilocally simply-connected space. Suppose $b_0 \in B$. Then, the map from the set of equivalence classes of path-connected covering spaces of B to the set of conjugacy classes of subgroups of $\pi_1(B, b_0)$, where

$$[p : E \to B] \mapsto \left[p_*(\pi_1(E, e_0)) \text{ for any choice of } e_0 \in p^{-1}(b_0)\right]$$

is a well-defined order-reversing bijection.

We compare Theorem 4.4 with the Galois correspondence (Theorem 4.5) that one would encounter in MA4203 Galois Theory. Briefly recall the definition of $\mathrm{Gal}(L/K)$.

Theorem 4.5 (the Galois correspondence). Let k be a field and k^s be a separable algebraic closure of k. The map sending the set of k-isomorphism classes of algebraic separable extensions of k to conjugacy classes of subgroups of $\mathrm{Gal}(k^s/k)$, where

$$[k \hookrightarrow L] \mapsto [\mathrm{Gal}(k^s/i(L)) \text{ for any choice of } k\text{-embedding } i : L \hookrightarrow k^s]$$

is a well-defined order-reversing bijection.

4.3 Covering Transformations

Definition 4.4 (covering transformation/deck transformation). Let $p : E \to B$ be a covering space of B. A covering transformation, also known as a deck transformation, is an equivalence of $p : E \to B$ with itself, i.e. an isomorphism of B-spaces.

Definition 4.5 (covering transformation action). Let $\mathcal{C}(E, p, B)$ be the group of all covering transformations of p. This is in fact equal to

$$\mathrm{Aut}(E/B) = \mathrm{Aut}(P) \quad \text{which acts from the left on the space } E.$$

Let $b_{\in}B$. Then, $\mathrm{Aut}(p)$ acts from the left on the fiber $p^{-1}(b_0)$ over b_0, i.e.

$$\mathrm{Aut}(p) \times p^{-1}(b_0) \to p^{-1}(b_0) \quad \text{given by} \quad (h, e_0) \mapsto h(e_0) \quad \text{is the covering transformation action.}$$

Lemma 4.4. Let $p : E \to B$ be a covering map, with $b_0 \in B$ and $e_0 \in p^{-1}(b_0)$. The covering transformation $\mathrm{Aut}(p)$ on $p^{-1}(b_0)$ is fixed-point free, i.e.

$$\psi : \mathrm{Aut}(p) \to p^{-1}(b_0) \quad \text{such that} \quad h \mapsto h(e_0) \quad \text{is injective.}$$

Let $H = p_*(\pi_1(E, e_0)) \subseteq \pi_1(B, b_0)$. Define

$$N(H) = \{g : gHg^{-1} = H\} \quad \text{to be the normaliser of } H \text{ in } \pi_1(B, b_0).$$

Then, the ψ-image of $\mathrm{Aut}(p)$ in $p^{-1}(b_0)$ is equal to the Φ-image of $H/N(H)$ in $p^{-1}(b_0)$.

Proof. Since $h \in \mathrm{Aut}(p)$ is a lifting of $p : E \to B$ over itself, by the general lifting lemma, h is uniquely determined by $h(e_0)$. As such, $\psi : \mathrm{Aut}(p) \to p^{-1}(b_0)$ is injective.

Now, we need to prove that

$$\text{the } \psi\text{-image of } \mathrm{Aut}(p) \text{ in } p^{-1}(b_0) = \Phi\text{-image of } H/N(H) \text{ in } p^{-1}(b_0).$$

We first prove the forward inclusion $\subseteq$. Suppose $h \in \mathrm{Aut}(p)$ and $e_1 = h(e_0)$ in $p^{-1}(b_0)$. Then, $h^{-1}(e_1)$, so by the general lifting lemma, we have $p_*(\pi_1(E, e_0)) = p_*(\pi_1(E, e_1))$.

Choose a path γ in E from e_0 to e_1. We can do so since E is path-connected. Also, let $\alpha = p \circ \gamma$ and a loop in B at b_0. Then,

$$[\alpha] * p_*(\pi_1(E, e_1)) * [\alpha]^{-1} = p_*(\pi_1(E, e_0)). \quad \text{as subgroups of } \pi_1(B, b_0).$$

Then,

$$[\alpha] \in N(H) \quad \text{in } \pi_1(B, b_0),$$

so

$$\Psi(h) = h(e_0) = e_1 = \gamma(1) = \Phi(H[\alpha]) \quad \text{lies in the } \Phi\text{-image of } H/N(H).$$

We briefly explain a couple of the equality signs. Note that $e_1 = \gamma(1)$ because of the choice of γ — it is a path in E from e_0 to e_1. As for $\gamma(1) = \Phi(H[\alpha])$, it follows by the monodromy action. We have thus proven $\subseteq$.

We then prove the reverse inclusion $\supseteq$. Suppose $[\alpha] \in N(H)$, represented by a loop α in B based at b_0. Let γ be the unique path lifting of α over p beginning at e_0. Let $e_1 = \gamma(1)$. Then, one has

$$[\alpha] * p_*(\pi_1(E, e_1)) * [\alpha]^{-1} = p_*(\pi_1(E, e_0)) \quad \text{since } [\alpha] \in N(H).$$

By the general lifting lemma, there exists a B-space map $h : E \to E$ with $h(e_0) = e_1$. By reversing roles, one sees that $h \in \mathrm{Aut}(p)$ is invertible. Then,

$$\Phi(H[\alpha]) = \gamma(1) = e_1 = h(e_0) = \Psi(h) \quad \text{lies in the } \Psi\text{-image of } \mathrm{Aut}(p).$$

The first equality follows by the definition of monodromy action. Also, note that the proof of the reverse inclusion mimics that of the forward inclusion. Anyway, the proof is complete. □

Theorem 4.6. Let $p : E \to B$ be a covering map, $b_0 \in B$ and $e_0 \in p^{-1}(b_0)$. Also, let

$$H = p_*(\pi_1(E, e_0)) \subseteq \pi_1(B, b_0).$$

The composite map

$$\Phi^{-1} \circ \psi : \mathrm{Aut}(p) \xrightarrow{\Psi} \Psi\text{-image in } p^{-1}(b_0) \xrightarrow{\Phi^{-1}} N(H)/H$$

is a group homomorphism, hence isomorphism.

Proof. Let $h, k \in \mathrm{Aut}(p)$ and let

$$e_1 = h(e_0) = \Psi(h) \quad e_2 = k(e_0) = \Psi(k) \quad e_3 = h(e_2) = h(k(e_0)) \quad \text{in } p^{-1}(b_0).$$

Choose paths γ, δ in E from e_0 to e_1, e_2 respectively. Let $\alpha = p \circ \gamma$ and $\beta = p \circ \delta$ be loops in B based at b_0, so

$$H[\alpha] = \Phi^{-1}(e_1), H[\beta] = \Phi^{-1}(e_2), H[\alpha * \beta] \quad \text{lie in } N(H)/H.$$

The path $h \circ \delta$ in E is from $h(e_0) = e_1$ to $h(e_2) = h(k(e_0)) = e_3$ and it is the unique path lifting of β over p beginning at e_1. Hence, $\gamma * (h \circ \delta)$ is the unique path lifting of $\alpha * \beta$ over p beginning at e_0, so it follows that

$$\Phi(H[\alpha * \beta]) = (\gamma * (h \circ \delta))(1) = e_3 = h(k(e_0)) = \Psi(h \circ k).$$

We briefly explain why $(\gamma * (h \circ \delta))(1) = e_3$. Recall that γ is a path from e_0 to $e_1 = h(e_0)$, $h \in \mathrm{Aut}(p)$ and δ is a path from e_1 to $e_2 = k(e_0)$. As such, the concatenation of paths γ with $h \circ \gamma$, denoted by $\gamma * (h \circ \delta)$, involves first considering a path that takes from e_0 to e_1 (by γ), then e_1 to e_3 (by $h \circ \delta$). Evaluation of the concatenation of paths at time $t = 1$ implies we should obtain e_3. □

Corollary 4.3. Let $p : E \to B$ be a universal covering space of B. Then, for any choice of $b_0 \in B$, $e_0 \in p^{-1}(b_0)$, the map

$$\mathrm{Aut}(p) \to \pi_1(B, b_0) \quad \text{given by} \quad h \mapsto [pi \circ \gamma]$$

for any path γ in B from e_0 to $h(e_0)$ is an isomorphism.

Definition 4.6 (regular/Galois covering map). Let $p : E \to B$ be a covering map. It is regular/Galois if and only if E is path-connected and $H = p_*(\pi_1(E, e_0)) \trianglelefteq \pi_1(B, b_0)$ for any choice of $b_0 \in B$ and $e_0 \in p^{-1}(b_0)$.

Corollary 4.4. A covering map p is Galois

$$\begin{aligned} &\text{if and only if} \quad N(H) = \pi_1(B, b_0) \\ &\text{if and only if} \quad \text{the } \Psi\text{-image of } \mathrm{Aut}(p) \text{ in } p^{-1}(b_0) \text{ is } p^{-1}(b_0) \end{aligned}$$

Example 4.12. Since $\pi_1(S^1, 1) \to \mathbb{Z}$ is Abelian (and isomorphic), then every covering of S^1 is Galois.

To see why, recall that the set of equivalence classes of path-connected covering spaces of S^1 is equivalent to the exponential map $p : \mathbb{R} \to S^1$ given by $x \mapsto e^{2\pi i x}$. Recall that $\mathbb{R}$ is contractible so $\mathrm{Aut}(p) \to \pi_1(S^1, 1) \to \mathbb{Z}$ are isomorphic, where $\mathrm{Aut}(p) \to \mathbb{Z}$ is given by $h \mapsto h(0)$.

Example 4.13. Consider the covering map

$$p_n : S^1 \to S^1 \quad \text{given by} \quad z \mapsto z^n, \quad \text{where } n \in \mathbb{Z}_{>0}.$$

We have $\mathrm{Aut}(p_n) \to \pi_1(S^1, 1)/p_{n_*}\pi_1(S^1, 1) \to \mathbb{Z}/n\mathbb{Z}$ being isomorphic. Here, $h \mapsto \alpha \in \mathbb{Z}/n\mathbb{Z}$, where α is unique, such that $h(1) = e^{2\pi i/n}$, which is an nth root of unity.

There is another example here. Remember to take from Lou Yi. Not sure if it is related to the fundamental group of the figure eight.

Definition 4.7 (set of homeomorphisms). Let X be a topological space. Define

$$\text{Homeo}(X) \text{ to be the set of homeomorphisms of } X.$$

Definition 4.8 (G-action). Let

$$X \text{ be a set} \quad \text{and} \quad G \leq \text{Aut}(X) = \text{Perm}(X).$$

The orbit space X/G is the quotient set of X modulo the equivalence relation $\sim$ of G-action, i.e.

$$x_1 \sim x_2 \quad \text{if and only if} \quad \text{there exists } g \in G \text{ such that } g(x_1) = x_2.$$

We have the quotient map

$$\pi : X \to X/G \quad \text{given by} \quad x \mapsto G \cdot x = G\text{-orbit of } x.$$

If

$$X \text{ is a topological space} \quad \text{and} \quad G \leq \text{Homeo}(X),$$

then X/G is given with the quotient topology.

Definition 4.9 (fixed-point free and properly discontinuous G-actions). Let X be a topological space and $G \leq \text{Homeo}(X)$.

- **Fixed-point free:** The G-action on X is fixed-point free if and only if

  $$\text{for any } x \in X, g \in G \setminus \{e\}, \quad \text{we have} \quad g(x) \neq x,$$

  i.e. no element of G other than the identity e has a fixed point
- **Properly discontinuous:** The G-action on X is properly discontinuous if and only if

  $$\text{for any } x \in X, \quad \text{there exists a neighbourhood } U \text{ of } x \in X \quad \text{such that}$$
  $$\text{for any } g \in G \setminus \{e\}, \text{ we have } g(U) \cap U = \emptyset,$$

  i.e. $g_0(U) \cap g_1(U) = \emptyset$ whenever $g_0 \neq g_1$

Remark 4.3. If a G-action on X is properly discontinuous, then it is fixed-point free. The converse holds when G is finite and X is Hausdorff.

The converse above appears in Exercise 81.4 of Munkres. To be discussed.

Example 4.14. $\mathbb{Z}$ acting on $\mathbb{R}$ by translation is properly discontinuous.

Recall that $\mathbb{Z} \leq \mathbb{R}$ with the group operation being addition. Here, the group $G = \mathbb{Z}$ acts on $X = \mathbb{R}$ by translation. Specifically, for any $n \in \mathbb{Z}$, the action is given by

$$n \cdot x = n + x \quad \text{for } x \in \mathbb{R}.$$

This means that n shifts $x \in \mathbb{R}$ by n units. So, we need to find a neighbourhood U of $x \in \mathbb{R}$ such that for any $n \in \mathbb{Z}$, where $n \neq 0$, the translated neighborhood does not intersect with U, i.e. $(U+n) \cap U = \emptyset$. Take U to be a sufficiently small ε-neighbourhood around x, where $\varepsilon < 1/2$. So,

$$U = (x - \varepsilon, x + \varepsilon) \quad \text{where } 0 < \varepsilon < 1/2.$$

Then,

$$U + n = (x + n - \varepsilon, x + n + \varepsilon).$$

Clearly,

$$(U+n) \cap U = (x+n-\varepsilon, x+n+\varepsilon) \cap (x-\varepsilon, x+\varepsilon) = \emptyset \quad \text{for } \varepsilon < 1/2.$$

The choice of $\varepsilon < 1/2$ is obvious because

$$x + n - \varepsilon > x + \varepsilon \text{ if } n \geq 1 \quad \text{and} \quad x + n + \varepsilon < x - \varepsilon \text{ if } n \leq -1.$$

As we have constructed such a neighbourhood U, it follows that the $\mathbb{Z}$-action on $\mathbb{R}$ is properly iscontinuous.

Example 4.15. $\mathbb{Q}$ acting on $\mathbb{R}$ by translation is fixed-point free but not properly discontinuous.

Again, we recall that $\mathbb{Q} \leq \mathbb{R}$ with the group operation being addition. Here, the group $\mathbb{Q}$ acts on $\mathbb{R}$ by translation. Specifically, for any $q \in \mathbb{Q}$, the $\mathbb{Q}$-action is given by

$$q \cdot x = q + x \quad \text{for } x \in \mathbb{R}.$$

Similarly, this means that q shifts $x \in \mathbb{R}$ by q units. We first prove that the $\mathbb{Q}$-action on $\mathbb{R}$ is not properly discontinuous. Let U be an ε-neighbourhood around x, where $\varepsilon > 0$. So,

$$U = (x - \varepsilon, x + \varepsilon) \quad \text{and} \quad U + q = (x + q - \varepsilon, x + q - \varepsilon).$$

Let $q > 0$ be arbitrary (the case where $q < 0$ can be argued similarly based on what we did towards the end of our analysis of Example 4.14). Then, there exists $\varepsilon > q/2 > 0$ such that $(U+q) \cap U \neq \emptyset$. In fact, this argument is analogous to saying that there are infinitely many $q \in \mathbb{Q}$ satisfying the inequality $q < 2\varepsilon$ as $\mathbb{Q}$ is dense in $\mathbb{R}$.

Having said that, the $\mathbb{Q}$-action on $\mathbb{R}$ by translation, as mentioned, is fixed-point free. To see why, note that $q + x = x$, then we must have $q = 0$, i.e. the only element of $\mathbb{Q}$ that fixes any $x \in \mathbb{R}$ is $q = 0$, but $0 \in \mathbb{Q}$ is the identity element of $\mathbb{Q}$. It implies that for any non-identity element of $\mathbb{Q}$, we have $q \cdot x \neq x$.

Lemma 4.5. Let $p : E \to B$ be a covering map. Then, the action of $\mathrm{Aut}(p) \subseteq \mathrm{Homeo}(E)$ on E is properly discontinuous.

Proof. Since $p : E \to B$ is a covering map, then there exists a neighbourhood V of b_0 in B that is evenly covered by p. Let $U \subseteq p^{-1}(V)$ be the unique slice such that $e_0 \in U$. The action of $\mathrm{Aut}(p)$ on each fiber is fixed-point free. Hence, for any $x \in U$ and $g \in \mathrm{Aut}(p) \setminus \{e\}$, one has $g(x) \neq x$, so $g(x)$ lies in a slice of $p^{-1}(V)$ different from U. Hence. $g(U) \cap U = \emptyset$. □

Theorem 4.7. Let X be path-connected and locally path-connected, with $G \subseteq \mathrm{Homeo}(X)$. Then,

$$\begin{aligned} &\text{the quotient map } \pi : X \to X/G \text{ is a covering map} \quad \text{if and only if} \\ &\text{the } G\text{-action on } X \text{ is properly discontinuous.} \end{aligned}$$

In this case, $\pi : X \to X/G$ is a Galois covering map and $\mathrm{Aut}(\pi) = G$ as subgroups of $\mathrm{Homeo}(X)$.

Proof. The proof of the forward direction follows from Lemma 4.5.

As for the reverse direction. Suppose the G-action on X is properly discontinuous. For any open $U \subseteq X$,

$$\pi^{-1}(\pi(U)) = \bigcup_{g \in G} g(U) \text{ open in } X \quad \text{so} \quad \pi(U) \text{ is open in } X/G.$$

Given $G \cdot x \in X/G$, represented by $x \in X$, there exists an open neighbourhood U of x in X such that for any $g_0 \neq g_1$ in G, we have $g_0(U) \cap g_1(U) = \emptyset$. Then,

$$\pi^{-1}(\pi(U)) = \bigsqcup_{g \in G} g(U) \quad \text{is a disjoint union of the } g(U)\text{'s.}$$

For each $g \in G$, the map

$$\pi : g(U) \to \pi(U) \quad \text{is bijective, continuous, open and hence a homeomorphism.}$$

Hence, $\pi : X \to X/G$ is a covering map. We have proven the reverse direction.

For any $x \in X$ and $g \in G$, one has $G \cdot g(x) = G \cdot x$ in X/G since x and $g(x)$ are in the same G-orbit. Hence, $\pi \circ g = \pi$, which implies $G \subseteq \mathrm{Aut}(X)$. For any $h \in \mathrm{Aut}(\pi)$, one has $\pi \circ h = \pi$ so $G \cdot h(x) = G \cdot x$ in X/G. As such, there exists $g \in G$ such that $g(x) = h(x)$ in X.

The action of $\mathrm{Aut}(\pi)$ on each fiber is fixed-point free so $h = g$ lies in G. Hence, $\mathrm{Aut}(\pi) = G$. Since $\mathrm{Aut}(\pi) = G$ acts simply transitively on $\pi^{-1}(\pi(x))$, then $\pi : X \to X/G$ is a Galois covering map. □

Theorem 4.8. Let $p : X \to B$ be a Galois covering map. Let $G = \mathrm{Aut}(p)$. Then, there exists a unique continuous map $k : X/G \to B$ such that the following diagram commutes:

$$\begin{tikzcd} & X \arrow[dl, "\pi"'] \arrow[dr, "p"] & \\ X/G \arrow[rr, "k"] & & B \end{tikzcd}$$

Here, $\pi : X \to X/G$ is a quotient map. Moreover, k is a homeomorphism.

Proof. For any $g \in G$, one has $p \circ g = p$, so p is constant on each G-orbit. By the universal property of X/G, there exists a unique continuous map $k : X/G \to B$ such that $k \circ \pi = p$.

Note that k is surjective since p is surjective. k is injective since p is Galois (so $G = \mathrm{Aut}(p)$ acts simply transitively on each fiber of p). Also, k is an open map as X/G is given with the quotient topology from $\pi : X \to X/G$ and p is an open map. Hence, k is a homeomorphism. □

Definition 4.10 (domination of covering space). Let $p_0 : X \to B$ be a Galois covering map and $G = \mathrm{Aut}(p_0)$. A covering space of B is dominated by X consists of

$$\text{a covering space } p : E \to B \quad \text{and} \quad \text{a } B\text{-space map } \pi : X \to E$$

Let X and X' be covering spaces of B dominated by X. A morphism from E to E' is a B-space map $k : E \to E'$ such that the following diagram commutes, i.e.

$$k \circ \pi = \pi' \quad \text{and} \quad p' \circ k = p$$

$$\begin{tikzcd} X \arrow[d, "\pi"'] & X \arrow[d, "\pi'"] \\ E \arrow[r] \arrow[d, "p"'] & E' \arrow[d, "p'"] \\ B & B \end{tikzcd}$$

Theorem 4.9. Let $p_0 : X \to B$ be a Galois covering map. Let $G = \mathrm{Aut}(p_0)$. Then, the map from the set of isomorphism classes of path-connected covering spaces of B dominated by X to the set of subgroups of G, where

$$(X \pi E p B) \mapsto \mathrm{Aut}(\pi)$$

is a well-defined order-reversing bijection.

Compare this with the following result:

Theorem 4.10. Let k be a field and let $k \hookrightarrow M$ be a Galois extension of k. Then, the map from the set of k-isomorphism classes of field extensions of k contained in M to the set of closed subgroups of $\mathrm{Gal}(M/k)$, where

$$M \mapsto \mathrm{Aut}(M/L),$$

is a well-defined order-reversing bijection. Here, $k \subseteq L \subseteq M$ are field extensions.

Chapter 5

Fundamental Groups Revisited

5.1 Fundamental Group of a Union

Example 5.1 (the theta space). Consider the theta space $X = A \cup B \cup C$, where A, B, C are three arcs, each homeomorphic to $[0,1]$, and all three arcs have the same endpoints p and q. Thus X is a graph with two vertices p, q and three edges joining them.

Choose points $a \in \mathrm{int}(A)$, $b \in \mathrm{int}(B)$, and $c \in \mathrm{int}(C)$. Let $U = X \setminus \{a\}$ and $V = X \setminus \{b\}$. Then $X = U \cup V$. We choose a base point $x_0 \in C$, so $x_0 \in U \cap V$.

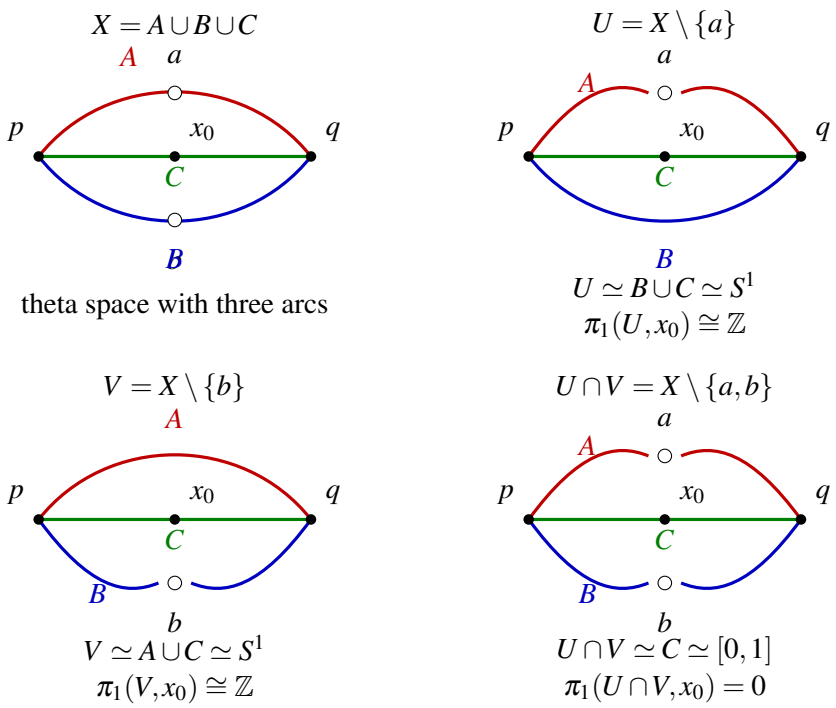


Figure 5.1: The theta space $X = A \cup B \cup C$, together with $U = X \setminus \{a\}$, $V = X \setminus \{b\}$, and $U \cap V = X \setminus \{a, b\}$

Since removing a breaks the arc A into two contractible pieces attached at p and q, the

space $U = X \setminus \{a\}$ deformation retracts onto the circle $B \cup C$. Hence

$$\pi_1(U, x_0) \cong \pi_1(B \cup C, x_0) \cong \mathbb{Z}.$$

Similarly, removing b breaks the arc B into two contractible pieces attached at p and q, so $V = X \setminus \{b\}$ deformation retracts onto the circle $A \cup C$. Hence

$$\pi_1(V, x_0) \cong \pi_1(A \cup C, x_0) \cong \mathbb{Z}.$$

On the other hand, $U \cap V = X \setminus \{a, b\}$. In this space, both A and B have been broken, so the remaining pieces of A and B deformation retract onto their endpoints p and q. Thus $U \cap V$ deformation retracts onto the single arc C, which is contractible. Therefore $\pi_1(U \cap V, x_0) = 0$. By the Seifert-van Kampen theorem (Theorem 1.19),

$$\pi_1(X, x_0) \cong \pi_1(U, x_0) * \pi_1(V, x_0) \cong \mathbb{Z} * \mathbb{Z}.$$

Thus the theta space has free fundamental group on two generators.

Theorem 5.1 (adjoining a two-cell). Let X be a T_2 space and $A \subseteq X$ be a closed path-connected subspace. Let $h : B^2 \to X$ be a continuous map such that

$$h \text{ maps } \operatorname{Int} B^2 \text{ bijectively onto } X \to A \quad \text{and} \quad h \text{ maps } S^1 \text{ into } A.$$

So, $X = A \cup h\left(\operatorname{Int} B^2\right)$.

Let $a = h(p) \in A$. Then,

the homomorphism induced by inclusion $\quad i_* : \pi_1(A, a) \to \pi_1(X, a) \quad$ is surjective.

We have $\ker i_* \trianglelefteq \pi_1(A, a)$, generated by

$$\text{the image of} \quad h_* : \pi_1\left(S^1, p\right) \to \pi_1(A, a).$$

Proof. Consider the open sets

$$V = X \setminus A = h\left(\operatorname{Int} B^2\right) \quad \text{and} \quad U = X \setminus x_0.$$

Here, $x_0 = h(0) \in V$. The sets U and V are open since X is Hausdorff. Also,

$$U \cap V = V \setminus \{x_0\} = h\left(\operatorname{Int} B^2 \setminus \{\mathbf{0}\}\right) \quad \text{is path-connected.}$$

The map

$$h \times \mathrm{id} : B^2 \times I \to h\left(B^2\right) \times I \quad \text{is closed}$$

since this is a map from a compact space to a subset of a Hausdorff space (since $h\left(B^2\right) \times I \subseteq X \times I$). So,

$$\begin{aligned} \text{the topology of } h\left(B^2\right) \times I \quad &\text{is the quotient topology from} \quad B^2 \times I \quad \text{and} \\ \text{the topology of } \left(h\left(B^2\right) \setminus \{x_0\}\right) \times I \quad &\text{is the quotient topology from} \quad \left(B^2 \setminus \{\mathbf{0}\}\right) \times I \end{aligned}$$

The deformation retraction of the punctured closed disc $B^2 \setminus \{\mathbf{0}\}$ onto the unit circle S^1 then induces a deformation retraction of $h\left(B^2\right) \setminus \{x_0\}$ onto $h\left(S^1\right) \subseteq A$. We can extend this to a deformation retraction of $X \setminus \{x_0\}$ onto A by keeping each point of A fixed. So, A is a deformation retract of $U = X \setminus \{x_0\}$. It follows that

$$i_* : \pi_1(A,a) \to \pi_1(U,a) \quad \text{is an isomorphism induced by inclusion.}$$

Choose any $b \in U \cap V$ and $q = h^{-1}(b) \in \operatorname{Int}\left(B^2 \setminus \{\mathbf{0}\}\right)$. Let γ be any path in $B^2 \setminus \{\mathbf{0}\}$ from q to $p \in S^1$. Then, $\delta = h \circ \gamma$ is a path in $U \cap V$ from b to $a \in A$.

Let f be a loop of S^1 such that $[f]$ generates $\pi_1\left(S^1, p\right)$. Then, $g = h \circ f$ is a loop in U based at a and $\delta * g * \widetilde{\delta} = h \circ \left(\gamma * f * \widetilde{f}\right)$ is a loop in U based at b. Hence,

$$\widetilde{\delta}^{-1}[g] = \left[\delta * g * \widetilde{\delta}\right] \in \pi_1(U_b) \quad \text{is the image of a generator of } \pi_1(U \cap V, b).$$

The result follows by the Seifert-van Kampen theorem (Theorem 1.19). □

Theorem 5.2. The fundamental group of the torus has the presentation

$$\langle \alpha, \beta \mid \alpha\beta\alpha^{-1}\beta^{-1} \rangle \quad \text{which is a free Abelian group of rank 2.}$$

Definition 5.1 (dunce cap). Let n be a positive integer with $n > 1$. Let

$$\mu_n = \left\{ e^{2k\pi i/n} : k \in \mathbb{Z} \right\} \subseteq S^1 \quad \text{be the finite subgroup of the } n^{\text{th}} \text{ roots of unity.}$$

This group acts on $\mathbb{C}$ by multiplication. The n-fold dunce cap is the quotient space

$$X_n = B^2/\mu_n \quad \text{of the unit ball } B^2 \text{ modulo the action of } \mu_n \text{ by rotation}$$

given with the quotient topology.

For any positive integer n with $n > 1$, let $r : S^1 \to S^1$ be rotation through the angle $2\pi/n$, i.e.

$$(\cos\theta, \sin\theta) \mapsto \left(\cos\left(\theta + \frac{2\pi}{n}\right), \sin\left(\theta + \frac{2\pi}{n}\right)\right).$$

Form a quotient space X from the unit ball by identifying each point $\mathbf{x}$ of S^1 with the points $r(\mathbf{x}), r^2(\mathbf{x}), \ldots, r^{n-1}(\mathbf{x})$. Then, the compact Hausdorff space X is known as the n-fold dunce cap.

Theorem 5.3. The fundamental group of the n-fold dunce cap is cyclic of order n, i.e. it has the presentation

$$\langle \alpha \mid \alpha^n \rangle.$$

Chapter 6

Classification of Surfaces

6.1 Fundamental Groups of Surfaces

Definition 6.1 (surface). A surface is a connected topological 2-dimensional manifold.

We shall give a list of compact surfaces such that **(i)** we can generate a good list of examples; **(ii)** no two surfaces on the list are homeomorphic; **(iii)** every compact surface is homeomorphic to one of them. So, we shall construct examples of compact connected surfaces as the quotient space obtained from a polygonal region in the plane by pasting its edges together.

Recall that for any $n \geq 3$, the regular n-gon P_n is the convex hull in $\mathbb{R}^2$ of the points

$$p_k = \left(\cos\frac{2k\pi}{n}, \sin\frac{2k\pi}{n}\right) \quad \text{where } 0 \leq k \leq n \text{ and } p_n = p_0.$$

Next, an n-sided polygonal region in the plane is any subset P of $\mathbb{R}^2$ which is given with a homeomorphism from a regular n-gon P_n satisfying the following properties: **(i)** the vertices/edges of P are the images of those of P_n; **(ii)** the orientation of an edge e of P is an ordering ∂e from the initial point to the final point; **(iii)** $\operatorname{Bd} P$ refers to the union of the edges of P, or equivalently, the images of the boundary of P_n; **(iv)** $\operatorname{Int} P = P \setminus \operatorname{Bd} P$, which is the images of the interior of P_n. Up to homeomorphism, we may assume that $P = P_n$ without loss of generality.

Definition 6.2 (labels). Let $P = P_n$ be an n-sided polygonal region in the plane. Let S be a set, where elements of S are known as labels.

(i) A labelling of the edges of P is a map from the set of edges of P to S

(ii) Given an orientation of each edge of P and a labelling of the edges of P, the corresponding labelling scheme w of length n is the sequence of labels with

exponents $+1$ or -1

$$w = (a_{i_1})^{\varepsilon_1} (a_{i_2})^{\varepsilon_2} \dots (a_{i_n})^{\varepsilon_n}$$

where for each $1 \leq k \leq n$,

$$a_{i_k} = \text{label assigned to the edge } p_{k-1}p_k \quad \text{and} \quad \varepsilon_k = \pm 1.$$

We define ε as either 1 or -1, depending on the assigned orientation of the edge. Specifically, $\varepsilon = 1$ if the edge is oriented from p_{k-1} to p_k, and $\varepsilon = -1$ if the orientation is reversed.

Example 6.1. Figure 6.1 shows a labelled polygonal region $P = P_6$ (in fact, it is a hexagonal region) with vertices $p_0, p_1, p_2, p_3, p_4, p_5$. arranged cyclically around the boundary. Each edge of the polygon is assigned a label from the set S. For example, the edge p_0p_1 is labelled a, the edge p_1p_2 is labelled b, the edge p_2p_3 is labelled c, the edge p_3p_4 is labelled a, the edge p_4p_5 is labelled d, and the edge p_5p_0 is labelled b.

The blue arrows indicate the chosen orientation of each edge. When we traverse the boundary in the order

$$p_0p_1, p_1p_2, p_2p_3, p_3p_4, p_4p_5, p_5p_0,$$

we compare the boundary direction with the chosen orientation of each edge. If the chosen orientation agrees with the boundary direction, the corresponding label is written with exponent $+1$. If the chosen orientation is opposite to the boundary direction, the corresponding label is written with exponent -1.

Therefore, the labelling scheme represented by Figure 6.1 is

$$w = ab^{-1}ca^{-1}db^{-1}.$$

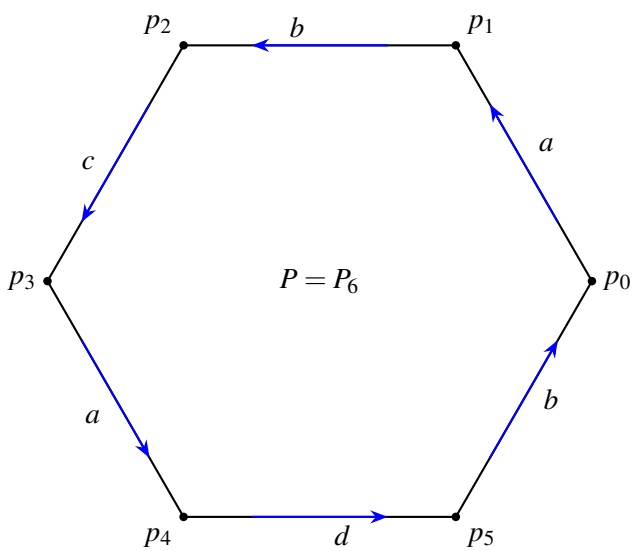


Figure 6.1: A labelled polygon with oriented edges

Define the equivalence relation $\sim_w$ on P as follows:

$$x \sim_w y \quad \text{if and only if} \quad x = y.$$

This is in fact equivalent to saying that there exist edges e_x, e_y of P with the same label such that $x \in e_x, y \in e_y$ and $y = h(x)$, where h is the unique orientation-preserving linear map from e_x to e_y.

The space obtained by pasting the edges of P together according to the given orientations and labelling the given labelling scheme w is the quotient set $X = P/\sim_w$ of P modulo the equivalence relation $\sim_w$ given with the quotient topology. Note that a cyclic permutation of the terms in a labelling scheme will change the space X only up to homeomorphism.

Example 6.2 (constructing a circle from a triangle). In Figure 6.2, the triangular shape (left) has edges labelled a and b. The edges are oriented (with arrows) and identified as follows: two sides labelled a and one side labelled b. The path around the triangle corresponds to the word $a \cdot a^{-1} \cdot b$, which, when simplified, effectively identifies certain edges and points.

The triangle is folded according to the identification of edges. Since both a's point towards the top vertex, we fold along the vertical line, thus obtaining the quotient map (collapsing identified points). In fact, the quotient map (centre) forms a cone-like structure, where the vertices meet at a single point (indicated by its apex). The cone is shown to be homeomorphic to a closed disc (right). To see why,

the apex of the cone	corresponds to	the centre of the disc	and
the boundary of the cone	corresponds to	the boundary of the disc	

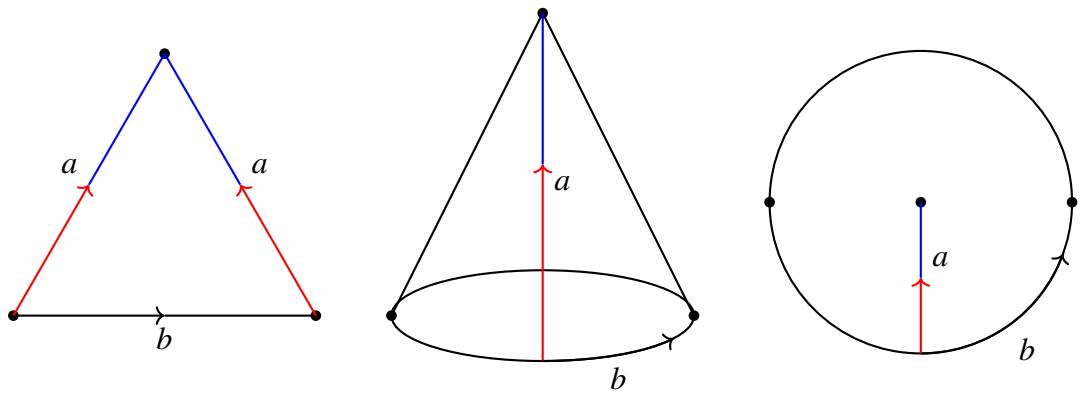


Figure 6.2: Constructing a circle from a triangle

This means that by identifying edges in this way, we get a shape that can be continuously deformed into a disc.

Example 6.3 (constructing a sphere from a square). In Figure 6.3, the square has four edges labelled with letters a and b, with pairs of opposite edges identified. Specifically, opposite a-edges and opposite b-edges are identified. The path around the boundary corresponds to $a^{-1}bb^{-1}a$ or equivalently $aa^{-1}bb^{-1}$, suggesting that opposite sides are glued together.

The square is transformed by identifying opposite edges, leading to a shape resembling a *lens* with two points (top and bottom) coming together. The resulting shape from identifying opposite sides in this way is homeomorphic to a sphere. This shows that a square with this particular edge identification can be transformed into a sphere through continuous deformation.

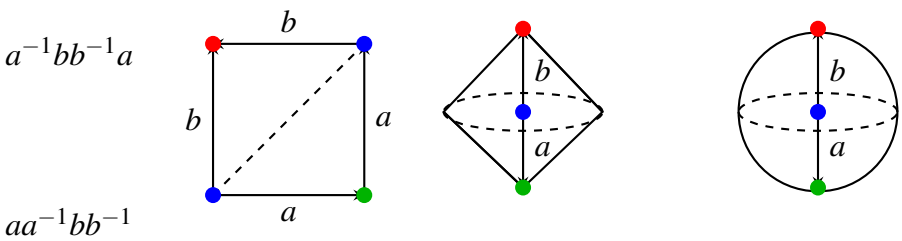


Figure 6.3: Constructing a sphere from a square

Example 6.4 (constructing a torus). We shall construct a torus from a square. This square has edges labelled a and b, with arrows indicating specific identifications.

By first identifying the two edges labelled a, we obtain a cylindrical shape. The two remaining edges labelled b are on the ends of the cylinder. Identifying the two b-edges results in a torus, a surface with a single hole. The torus is obtained by gluing the opposite edges of the square as per the labelling, which wraps the shape around to form a continuous loop with no boundary.

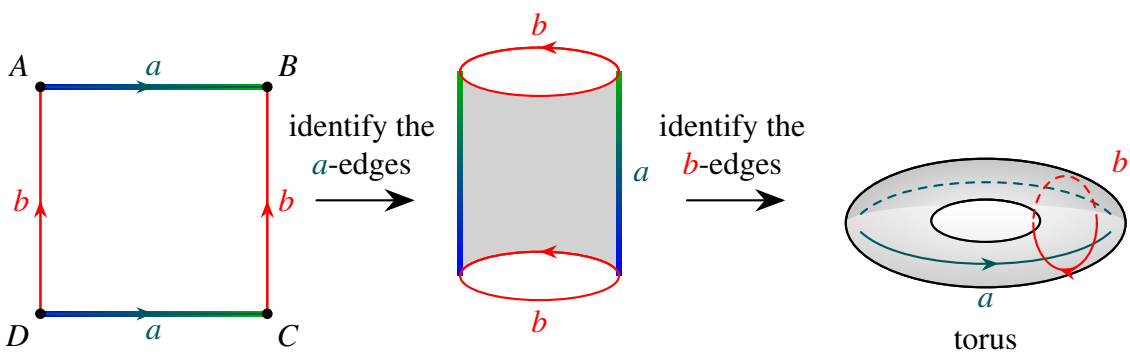


Figure 6.4: Constructing a torus from a square

Example 6.5 (constructing the dunce cap). The bottom diagram of Figure 6.5 shows that we can construct the dunce cap from a square. This square also has edges labelled, but the path around the boundary is $abab$. Here, adjacent edges are identified in a way that does not preserve orientation.

When identifying the edges in this specific manner, the shape that emerges is the dunce cap, which is a type of quotient space with interesting topological properties. The dunce cap has a relation that can be thought of as $\mathbb{P}^2 = S^2/\{\pm 1\}$, indicating a certain symmetry, but it does not embed into $\mathbb{R}^3$ smoothly.

The fundamental group of this space, $\pi_1\left(P^2, x_0\right) = \langle c | c^2 \rangle$, which is not isomorphic to $\mathbb{Z} \oplus \mathbb{Z}$. This implies that the dunce cap has different homotopy properties compared to spaces like the torus, whose fundamental group is $\mathbb{Z} \oplus \mathbb{Z}$.

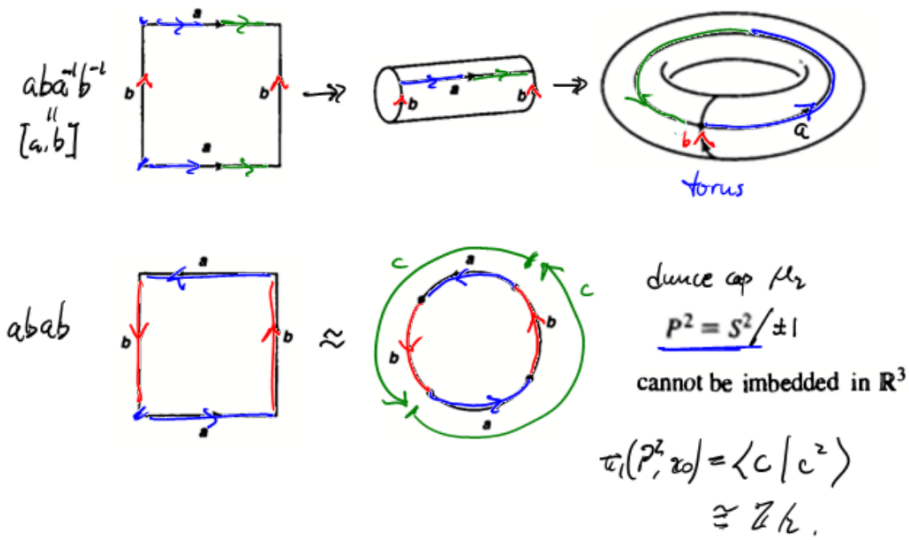


Figure 6.5: Constructing a torus and the dunce cap

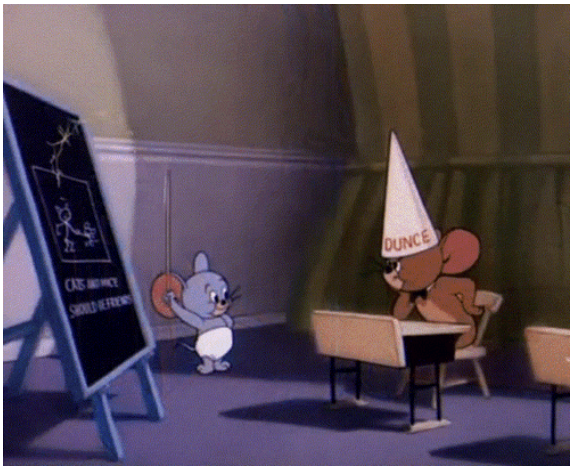


Figure 6.6: Dunce

Given a finite number $P_1, \ldots, P_k$ of disjoint polygonal regions, along with orientation and labelling of their edges, one can form a quotient space X by pasting the edges of the regions together in exactly the same way.

Example 6.6 (constructing a Möbius band and a cylinder). Consider Figure 6.7. It shows a rectangle with four edges labelled a, b, c, and d. The edges are oriented, and we have expressions that describe how they should be identified. The relation $dbc^{-1}a^{-1}$ indicates specific edge pairings. Another relation, $fa^{-1}e^{-1}b^{-1}$, suggests additional identifications.

The rectangle's edges are manipulated to identify certain pairs. Specifically, edge a on one side is identified with edge a on the opposite side, but with a twist (indicated by the reversal of the arrow direction). By joining the edges a and b with a half-twist, we obtain a Möbius band. This is a non-orientable surface, meaning it has only one side. The twisting of the edges creates a unique property: if you follow a path along the surface, you'll eventually return to your starting point but on the other side of the band.

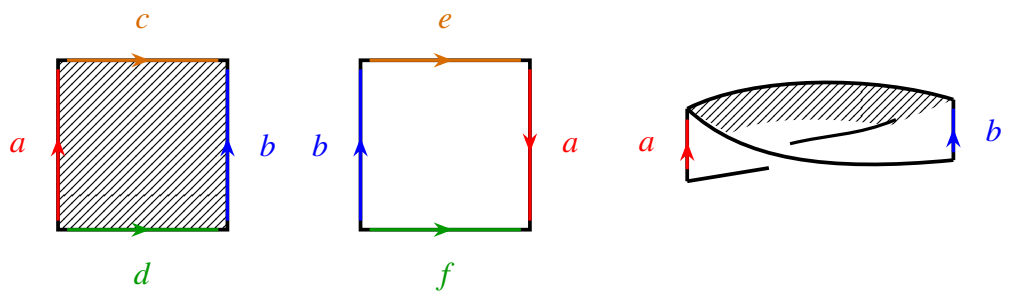


Figure 6.7: Constructing a Möbius band

Example 6.7. For Figure 6.8, the rectangle again has labelled edges (a, b, c, and d) with orientations. The relations here are $ac^{-1}a^{-1}d$ and $be^{-1}b^{-1}f$, describing how the edges should be glued. This time, edge a is identified with itself without any twists, resulting in a straightforward gluing. Similarly, edge b is identified with itself in a way that does not introduce any twists.

Identifying the edges in this way produces a cylinder, a surface with two boundaries (top and bottom) and an orientable structure. Unlike the Möbius band, the cylinder does not involve a twist, so it has two distinct sides.

To summarise, by identifying opposite edges with a twist, we obtain a non-orientable surface, the Möbius band, which has only one side; by identifying edges without a twist, we get a cylinder, an orientable surface with two distinct sides and boundaries.

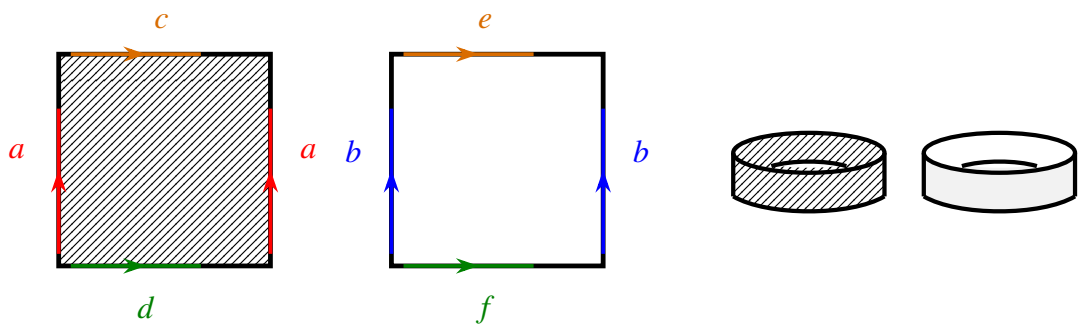


Figure 6.8: Constructing a cylinder

Theorem 6.1 (fundamental group of a polygonal edge-identification space). Let $P = P_n$ be an n-sided polygonal region in the plane, and let $X = P/\sim_w$ be the quotient space obtained by identifying the edges of P according to a labelling scheme

w. Suppose that all vertices of P are identified to a single point $x_0 \in X$. Let $a_1, \ldots, a_k$ be the distinct edge labels appearing in w. If the boundary word of P is

$$w = a_{i_1}^{\varepsilon_1} a_{i_2}^{\varepsilon_2} \cdots a_{i_n}^{\varepsilon_n},$$

where each $\varepsilon_j \in \{1, -1\}$ records whether the orientation of the jth edge agrees with the chosen orientation of the label a_{i_j}, then

$$\pi_1(X, x_0) \cong \left\langle a_1, \ldots, a_k \mid a_{i_1}^{\varepsilon_1} a_{i_2}^{\varepsilon_2} \cdots a_{i_n}^{\varepsilon_n} = 1 \right\rangle \cong \langle a_1, \ldots, a_k \mid w \rangle .$$

Definition 6.3 (n-fold torus). For any $n \geq 1$, the n-fold torus is the space obtained by pasting the edges of a $4n$-sided regular polygonal region P according to the labelling scheme

$$\left(a_1 b_1 a_1^{-1} b_1^{-1}\right) \cdots \left(a_n b_n a_n^{-1} b_n^{-1}\right).$$

This is also called the connected sum of n tori, denoted by $\mathbb{T}\#\ldots\#\mathbb{T}$.

Recall from Definition 3.13 that $[a_i, b_i] = a_i b_i a_i^{-1} b_i^{-1}$ is the commutator of a_i and b_i.

Example 6.8 (2-fold torus). The 2-fold torus, also called the genus two surface, is the connected sum $\mathbb{T}\#\mathbb{T}$.[1] Geometrically, this means that we take two ordinary tori, remove one open disk from each of them, and then glue the two resulting boundary circles together. The resulting surface has two holes, and is therefore a surface of genus two (Figure 6.10.

Figure 6.9 gives a polygonal schema for the genus two surface. The edges are labelled a_1, b_1, a_2, b_2. Each pair of edges with the same label is to be identified according to the indicated arrows. After all these identifications are made, every vertex of the polygon is identified to a single point, which becomes the base point of the quotient space.

The labels a_1, b_1 describe the two standard loops on the first torus, while a_2, b_2 describe the two standard loops on the second torus. Thus the 1-skeleton of the quotient space is a wedge of four circles $S^1 \vee S^1 \vee S^1 \vee S^1$ with generators a_1, b_1, a_2, b_2. The interior of the polygon becomes a single 2-cell attached along the boundary word of the polygon.

Using the usual convention in Definition 3.13, the boundary word of the genus two polygon is

$$[a_1, b_1][a_2, b_2] = a_1 b_1 a_1^{-1} b_1^{-1} a_2 b_2 a_2^{-1} b_2^{-1}.$$

Therefore, attaching the 2-cell imposes the single relation $[a_1, b_1][a_2, b_2] = 1$. Hence the fundamental group of the genus two surface is

$$\pi_1\left(\mathbb{T}^2\#\mathbb{T}^2, x_0\right) \cong \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = 1 \rangle .$$

[1] Here is a YouTube video which shows how one can visualise the construction of a 2-fold torus from an octagon.

The dashed curve c in Figure 6.9 should be understood as a cut separating the polygon into two pieces, each corresponding to a once-punctured torus. The first piece has boundary word $[a_1, b_1]$, while the second has boundary word $[a_2, b_2]$. When the two pieces are glued together along the cut c, the two boundary components are identified with opposite orientations.

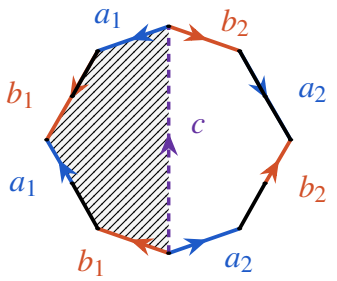


Figure 6.9: Polygonal schema for the 2-fold torus

Figure 6.10 gives a geometric picture of the same surface. It shows two tori joined together by the connected sum operation. Thus the polygonal diagram and the geometric diagram describe the same topological space: the genus two surface.

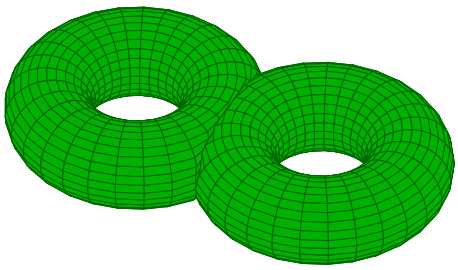


Figure 6.10: The 2-fold torus $\mathbb{T}\#\mathbb{T}$

Example 6.9 (3-fold torus). Similarly, we can obtain the 3-fold torus from a regular dodecagon (12-sided polygon) as shown in Figure 6.11.

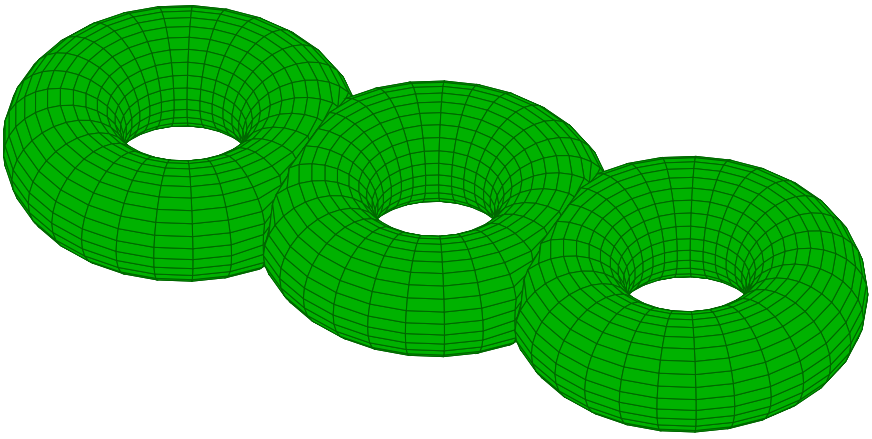


Figure 6.11: The 3-fold torus $\mathbb{T}\#\mathbb{T}\#\mathbb{T}$

Theorem 6.2 (fundamental group of n-fold torus). Let X denote the n-fold torus. Then, $\pi_1(X, x_0)$ has the presentation

$$\langle \alpha_1, \beta_1, \ldots, \alpha_n, \beta_n \mid [\alpha_1, \beta_1] \cdots [\alpha_n, \beta_n] \rangle .$$

Here, the presentation has $2n$ generators and a single relation.

Definition 6.4 (m-fold projective plane). For any $m > 1$, the m-fold projective plane is the space obtained by pasting the edges of a $2m$-sided polygonal region P according to the labelling scheme

$$(a_1 a_1)(a_2 a_2)\cdots(a_m a_m).$$

This is also called the connected sum of p projective planes, denoted by $\mathbb{P}^2\#\ldots\#\mathbb{P}^2$.

Example 6.10 (2-fold projective plane). Figure 6.12 illustrates the construction of the connected sum of two real projective planes. Recall that $\mathbb{P}^2$ may be represented by a polygonal region whose boundary labelling scheme is aa. To form the connected sum $\mathbb{P}^2\#\mathbb{P}^2$, we remove an open disc from each copy of $\mathbb{P}^2$, and then identify the two resulting boundary circles.

Equivalently, this construction can be described using labelling schemes. The connected sum of two projective planes is represented by the polygonal scheme $aabb$. Here, the two adjacent a-edges represent one copy of $\mathbb{P}^2$, while the two adjacent b-edges represent the other. Cutting the square along a diagonal labelled c separates the square into two polygonal regions with schemes aac^{-1} and cbb. The label c records the new boundary edge introduced by the cut. Thus the first piece is a projective plane with an open disc removed, and the second piece is another projective plane with an open disc removed. Pasting these two pieces together along the edges labelled c and c^{-1} recovers the original polygonal scheme $aabb$.

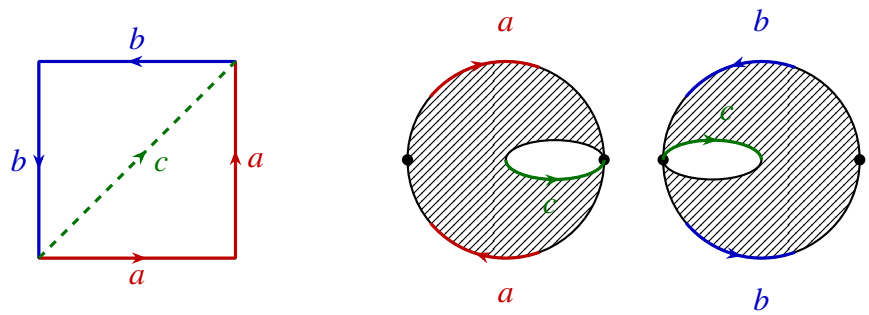


Figure 6.12: The connected sum of two projective planes $\mathbb{P}^2\#\mathbb{P}^2$

As with $\mathbb{P}^2$ itself, this surface does not admit a convenient faithful picture as an embedded surface in $\mathbb{R}^3$. In fact, $\mathbb{P}^2\#\mathbb{P}^2$ is homeomorphic to the Klein bottle (see Example 6.11), which cannot be embedded in $\mathbb{R}^3$ without self-intersection.

Example 6.11 (Klein bottle). The Klein bottle can be constructed from a square by identifying its boundary edges in pairs. Figure **??** illustrates the construction. In the left diagram, the square is regarded as a polygonal schema. The top and bottom edges are labelled a and are oriented in the same direction. The left and right edges are labelled b, but they are oriented in opposite directions. If we traverse the boundary of the square,

starting from the bottom-left vertex and moving anticlockwise, the corresponding boundary word is $aba^{-1}b$. Thus the Klein bottle has polygonal schema $aba^{-1}b$. Equivalently, its fundamental group admits the presentation

$$\pi_1(K) \cong \langle a, b \mid aba^{-1}b = 1 \rangle. \tag{6.1}$$

Geometrically, this says that conjugation by a reverses the direction of the loop b. This reversal is the algebraic shadow of the twist which appears in the construction of the Klein bottle.

We first glue the two edges labelled a. Since these two edges are oriented in the same direction, this gluing produces a cylinder. After this identification, the two remaining boundary components of the cylinder come from the two edges labelled b. The arrows on these two boundary circles point in opposite directions, indicating that the final gluing of the b-edges must be performed with a twist.

To complete the construction, we identify the two circular boundary components of the cylinder by an orientation-reversing identification. This twisted gluing is what distinguishes the Klein bottle from the torus. For the torus, both pairs of opposite edges are glued with compatible orientations. For the Klein bottle, one pair is glued with a reversal, and this is what makes the resulting surface non-orientable[2].

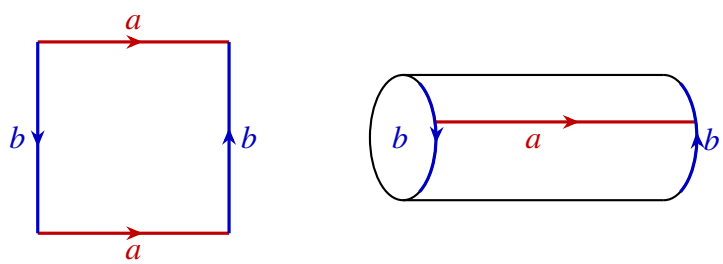


Figure 6.13: Identifying the two a-edges of the square produces a cylinder

The usual three-dimensional picture of the Klein bottle, shown in Figure 6.14, is only an immersed model in $\mathbb{R}^3$. The apparent self-intersection is not part of the actual topological space; it is an artefact of trying to draw the Klein bottle in three-dimensional space. The Klein bottle cannot be embedded in $\mathbb{R}^3$ without self-intersection, although it can be embedded in $\mathbb{R}^4$.

[2]Here is a YouTube video which shows how one can visualise the construction of the Klein bottle from a square.

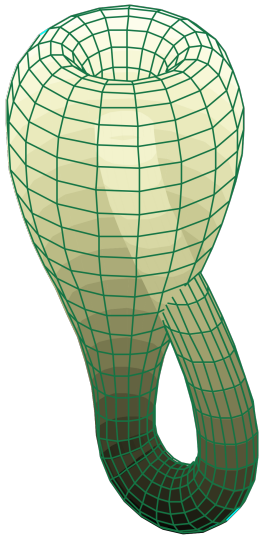


Figure 6.14: The Klein bottle

In group theory, the Baumslag-Solitar groups (Definition 6.5) are examples of two-generators one-relation groups.

Definition 6.5 (Baumslag-Solitar group). The Baumslag-Solitar groups have the presentation

$$\mathrm{BS}(m,n) = \langle a,b \mid ba^m b^{-1} = a^n \rangle .$$

Example 6.12. $\mathrm{BS}(1,1)$ is the free Abelian group on two generators; $\mathrm{BS}(1,-1)$ is the fundamental group of the Klein bottle (see Example 6.13 where the Munkres textbook presents a question on this).

Example 6.13 (Munkres p. 454 Question 3). The Klein bottle K is the space obtained from a square by means of the labelling scheme $aba^{-1}b$. Figure 6.14 indicates how K can be pictured as an immersed surface in $\mathbb{R}^3$.

(a) Find a presentation for the fundamental group of K.

(b) Find a double covering map $p : \mathbb{T} \to K$, where $\mathbb{T}$ is the torus. Describe the induced homomorphism of fundamental groups.

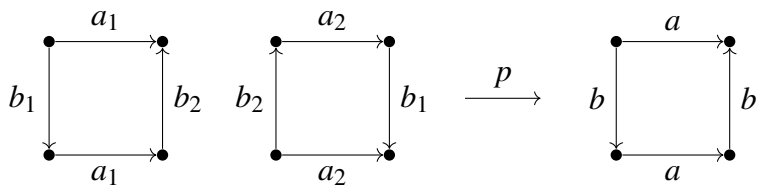
Solution.

(a) As shown in Figure 6.13, in a presentation of the fundamental group of the Klein bottle, it should contain the generators are a and b with the relation $aba^{-1}b = 1$. As such, we obtain (6.1).

(b) Each diagram on the left represents a torus $\mathbb{T}$, whereas the diagram on the right represents the Klein bottle K. So, we can define the covering map $p : \mathbb{T} \to K$ as follows:

$$p([s,t]) = \begin{cases} [2s,t] & \text{if } 0 \le t \le 1/2; \\ [2s-1, 1-t] & \text{if } 1/2 \le t \le 1. \end{cases}$$

So, the map p takes the first square directly to the square on the right, and it flips the second square vertically (along the middle horizontal line). By the pasting lemma, this map p is continuous on the image of each of the two squares and agrees on the overlap.



We then compute the induced homomorphism p_*, where

$$p_* : \pi_1(\mathbb{T}, x_0) \to \pi_1(K, y_0).$$

Observe that

$$p_*(a_1) = p_*(a_2) = a \quad \text{and} \quad p_*(b_1) = p_*(b_2) = b.$$

Note that $\alpha = a_1 a_2$ and $\beta = b_1$ are loops. As $p_*(\alpha) = a^2$ and $p_*(\beta) = b$ and the relation $aba^{-1}b = 1$ in the presentation of K ensures that the two equations induce a homomorphism on $\pi_1(\mathbb{T}, x_0)$, i.e. $p_*(\alpha) \cdot p_*(\beta) = p_*(\beta) \cdot p_*(\alpha)$. □

Theorem 6.3 (fundamental group of m-fold projective plane). Let X denote the m-fold projective plane. Then, $\pi_1(X, x_0)$ has the presentation

$$\left\langle \alpha_1, \ldots, \alpha_m \mid (\alpha_1)^2 \ldots (\alpha_m)^2 \right\rangle.$$

Here, the presentation is given by m generators and a single relation.

6.2 Homology of Surfaces

We wish to give a good classification, i.e. give a list of compact surfaces such that the surfaces obtained by pasting the edges of disjoint polygonal regions, according to labelling schemes $w_1, \ldots, w_m$, are such that no two surfaces on the list are homeomorphic and every compact surface is homeomorphic to one of them.

The spaces of interest are the 2-sphere S^2, the n-fold torus $\mathbb{T}\#\cdots\#\mathbb{T}$ where $n \geq 1$, and the m-fold projective plane $\mathbb{P}^2\#\cdots\#\mathbb{P}^2$ where $m \geq 1$.

Recall from MA2202 Algebra I that there is no effective procedure in general for the isomorphism problem for groups, i.e. given two finite presentations of groups, it is very difficult to determine whether or not the groups are isomorphic. However, for finitely generated groups with one relation, we can consider the abelianization of a group, which is defined as follows:

Definition 6.6 (abelianization). The abelianization of a group G is the maximal abelian quotient $G^{\mathrm{ab}} = G/[G,G]$ of G.

Definition 6.7 (abelianized fundamental group). Let X be a path-connected topological space. The abelianized fundamental group of X is

$$\pi_1^{\mathrm{ab}}(X) = \pi_1(X,x_0)^{\mathrm{ab}} = \pi_1(X,x_0)/[\pi_1(X,x_0),\pi_1(X,x_0)]$$

for any choice of base point $x_0 \in X$.

Example 6.14 (abelian groups). If G is already abelian, then every commutator is trivial. Hence, $[G,G] = \{e\}$ so $G^{\mathrm{ab}} \cong G$. One can take G to be $\mathbb{Z}$ or $\mathbb{Z}/n\mathbb{Z}$ for example.

Example 6.15 (the symmetric group S_3). Consider the symmetric group S_3. Its commutator subgroup is the alternating group A_3. Hence $[S_3,S_3] = A_3$. As such, $S_3^{\mathrm{ab}} = S_3/A_3$. Since Since A_3 has index 2 in S_3, we get $S_3^{\mathrm{ab}} \cong \mathbb{Z}/2\mathbb{Z}$. Equivalently, the abelianization remembers only the sign of a permutation.

Lemma 6.1. For any $x_1 \in X$, choose a path α in X from x_0 to x_1. The isomorphism

$$\widehat{\alpha} : \pi_1(X,x_0) \to \pi_1(X,x_1) \quad \text{where} \quad [f] \mapsto \alpha([f]) = [\widetilde{\alpha}] * [f] * [\alpha]$$

induces the same isomorphism

$$\pi_1(X,x_0)^{\mathrm{ab}} \cong \pi_1(X,x_1)^{\mathrm{ab}} \quad \text{independent of the choice of the path } \alpha.$$

Proof. Suppose β is another path in X from x_0 to x_1. Then, $g = \alpha * \widetilde{\beta}$ is a loop in X at x_0, so $[g] \in \pi_1(X,x_0)$. As such, $\widehat{g} : \pi_1(X,x_0) \to \pi_1(X,x_0)$ given by $[f] \mapsto \widehat{g}[f] = [g]^{-1} * [f] * [g]$ (conjugation by $[g]^{-1}$) induces the identity isomorphism on $\pi_1(X,x_0)^{\mathrm{ab}}$. Hence, $\widehat{\alpha} = \widehat{\beta}$. □

The homology groups $H_n(X)$ of X for all $n \geq 0$ are defined in homology theory which would eventually be covered in MA5209 Algebraic Topology. Moreover, such graduate course would cover theorems such as Hurewicz theorem, which states that $H_1(X) \cong \pi_1^{\mathrm{ab}}(X)$ canonically.

Theorem 6.4 (universal property of abelianization). Let F be a group and $N \trianglelefteq F$. Let

$$p : F \to F^{\mathrm{ab}} = F/[F,F] \quad \text{be the quotient homomorphism.}$$

Then,

$$F^{\mathrm{ab}}/p(N) \cong (F/N)^{\mathrm{ab}} \quad \text{canonically.}$$

Proof. Both $F^{\mathrm{ab}}/p(N)$ and $(F/N)^{\mathrm{ab}}$ are Abelian groups with the same universal property. For any homomorphism $h : F \to H$ from any group F to an Abelian group H such that $N \subseteq \ker h$, there exists unique homomorphism

$$F^{\mathrm{ab}}/p(N) \to H \quad \text{and} \quad (F/N)^{\mathrm{ab}} \to H$$

such that the following diagram commutes:

$$\begin{tikzcd}
 & F^{\mathrm{ab}} \arrow[r, two heads] & F^{\mathrm{ab}}/p(N) \arrow[dr, dotted] & \\
F \arrow[ur, two heads, "p"] \arrow[rr, "h"] \arrow[dr, two heads, "q"'] & & & H \\
 & F/N \arrow[r, two heads] & (F/N)^{\mathrm{ab}} \arrow[ur, dotted] &
\end{tikzcd}$$

The result follows. □

Corollary 6.1. Let G be a group with the presentation $\langle \alpha_1, \ldots, \alpha_n \mid x \rangle$, where $\alpha, \ldots, \alpha_n$ are the n generators and x is the relation. Let

$$F = \langle \alpha_1, \ldots, \alpha_n \rangle \quad \text{be the free group generating } G.$$

Define $p : F \to F^{\mathrm{ab}} = \mathbb{Z}^n$ to be the quotient homomorphism. Then, $G^{\mathrm{ab}} \cong \mathbb{Z}^n/p(x)$ canonically.

Proof. Write $G = F/N$, where $N = \langle x \rangle \trianglelefteq F$ is the normal subgroup of F generated by $\langle x \rangle$. The result follows by Theorem 6.4. □

Theorem 6.5 (fundamental group of n-fold torus). Let X denote the n-fold torus $\mathbb{T}\#\ldots\#\mathbb{T}$. Then,

$$\pi_1^{\mathrm{ab}}(X) \cong \mathbb{Z}^{2n} \quad \text{is free Abelian of rank } 2n.$$

Proof. Recall that $\pi_1(X, x)$ has the presentation

$$\langle \alpha_1, \beta_1, \ldots, \alpha_n, \beta_n \mid [\alpha_1, \beta_1] \ldots [\alpha_n, \beta_n] \rangle.$$

The element $[\alpha_1, \beta_1] \ldots [\alpha_n, \beta_n] \in \langle \alpha_1, \beta_1, \ldots, \alpha_n, \beta_n \rangle$ maps to 0 in $\mathbb{Z}^{2n}$. □

Theorem 6.6. Let X denote the m-fold projective plane $\mathbb{P}^2\#\ldots\#\mathbb{P}^2$. Then,

$$\pi_1^{\mathrm{ab}}(X) \cong \mathbb{Z}^{m-1} \oplus \mathbb{Z}/2\mathbb{Z}$$

has torsion subgroup of order 2 and it is free quotient of rank $m-1$ (Copy phrasing from Munkres).

Proof. Recall that $\pi_1(X, x_0)$ has the presentation

$$\left\langle \alpha_1, \ldots, \alpha_m \mid (\alpha_1)^2 \ldots (\alpha_m)^2 \right\rangle.$$

The element $(\alpha_1)^2 \ldots (\alpha_m)^2$ maps to $p(x) = 2(\alpha_1 + \alpha_m) \in \mathbb{Z}^m$. Changing bases in $\mathbb{Z}^m$ to $\alpha_1, \ldots, \alpha_{m-1}$, with $\beta = \alpha_1 + \ldots + \alpha_m$, we have $p(x) = 2\beta$, so $\pi_1^{\mathrm{ab}}(X) \cong (\mathbb{Z}\alpha_1 \oplus \ldots \oplus \mathbb{Z}\alpha_{m-1}) \oplus \mathbb{Z}\beta/2\beta$.

Here, we note that $\mathbb{Z}\alpha_1 \oplus \ldots \oplus \mathbb{Z}\alpha_{m-1} \cong \mathbb{Z}^{m-1}$ and $\mathbb{Z}\beta/2\beta \cong \mathbb{Z}/2\mathbb{Z}$. □

Theorem 6.7. Let $\mathbb{T}_n$ and $\mathbb{P}_m$ denote the n-fold connected sum of tori and the m-fold connected sum of projective planes, respectively. Then the surfaces $S^2; \mathbb{T}_1, \mathbb{T}_2, \ldots; \mathbb{P}_1, \mathbb{P}_2, \ldots$ are topologically distinct.

6.3 Cutting and Pasting

The method of cutting and pasting is one of the most useful ways of understanding compact surfaces. Instead of trying to visualise a surface directly in three-dimensional space, we describe it by a polygon whose edges are identified in pairs according to a prescribed labelling scheme. The resulting quotient space often gives familiar surfaces such as the torus, the Klein bottle, the projective plane, and higher genus surfaces.

The advantage of this approach is that topological information can be encoded combinatorially. A surface is represented by a word in edge labels, and operations on the surface become operations on this word. Cutting a polygon introduces a new pair of identified edges, while pasting reverses this process. Similarly, relabelling, cyclically permuting, or reversing the orientation of a polygon changes only the description of the surface, not the surface itself.

We now describe these elementary operations precisely. As shown in Figure 6.15, let P be a polygonal region with successive vertices $p_0, \ldots, p_n$, where $p_n = p_0$. For $1 < k < n-1$, let

$$\begin{aligned} Q_1 &= \text{the polygonal region with successive vertices} \quad p_0, p_1, \ldots, p_k, p_0 \\ Q_2 &= \text{the polygonal region with successive vertices} \quad p_0, p_k, \ldots, p_n = p_0 \end{aligned}$$

where they have the edge $p_0 p_k$ in common. Define Q_1' to be any translation of Q_1 disjoint from Q_2 with successive vertices $q_0, q_1, \ldots, q_k, q_0$.

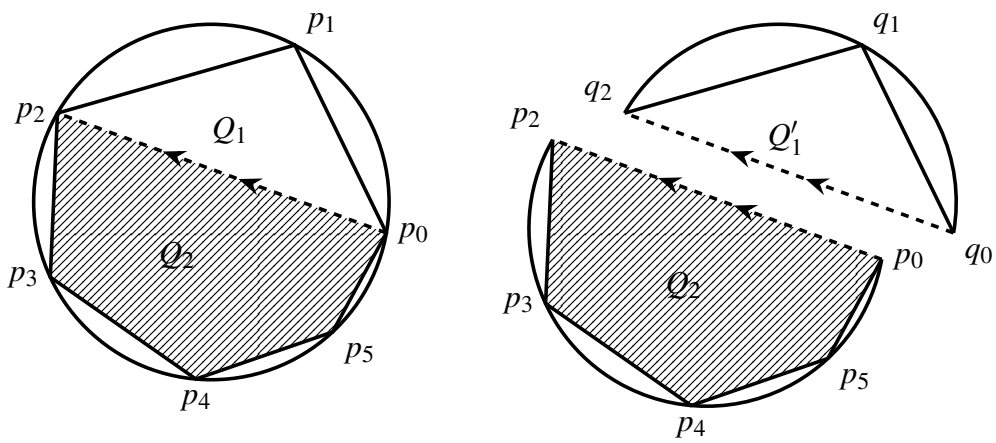


Figure 6.15

Then, P is homeomorphic to the quotient space of Q_1', Q_2 obtained by pasting the edge q_0q_k of Q_2 to the edge p_0p_k of Q_1'.

Say that the polygonal region P has labelling scheme $w = y_0y_1$, where y_0 denotes the first k consecutive labels along the boundary of P and y_1 denotes the remaining consecutive labels. Cutting P along a diagonal from the initial vertex of y_0 to the terminal vertex of y_0 separates P into two polygonal regions, say Q_1' and Q_2. If c is a new label which does not occur anywhere else in the total labelling scheme, then the new edge introduced by the cut is labelled c on one side and c^{-1} on the other side. Thus Q_1' has labelling scheme y_0c^{-1}, while Q_2 has labelling scheme cy_1. The opposite orientations of c and c^{-1} indicate that these two new edges are to be identified with opposite orientations when the two pieces are pasted back together.

Elementary operations on labelling schemes of disjoint polygonal regions are the following.

(i) Cut: Suppose one of the polygonal regions has labelling scheme $w_i = y_0y_1$, where y_0 and y_1 each have length at least two. We may cut the polygonal region along a diagonal joining the initial vertex of y_0 to the terminal vertex of y_0. This replaces the single scheme y_0y_1 by the two schemes

$$y_0c^{-1} \quad \text{and} \quad cy_1,$$

where c is a new label which does not appear anywhere else in the total labelling scheme. The labels c and c^{-1} represent the two sides of the newly created cut edge, with opposite orientations. Pasting these two new edges together recovers the original polygonal region. Hence the quotient space is unchanged.

(ii) Relabel: One may replace every occurrence of a label a by a new label b, provided b does not already occur elsewhere in the total labelling scheme. This is simply a change of notation and does not alter the pasting map.

Similarly, one may replace every occurrence of a by a^{-1}, and every occurrence of a^{-1} by a. Geometrically, this reverses the chosen orientation of all edges labelled a. Since all occurrences are changed simultaneously, the resulting quotient space is unchanged.

(iii) **Permute:** One may replace a scheme w_i by any cyclic permutation of itself. More explicitly, if $w_i = y_0 y_1$, then one may replace w_i with $y_1 y_0$. This corresponds to choosing a different starting vertex when reading the boundary labels of the polygonal region P_i. Since the polygon and its edge identifications have not changed, the quotient space is unchanged.

(iv) **Flip:** One may replace a scheme

$$w_i = (a_{i_1})^{\varepsilon_1}(a_{i_2})^{\varepsilon_2}\cdots(a_{i_n})^{\varepsilon_n},$$

where each $\varepsilon_j \in \{\pm 1\}$, by its formal inverse

$$w_i^{-1} = (a_{i_n})^{-\varepsilon_n}(a_{i_{n-1}})^{-\varepsilon_{n-1}}\cdots(a_{i_1})^{-\varepsilon_1}.$$

Geometrically, this amounts to flipping the polygonal region P_i over. The order in which the boundary edges are read is reversed, and the orientation of every edge is also reversed. The pasting map, and hence the quotient space, is unchanged.

(v) **Cancel:** Suppose one of the schemes has the form $w_i = y_0 a a^{-1} y_1$, where a is a label which does not appear anywhere else in the total labelling scheme, and where y_0 and y_1 each have length at least two. Then the adjacent edges labelled a and a^{-1} are pasted together with opposite orientations. These two edges therefore cancel each other geometrically, and we may replace the scheme w_i by $y_0 y_1$. This operation removes a pair of adjacent edges which are identified with each other and does not change the quotient space.

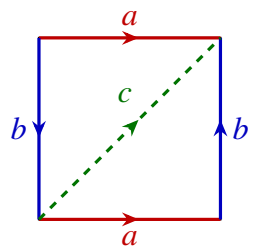
(vi) **Uncancel:** This is the reverse of the cancellation operation. Namely, one may replace a scheme $y_0 y_1$ with $y_0 a a^{-1} y_1$, where a is a new label which does not appear elsewhere in the total labelling scheme. Geometrically, this inserts a pair of adjacent edges which are immediately identified with each other in opposite orientations. Since these two inserted edges cancel after pasting, the quotient space is unchanged.

Definition 6.8 (equivalent labelling schemes). Two labelling schemes for disjoint polygonal regions are equivalent if and only if one can be obtained from the other by a finite sequence of elementary operations.

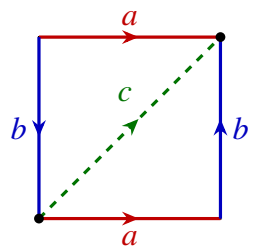
Note that the spaces obtained by pasting the edges of disjoint polygonal regions according to equivalent labelling schemes are homeomorphic to each other.

Example 6.16 (Klein bottle is homeomorphic to the connected sum of projective planes). We note that the Klein bottle K is homeomorphic to the connected sum of the projective planes. That is to say, K is homeomorphic to $\mathbb{P}^2 \# \mathbb{P}^2$. This is because their labelling schemes are equivalent. Recall the respective labelling schemes in Figures 6.13 and 6.12.

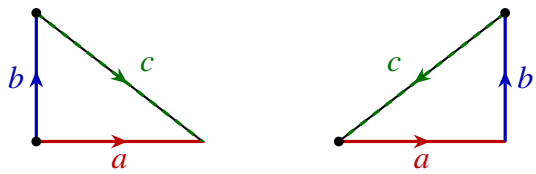
Start with the labelled square with labelling scheme $aba^{-1}b^{-1}$.



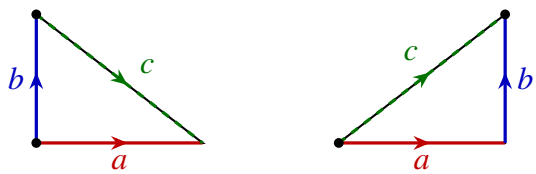
We then cut along the diagonal c to obtain the labelling schemes abc^{-1} and $ca^{-1}b$.



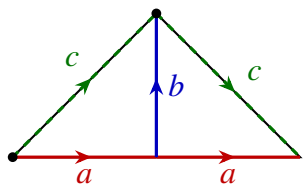
Then, cyclically permute the first boundary word to obtain the labelling schemes $c^{-1}ab$ and $ca^{-1}b$.



We flip the second triangle to obtain the labelling schemes $c^{-1}ab$ and $b^{-1}ac^{-1}$.



Pasting along the b-edges, we obtain the following:



After cyclic permutation, we obtain the labelling scheme $aac^{-1}c^{-1}$, which is the labelling scheme of the connected sum of projective planes as shown in Figure 6.12.

6.4 The Classification Theorem

Recall that the examples $S^2, \mathbb{T}_n$ (where $n \geq 1$) and $\mathbb{P}_m$ (where $m \geq 1$) are obtained by pasting the edges in pairs from a single polygonal region. Our current goal is to provide a topological classification of surfaces obtained by quotients of a polygonal region.

Theorem 6.8 (classification theorem). Let X be the space obtained by pasting the edges in pairs from a single polygonal region. Then, X is homeomorphic to either of the following:

$$\text{the 2-sphere } S^2 \quad \text{or} \quad \text{the } n\text{-fold torus } \mathbb{T}_n \quad \text{or} \quad \text{the } m\text{-fold projective plane } \mathbb{P}_m,$$

where $n, m \geq 1$.

Definition 6.9 (proper labelling scheme). A labelling scheme $w_1, \ldots, w_k$ of disjoint polygonal regions $P_1, \ldots, P_k$ is proper if and only if each label appears exactly twice in $w_1, \ldots, w_k$.

Theorem 6.9 (classification of proper labelling schemes). Let w be a proper labelling scheme of a single polygonal region of length ≥ 4. Then, w is equivalent to either of the following:

(i) $aa^{-1}bb^{-1}$

(ii) $abab$

(iii) $(a_1a_1)\cdots(a_ma_m)$ where $m \geq 2$

(iv) $\left(a_1b_1a_1^{-1}b_1^{-1}\right)\cdots\left(a_nb_na_n^{-1}b_n^{-1}\right)$ where $n \geq 1$

Definition 6.10 (torus labelling scheme). Let w be a proper labelling scheme of a single polygonal region. Then, w is of torus type if and only if each label appears exactly once with exponent ± 1. Otherwise, w is of projective type.

Lemma 6.2. If $w = [y_0]\,a\,[y_1]\,a\,[y_2]$, then $w \sim aa\left[y_0y_1^{-1}y_2\right]$. Here, w denotes a proper labelling scheme of a single polygonal scheme, and y_i denotes labelling schemes that may be empty, whereas a denotes a label.

Corollary 6.2. If w is a proper labelling scheme of projective type, then w is equivalent to a scheme of the same length of the form

$$(a_1a_1)\cdots(a_ka_k)\,w_1 \quad \text{where } k \geq 1 \text{ with } w_1 \text{ of torus type or empty.}$$

Corollary 6.3. If $w = w_0\,(cc)\left(aba^{-1}b^{-1}\right)w$, then $w \sim w_0\,(aabbcc)\,w_1$.

Lemma 6.3. If $w = w_0w_1$, where w_1 is of torus type with no two adjacent terms having the same label, then $w \sim w_0w_2$, where $w_2 = aba^{-1}b^{-1}w_3$ with w_2 having the same length as w_1 and w_3 of torus type or empty.

6.5 Constructing Compact Surfaces

The main goal of this section is to show that every compact connected surface admits a polygonal description. More precisely, we shall prove that if X is a compact connected surface, then X is homeomorphic to a quotient space obtained from a single polygonal region by identifying its edges in pairs (Theorem 6.10).

This result is the first major step towards the classification of compact surfaces. Instead of studying an arbitrary compact surface directly, we reduce it to a finite combinatorial object: a polygon together with a proper labelling scheme on its boundary. Once such a polygonal model has been obtained, the topological information of the surface is encoded in the cyclic word determined by the labelled and oriented edges of the polygon.

The strategy is as follows. First, we use Radó's triangulation theorem (Theorem 6.11), which says that every compact surface can be described using finitely many triangular regions whose edges are pasted together in pairs. Then, using the connectedness of X, we paste these triangles together one at a time whenever two distinct triangular regions have edges with the same label. Repeating this process reduces the finite collection of triangular regions to a single polygonal region. The resulting edge identifications give the desired polygonal presentation of X.

As mentioned, it suffices to establish Theorem 6.10.

Theorem 6.10. Let X be a compact connected surface. Then, X is homeomorphic to the space obtained by pasting the edges in pairs from a single polygonal region.

Theorem 6.11 (Radó's triangulation theorem). Let X be a compact surface. Then, X is homeomorphic to the space obtained by pasting the edges in pairs from a finite collection of disjoint triangular regions.

We first give a proof of Theorem 6.10 before proving Radó's triangulation theorem (Theorem 6.11).

Proof. By Radó's theorem (Theorem 6.11), there exists a finite collection of disjoint polygonal triangular regions $T_1, \ldots, T_n$, and a proper labelling scheme of these regions such that X is homeomorphic to the space obtained by pasting the edges of these regions according to the labelling scheme.

If there exist two edges in distinct regions with the same label, then by flipping one of the regions if necessary, we can paste these regions along these two edges. As such, the number of polygonal regions is reduced by 1, making the labelling scheme still proper.

By induction, this is reduced to the situation where either one has only a single polygonal

region with proper labelling scheme (for which we are done) or one has more than one polygonal region such that no two distinct regions missing! □

Definition 6.11 (curved triangle). Let T be the regular 3-gon in $\mathbb{R}^2$ and X be a topological space. A curved triangle in X is a subspace $A \subseteq X$ equipped with a homeomorphism $h : T \to A$. The vertices and edges of A are the images of those of T.

Example 6.17 (a curved triangle in $\mathbb{R}^2$). Let $X = \mathbb{R}^2$ and

$$T = \left\{(u,v) \in \mathbb{R}^2 : u \geq 0, v \geq 0, u+v \leq 1\right\}.$$

Consider the homeomorphism

$$h : T \to \mathbb{R}^2 \quad \text{where} \quad h(u,v) = \left(u, v + \frac{1}{4}u(1-u)\right).$$

Then, $A = h(T) \subseteq \mathbb{R}^2$. So, A is a curved triangle in $\mathbb{R}^2$. The vertices of A are $(0,0)$, $(1,0)$, and $(0,1)$. The edges of A are the images under h of the three edges of T. For instance, as shown in Figure 6.16, the bottom edge of T is mapped to the curved edge

$$\left\{\left(u, \frac{1}{4}u(1-u)\right) : 0 \leq u \leq 1\right\}.$$

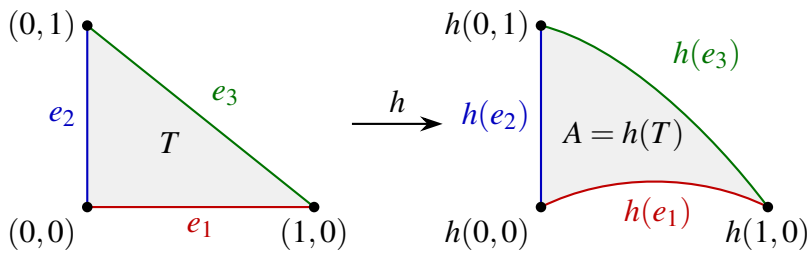


Figure 6.16: A curved triangle A is the homeomorphic image of an ordinary triangle T

Definition 6.12 (triangulation). Let X be a topological space. A triangulation of X is a collection $\{A_\alpha\}_\alpha$ of subspaces of X satisfying the following properties:

(i) for each α, the subspace A_α is a curved triangle in X, so homeomorphic to T (Definition 6.11)

(ii) for each $\alpha \neq \beta$, we have

$$A_\alpha \cap A_\beta = \begin{cases} \emptyset; \\ \text{a vertex of both } A_\alpha, A_\beta; \\ \text{an edge of both } A_\alpha, A_\beta \end{cases}$$

and if $e = A_\alpha \cap A_\beta$ is an edge of both and

$$h_\alpha : T \to A_\alpha \text{ and } h_\beta : T \to A_\beta \quad \text{are the given homeomorphisms,}$$

then $h_\beta^{-1} h_\alpha : h_\alpha^{-1}(e) \to h_\beta^{-1}(e)$ is a linear homeomorphism between the edges of T

(iii) $X = \bigcup_\alpha A_\alpha$ and it is given with the colimit topology

Definition 6.13 (triangularizable). Let X be a topological space. Then,

$$X \text{ is triangularizable} \quad \text{if and only if} \quad \text{there exists a triangulation of } X.$$

Example 6.18 (triangulation of a square). Let $X = [0,1] \times [0,1] \subseteq \mathbb{R}^2$. Also, we define $\operatorname{conv}\{p_1, \ldots, p_n\}$ to be the smallest convex set containing the points $p_1, \ldots, p_n$. Consider the two triangular subspaces

$$A_1 = \operatorname{conv}\{(0,0),(1,0),(1,1)\} \quad \text{and} \quad A_2 = \operatorname{conv}\{(0,0),(1,1),(0,1)\}.$$

Then A_1 and A_2 are both homeomorphic to the standard triangle T. Their intersection is precisely the diagonal edge $A_1 \cap A_2 = \operatorname{conv}\{(0,0),(1,1)\}$ which is an edge of both triangles. Moreover, $X = A_1 \cup A_2$. Hence, $\{A_1, A_2\}$ is a triangulation of the square X.

Intuitively, this says that a square can be regarded as a space made by gluing two triangles along one common edge (Figure 6.17).

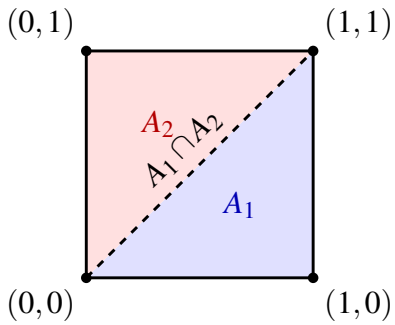


Figure 6.17: A triangulation of the square by two triangles

We now prove Radó's theorem (Theorem 6.11).

Proof. Let X be a compact surface. Choose a finite triangulation $X = \bigcup_{i=1}^n A_i$, where each A_i is a curved triangle. For each i, let $h_i : T_i \to A_i$ be the corresponding homeomorphism, where the T_i's are pairwise disjoint triangular regions in $\mathbb{R}^2$.

Let

$$E = \bigsqcup_{i=1}^n T_i \quad \text{be} \quad \text{the disjoint union of these triangular regions.}$$

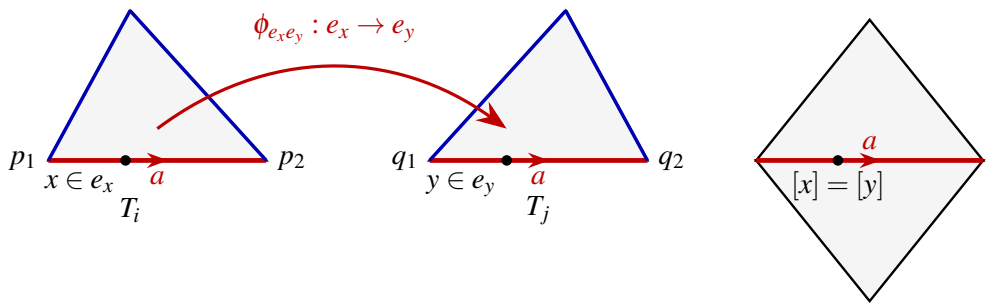
The maps h_i define a continuous surjective map $h : E \to X$ where $h|_{T_i} = h_i$. Since E is compact and X is Hausdorff, the map h is closed. Hence h is a quotient map. Therefore X has the quotient topology induced by h.

We now describe the equivalence relation induced by h. Let e be an edge of one of the curved triangles A_i. Since X is a surface without boundary, every point in the interior of e has a neighbourhood homeomorphic to an open disc. Hence e must be shared by exactly one other curved triangle A_j. Therefore the map $h : E \to X$ identifies the edges of the triangles in pairs.

Label each pair of identified edges with the same label, with arrows chosen so that the corresponding edge-identification is linear. This gives a proper labelling scheme w on the edges of E. Let $\sim_w$ be the equivalence relation on E generated by the following elementary identifications:

$$x \sim_w y$$

whenever either $x = y$, or there exist two edges e_x, e_y with the same label such that $x \in e_x$, $y \in e_y$, and $y = \phi_{e_x e_y}(x)$, where $\phi_{e_x e_y} : e_x \to e_y$ is the linear homeomorphism determined by the labelling and the chosen orientations.



By construction,

$$x \sim_w y \quad \text{implies} \quad h(x) = h(y).$$

Thus h factors through the quotient $E/\sim_w$. Hence, there exists a continuous map $\overline{h} : E/\sim_w \longrightarrow X$ such that $h = \overline{h} \circ q$, where $q : E \to E/\sim_w$ is the quotient map.

It remains to prove that $\overline{h}$ is injective. Suppose $h(x) = h(y)$. If x and y lie in the interiors of triangles, then $x = y$, because each $h_i : T_i \to A_i$ is a homeomorphism. If x and y lie in the interiors of edges, then they are identified by the edge-pairing, and hence $x \sim_w y$. Thus the only remaining case is when x and y are vertices of triangles.

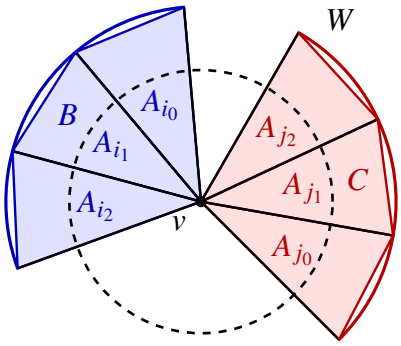
Let $v = h(x) = h(y) \in X$. We claim that all curved triangles having v as a vertex are connected through a chain of edge-adjacencies at v. More precisely, if A_i and A_j both have v as a vertex, then there exists a sequence

$$A_i = A_{i_0}, A_{i_1}, \ldots, A_{i_m} = A_j$$

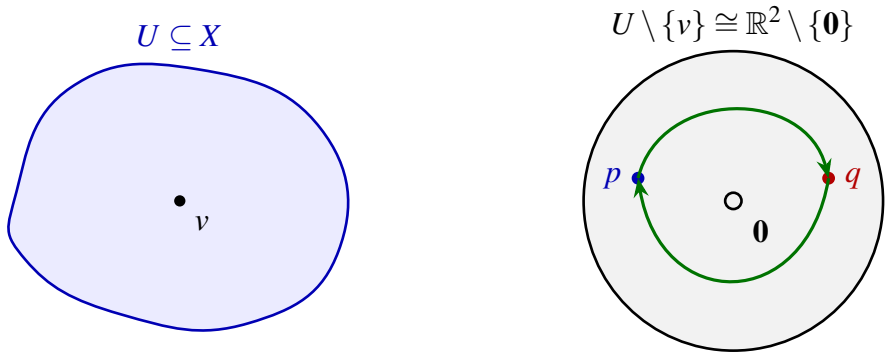
of curved triangles, all having v as a vertex, such that $A_{i_r} \cap A_{i_{r+1}}$ is an edge of both triangles containing v, for every $r = 0, \ldots, m-1$. Suppose otherwise. Define a relation on the set of curved triangles having v as a vertex by declaring $A_i \sim A_j$ if and only if there exists such a sequence of curved triangles from A_i to A_j. This is clearly an equivalence relation. If there is more than one equivalence class, let B be the union of the curved triangles in

one equivalence class, and let C be the union of the curved triangles in all the remaining equivalence classes.

Then no triangle in B shares an edge containing v with a triangle in C. Hence, near v, the sets B and C meet only at v. Therefore, for every sufficiently small neighbourhood W of v contained in $B \cup C$, we have $W \setminus \{v\}$ disconnected.



However, since X is a surface, there exists a neighbourhood U of v homeomorphic to an open disc in $\mathbb{R}^2$. We may choose U sufficiently small. Then $U \setminus \{v\} \cong \mathbb{R}^2 \setminus \{\mathbf{0}\}$. But $\mathbb{R}^2 \setminus \{\mathbf{0}\}$ is connected. This is a contradiction. Hence all triangles incident to v lie in a single edge-adjacency chain.



It follows that the preimage vertices x and y are identified by a finite chain of edge-identifications. Hence $x \sim_w y$. Therefore, $h(x) = h(y)$ implies $x \sim_w y$. Thus, the induced map $\overline{h} : E/\sim_w \to X$ is bijective.

Finally, $E/\sim_w$ is compact because it is the quotient of the compact space E, and X is Hausdorff. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism. Therefore $E/\sim_w \cong X$. Hence X is obtained by pasting together finitely many triangular regions according to a proper labelling scheme. $\square$

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